

# PROPERTY TABLES AND CHARTS (SI UNITS)

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TABLE A-1

Molar mass, gas constant, and critical-point properties

| Substance                      | Formula                           | Molar mass,<br><i>M</i> kg/kmol | Gas<br>constant,<br><i>R</i> kJ/kg·K* | Critical-point properties |                  |                                 |
|--------------------------------|-----------------------------------|---------------------------------|---------------------------------------|---------------------------|------------------|---------------------------------|
|                                |                                   |                                 |                                       | Temperature,<br>K         | Pressure,<br>MPa | Volume,<br>m <sup>3</sup> /kmol |
| Air                            | —                                 | 28.97                           | 0.2870                                | 132.5                     | 3.77             | 0.0883                          |
| Ammonia                        | NH <sub>3</sub>                   | 17.03                           | 0.4882                                | 405.5                     | 11.28            | 0.0724                          |
| Argon                          | Ar                                | 39.948                          | 0.2081                                | 151                       | 4.86             | 0.0749                          |
| Benzene                        | C <sub>6</sub> H <sub>6</sub>     | 78.115                          | 0.1064                                | 562                       | 4.92             | 0.2603                          |
| Bromine                        | Br <sub>2</sub>                   | 159.808                         | 0.0520                                | 584                       | 10.34            | 0.1355                          |
| <i>n</i> -Butane               | C <sub>4</sub> H <sub>10</sub>    | 58.124                          | 0.1430                                | 425.2                     | 3.80             | 0.2547                          |
| Carbon dioxide                 | CO <sub>2</sub>                   | 44.01                           | 0.1889                                | 304.2                     | 7.39             | 0.0943                          |
| Carbon monoxide                | CO                                | 28.011                          | 0.2968                                | 133                       | 3.50             | 0.0930                          |
| Carbon tetrachloride           | CCl <sub>4</sub>                  | 153.82                          | 0.05405                               | 556.4                     | 4.56             | 0.2759                          |
| Chlorine                       | Cl <sub>2</sub>                   | 70.906                          | 0.1173                                | 417                       | 7.71             | 0.1242                          |
| Chloroform                     | CHCl <sub>3</sub>                 | 119.38                          | 0.06964                               | 536.6                     | 5.47             | 0.2403                          |
| Dichlorodifluoromethane (R-12) | CCl <sub>2</sub> F <sub>2</sub>   | 120.91                          | 0.06876                               | 384.7                     | 4.01             | 0.2179                          |
| Dichlorofluoromethane (R-21)   | CHCl <sub>2</sub> F               | 102.92                          | 0.08078                               | 451.7                     | 5.17             | 0.1973                          |
| Ethane                         | C <sub>2</sub> H <sub>6</sub>     | 30.070                          | 0.2765                                | 305.5                     | 4.48             | 0.1480                          |
| Ethyl alcohol                  | C <sub>2</sub> H <sub>5</sub> OH  | 46.07                           | 0.1805                                | 516                       | 6.38             | 0.1673                          |
| Ethylene                       | C <sub>2</sub> H <sub>4</sub>     | 28.054                          | 0.2964                                | 282.4                     | 5.12             | 0.1242                          |
| Helium                         | He                                | 4.003                           | 2.0769                                | 5.3                       | 0.23             | 0.0578                          |
| <i>n</i> -Hexane               | C <sub>6</sub> H <sub>14</sub>    | 86.179                          | 0.09647                               | 507.9                     | 3.03             | 0.3677                          |
| Hydrogen (normal)              | H <sub>2</sub>                    | 2.016                           | 4.1240                                | 33.3                      | 1.30             | 0.0649                          |
| Krypton                        | Kr                                | 83.80                           | 0.09921                               | 209.4                     | 5.50             | 0.0924                          |
| Methane                        | CH <sub>4</sub>                   | 16.043                          | 0.5182                                | 191.1                     | 4.64             | 0.0993                          |
| Methyl alcohol                 | CH <sub>3</sub> OH                | 32.042                          | 0.2595                                | 513.2                     | 7.95             | 0.1180                          |
| Methyl chloride                | CH <sub>3</sub> Cl                | 50.488                          | 0.1647                                | 416.3                     | 6.68             | 0.1430                          |
| Neon                           | Ne                                | 20.183                          | 0.4119                                | 44.5                      | 2.73             | 0.0417                          |
| Nitrogen                       | N <sub>2</sub>                    | 28.013                          | 0.2968                                | 126.2                     | 3.39             | 0.0899                          |
| Nitrous oxide                  | N <sub>2</sub> O                  | 44.013                          | 0.1889                                | 309.7                     | 7.27             | 0.0961                          |
| Oxygen                         | O <sub>2</sub>                    | 31.999                          | 0.2598                                | 154.8                     | 5.08             | 0.0780                          |
| Propane                        | C <sub>3</sub> H <sub>8</sub>     | 44.097                          | 0.1885                                | 370                       | 4.26             | 0.1998                          |
| Propylene                      | C <sub>3</sub> H <sub>6</sub>     | 42.081                          | 0.1976                                | 365                       | 4.62             | 0.1810                          |
| Sulfur dioxide                 | SO <sub>2</sub>                   | 64.063                          | 0.1298                                | 430.7                     | 7.88             | 0.1217                          |
| Tetrafluoroethane (R-134a)     | CF <sub>3</sub> CH <sub>2</sub> F | 102.03                          | 0.08149                               | 374.2                     | 4.059            | 0.1993                          |
| Trichlorofluoromethane (R-11)  | CCl <sub>3</sub> F                | 137.37                          | 0.06052                               | 471.2                     | 4.38             | 0.2478                          |
| Water                          | H <sub>2</sub> O                  | 18.015                          | 0.4615                                | 647.1                     | 22.06            | 0.0560                          |
| Xenon                          | Xe                                | 131.30                          | 0.06332                               | 289.8                     | 5.88             | 0.1186                          |

\*The unit kJ/kg·K is equivalent to kPa·m<sup>3</sup>/kg·K. The gas constant is calculated from  $R = R_u/M$ , where  $R_u = 8.31447$  kJ/kmol·K and  $M$  is the molar mass.

Source of Data: K. A. Kobe and R. E. Lynn, Jr., *Chemical Review* 52 (1953), pp. 117–236; and ASHRAE, *Handbook of Fundamentals* (Atlanta, GA: American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc., 1993), pp. 16.4 and 36.1.

TABLE A-2

Ideal-gas specific heats of various common gases

(a) At 300 K

| Gas             | Formula                        | Gas constant, $R$<br>kJ/kg·K | $c_p$<br>kJ/kg·K | $c_v$<br>kJ/kg·K | $k$   |
|-----------------|--------------------------------|------------------------------|------------------|------------------|-------|
| Air             | —                              | 0.2870                       | 1.005            | 0.718            | 1.400 |
| Argon           | Ar                             | 0.2081                       | 0.5203           | 0.3122           | 1.667 |
| Butane          | C <sub>4</sub> H <sub>10</sub> | 0.1433                       | 1.7164           | 1.5734           | 1.091 |
| Carbon dioxide  | CO <sub>2</sub>                | 0.1889                       | 0.846            | 0.657            | 1.289 |
| Carbon monoxide | CO                             | 0.2968                       | 1.040            | 0.744            | 1.400 |
| Ethane          | C <sub>2</sub> H <sub>6</sub>  | 0.2765                       | 1.7662           | 1.4897           | 1.186 |
| Ethylene        | C <sub>2</sub> H <sub>4</sub>  | 0.2964                       | 1.5482           | 1.2518           | 1.237 |
| Helium          | He                             | 2.0769                       | 5.1926           | 3.1156           | 1.667 |
| Hydrogen        | H <sub>2</sub>                 | 4.1240                       | 14.307           | 10.183           | 1.405 |
| Methane         | CH <sub>4</sub>                | 0.5182                       | 2.2537           | 1.7354           | 1.299 |
| Neon            | Ne                             | 0.4119                       | 1.0299           | 0.6179           | 1.667 |
| Nitrogen        | N <sub>2</sub>                 | 0.2968                       | 1.039            | 0.743            | 1.400 |
| Octane          | C <sub>8</sub> H <sub>18</sub> | 0.0729                       | 1.7113           | 1.6385           | 1.044 |
| Oxygen          | O <sub>2</sub>                 | 0.2598                       | 0.918            | 0.658            | 1.395 |
| Propane         | C <sub>3</sub> H <sub>8</sub>  | 0.1885                       | 1.6794           | 1.4909           | 1.126 |
| Steam           | H <sub>2</sub> O               | 0.4615                       | 1.8723           | 1.4108           | 1.327 |

Note: The unit kJ/kg·K is equivalent to kJ/kg·°C.

Source of Data: B. G. Kyle, *Chemical and Process Thermodynamics*, 3rd ed. (Upper Saddle River, NJ: Prentice Hall, 2000).

TABLE A-2

Ideal-gas specific heats of various common gases (*Continued*)

(b) At various temperatures

| Temperature,<br>K | $c_p$<br>kJ/kg·K               | $c_v$<br>kJ/kg·K | $k$   | $c_p$<br>kJ/kg·K                      | $c_v$<br>kJ/kg·K | $k$   | $c_p$<br>kJ/kg·K             | $c_v$<br>kJ/kg·K | $k$   |
|-------------------|--------------------------------|------------------|-------|---------------------------------------|------------------|-------|------------------------------|------------------|-------|
|                   | <i>Air</i>                     |                  |       | <i>Carbon dioxide, CO<sub>2</sub></i> |                  |       | <i>Carbon monoxide, CO</i>   |                  |       |
| 250               | 1.003                          | 0.716            | 1.401 | 0.791                                 | 0.602            | 1.314 | 1.039                        | 0.743            | 1.400 |
| 300               | 1.005                          | 0.718            | 1.400 | 0.846                                 | 0.657            | 1.288 | 1.040                        | 0.744            | 1.399 |
| 350               | 1.008                          | 0.721            | 1.398 | 0.895                                 | 0.706            | 1.268 | 1.043                        | 0.746            | 1.398 |
| 400               | 1.013                          | 0.726            | 1.395 | 0.939                                 | 0.750            | 1.252 | 1.047                        | 0.751            | 1.395 |
| 450               | 1.020                          | 0.733            | 1.391 | 0.978                                 | 0.790            | 1.239 | 1.054                        | 0.757            | 1.392 |
| 500               | 1.029                          | 0.742            | 1.387 | 1.014                                 | 0.825            | 1.229 | 1.063                        | 0.767            | 1.387 |
| 550               | 1.040                          | 0.753            | 1.381 | 1.046                                 | 0.857            | 1.220 | 1.075                        | 0.778            | 1.382 |
| 600               | 1.051                          | 0.764            | 1.376 | 1.075                                 | 0.886            | 1.213 | 1.087                        | 0.790            | 1.376 |
| 650               | 1.063                          | 0.776            | 1.370 | 1.102                                 | 0.913            | 1.207 | 1.100                        | 0.803            | 1.370 |
| 700               | 1.075                          | 0.788            | 1.364 | 1.126                                 | 0.937            | 1.202 | 1.113                        | 0.816            | 1.364 |
| 750               | 1.087                          | 0.800            | 1.359 | 1.148                                 | 0.959            | 1.197 | 1.126                        | 0.829            | 1.358 |
| 800               | 1.099                          | 0.812            | 1.354 | 1.169                                 | 0.980            | 1.193 | 1.139                        | 0.842            | 1.353 |
| 900               | 1.121                          | 0.834            | 1.344 | 1.204                                 | 1.015            | 1.186 | 1.163                        | 0.866            | 1.343 |
| 1000              | 1.142                          | 0.855            | 1.336 | 1.234                                 | 1.045            | 1.181 | 1.185                        | 0.888            | 1.335 |
| Temperature,<br>K | <i>Hydrogen, H<sub>2</sub></i> |                  |       | <i>Nitrogen, N<sub>2</sub></i>        |                  |       | <i>Oxygen, O<sub>2</sub></i> |                  |       |
|                   | $c_p$<br>kJ/kg·K               | $c_v$<br>kJ/kg·K | $k$   | $c_p$<br>kJ/kg·K                      | $c_v$<br>kJ/kg·K | $k$   | $c_p$<br>kJ/kg·K             | $c_v$<br>kJ/kg·K | $k$   |
| 250               | 14.051                         | 9.927            | 1.416 | 1.039                                 | 0.742            | 1.400 | 0.913                        | 0.653            | 1.398 |
| 300               | 14.307                         | 10.183           | 1.405 | 1.039                                 | 0.743            | 1.400 | 0.918                        | 0.658            | 1.395 |
| 350               | 14.427                         | 10.302           | 1.400 | 1.041                                 | 0.744            | 1.399 | 0.928                        | 0.668            | 1.389 |
| 400               | 14.476                         | 10.352           | 1.398 | 1.044                                 | 0.747            | 1.397 | 0.941                        | 0.681            | 1.382 |
| 450               | 14.501                         | 10.377           | 1.398 | 1.049                                 | 0.752            | 1.395 | 0.956                        | 0.696            | 1.373 |
| 500               | 14.513                         | 10.389           | 1.397 | 1.056                                 | 0.759            | 1.391 | 0.972                        | 0.712            | 1.365 |
| 550               | 14.530                         | 10.405           | 1.396 | 1.065                                 | 0.768            | 1.387 | 0.988                        | 0.728            | 1.358 |
| 600               | 14.546                         | 10.422           | 1.396 | 1.075                                 | 0.778            | 1.382 | 1.003                        | 0.743            | 1.350 |
| 650               | 14.571                         | 10.447           | 1.395 | 1.086                                 | 0.789            | 1.376 | 1.017                        | 0.758            | 1.343 |
| 700               | 14.604                         | 10.480           | 1.394 | 1.098                                 | 0.801            | 1.371 | 1.031                        | 0.771            | 1.337 |
| 750               | 14.645                         | 10.521           | 1.392 | 1.110                                 | 0.813            | 1.365 | 1.043                        | 0.783            | 1.332 |
| 800               | 14.695                         | 10.570           | 1.390 | 1.121                                 | 0.825            | 1.360 | 1.054                        | 0.794            | 1.327 |
| 900               | 14.822                         | 10.698           | 1.385 | 1.145                                 | 0.849            | 1.349 | 1.074                        | 0.814            | 1.319 |
| 1000              | 14.983                         | 10.859           | 1.380 | 1.167                                 | 0.870            | 1.341 | 1.090                        | 0.830            | 1.313 |

Source of Data: Kenneth Wark, *Thermodynamics*, 4th ed. (New York: McGraw-Hill, 1983), p. 783, Table A-4M. Originally published in *Tables of Thermal Properties of Gases*, NBS Circular 564, 1955.

TABLE A-2

Ideal-gas specific heats of various common gases (*Concluded*)

(c) As a function of temperature

| $\bar{c}_p = a + bT + cT^2 + dT^3$ $(T \text{ in K, } c_p \text{ in kJ/kmol}\cdot\text{K})$ |                                 |        |                           |                          |                          |                      |  | % error |      |
|---------------------------------------------------------------------------------------------|---------------------------------|--------|---------------------------|--------------------------|--------------------------|----------------------|--|---------|------|
| Substance                                                                                   | Formula                         | $a$    | $b$                       | $c$                      | $d$                      | Temperature range, K |  | Max.    | Avg. |
| Nitrogen                                                                                    | N <sub>2</sub>                  | 28.90  | $-0.1571 \times 10^{-2}$  | $0.8081 \times 10^{-5}$  | $-2.873 \times 10^{-9}$  | 273–1800             |  | 0.59    | 0.34 |
| Oxygen                                                                                      | O <sub>2</sub>                  | 25.48  | $1.520 \times 10^{-2}$    | $-0.7155 \times 10^{-5}$ | $1.312 \times 10^{-9}$   | 273–1800             |  | 1.19    | 0.28 |
| Air                                                                                         | —                               | 28.11  | $0.1967 \times 10^{-2}$   | $0.4802 \times 10^{-5}$  | $-1.966 \times 10^{-9}$  | 273–1800             |  | 0.72    | 0.33 |
| Hydrogen                                                                                    | H <sub>2</sub>                  | 29.11  | $-0.1916 \times 10^{-2}$  | $0.4003 \times 10^{-5}$  | $-0.8704 \times 10^{-9}$ | 273–1800             |  | 1.01    | 0.26 |
| Carbon monoxide                                                                             | CO                              | 28.16  | $0.1675 \times 10^{-2}$   | $0.5372 \times 10^{-5}$  | $-2.222 \times 10^{-9}$  | 273–1800             |  | 0.89    | 0.37 |
| Carbon dioxide                                                                              | CO <sub>2</sub>                 | 22.26  | $5.981 \times 10^{-2}$    | $-3.501 \times 10^{-5}$  | $7.469 \times 10^{-9}$   | 273–1800             |  | 0.67    | 0.22 |
| Water vapor                                                                                 | H <sub>2</sub> O                | 32.24  | $0.1923 \times 10^{-2}$   | $1.055 \times 10^{-5}$   | $-3.595 \times 10^{-9}$  | 273–1800             |  | 0.53    | 0.24 |
| Nitric oxide                                                                                | NO                              | 29.34  | $-0.09395 \times 10^{-2}$ | $0.9747 \times 10^{-5}$  | $-4.187 \times 10^{-9}$  | 273–1500             |  | 0.97    | 0.36 |
| Nitrous oxide                                                                               | N <sub>2</sub> O                | 24.11  | $5.8632 \times 10^{-2}$   | $-3.562 \times 10^{-5}$  | $10.58 \times 10^{-9}$   | 273–1500             |  | 0.59    | 0.26 |
| Nitrogen dioxide                                                                            | NO <sub>2</sub>                 | 22.9   | $5.715 \times 10^{-2}$    | $-3.52 \times 10^{-5}$   | $7.87 \times 10^{-9}$    | 273–1500             |  | 0.46    | 0.18 |
| Ammonia                                                                                     | NH <sub>3</sub>                 | 27.568 | $2.5630 \times 10^{-2}$   | $0.99072 \times 10^{-5}$ | $-6.6909 \times 10^{-9}$ | 273–1500             |  | 0.91    | 0.36 |
| Sulfur                                                                                      | S <sub>2</sub>                  | 27.21  | $2.218 \times 10^{-2}$    | $-1.628 \times 10^{-5}$  | $3.986 \times 10^{-9}$   | 273–1800             |  | 0.99    | 0.38 |
| Sulfur dioxide                                                                              | SO <sub>2</sub>                 | 25.78  | $5.795 \times 10^{-2}$    | $-3.812 \times 10^{-5}$  | $8.612 \times 10^{-9}$   | 273–1800             |  | 0.45    | 0.24 |
| Sulfur trioxide                                                                             | SO <sub>3</sub>                 | 16.40  | $14.58 \times 10^{-2}$    | $-11.20 \times 10^{-5}$  | $32.42 \times 10^{-9}$   | 273–1300             |  | 0.29    | 0.13 |
| Acetylene                                                                                   | C <sub>2</sub> H <sub>2</sub>   | 21.8   | $9.2143 \times 10^{-2}$   | $-6.527 \times 10^{-5}$  | $18.21 \times 10^{-9}$   | 273–1500             |  | 1.46    | 0.59 |
| Benzene                                                                                     | C <sub>6</sub> H <sub>6</sub>   | -36.22 | $48.475 \times 10^{-2}$   | $-31.57 \times 10^{-5}$  | $77.62 \times 10^{-9}$   | 273–1500             |  | 0.34    | 0.20 |
| Methanol                                                                                    | CH <sub>4</sub> O               | 19.0   | $9.152 \times 10^{-2}$    | $-1.22 \times 10^{-5}$   | $-8.039 \times 10^{-9}$  | 273–1000             |  | 0.18    | 0.08 |
| Ethanol                                                                                     | C <sub>2</sub> H <sub>6</sub> O | 19.9   | $20.96 \times 10^{-2}$    | $-10.38 \times 10^{-5}$  | $20.05 \times 10^{-9}$   | 273–1500             |  | 0.40    | 0.22 |
| Hydrogen chloride                                                                           | HCl                             | 30.33  | $-0.7620 \times 10^{-2}$  | $1.327 \times 10^{-5}$   | $-4.338 \times 10^{-9}$  | 273–1500             |  | 0.22    | 0.08 |
| Methane                                                                                     | CH <sub>4</sub>                 | 19.89  | $5.024 \times 10^{-2}$    | $1.269 \times 10^{-5}$   | $-11.01 \times 10^{-9}$  | 273–1500             |  | 1.33    | 0.57 |
| Ethane                                                                                      | C <sub>2</sub> H <sub>6</sub>   | 6.900  | $17.27 \times 10^{-2}$    | $-6.406 \times 10^{-5}$  | $7.285 \times 10^{-9}$   | 273–1500             |  | 0.83    | 0.28 |
| Propane                                                                                     | C <sub>3</sub> H <sub>8</sub>   | -4.04  | $30.48 \times 10^{-2}$    | $-15.72 \times 10^{-5}$  | $31.74 \times 10^{-9}$   | 273–1500             |  | 0.40    | 0.12 |
| <i>n</i> -Butane                                                                            | C <sub>4</sub> H <sub>10</sub>  | 3.96   | $37.15 \times 10^{-2}$    | $-18.34 \times 10^{-5}$  | $35.00 \times 10^{-9}$   | 273–1500             |  | 0.54    | 0.24 |
| <i>i</i> -Butane                                                                            | C <sub>4</sub> H <sub>10</sub>  | -7.913 | $41.60 \times 10^{-2}$    | $-23.01 \times 10^{-5}$  | $49.91 \times 10^{-9}$   | 273–1500             |  | 0.25    | 0.13 |
| <i>n</i> -Pentane                                                                           | C <sub>5</sub> H <sub>12</sub>  | 6.774  | $45.43 \times 10^{-2}$    | $-22.46 \times 10^{-5}$  | $42.29 \times 10^{-9}$   | 273–1500             |  | 0.56    | 0.21 |
| <i>n</i> -Hexane                                                                            | C <sub>6</sub> H <sub>14</sub>  | 6.938  | $55.22 \times 10^{-2}$    | $-28.65 \times 10^{-5}$  | $57.69 \times 10^{-9}$   | 273–1500             |  | 0.72    | 0.20 |
| Ethylene                                                                                    | C <sub>2</sub> H <sub>4</sub>   | 3.95   | $15.64 \times 10^{-2}$    | $-8.344 \times 10^{-5}$  | $17.67 \times 10^{-9}$   | 273–1500             |  | 0.54    | 0.13 |
| Propylene                                                                                   | C <sub>3</sub> H <sub>6</sub>   | 3.15   | $23.83 \times 10^{-2}$    | $-12.18 \times 10^{-5}$  | $24.62 \times 10^{-9}$   | 273–1500             |  | 0.73    | 0.17 |

Source of Data: B. G. Kyle, *Chemical and Process Thermodynamics* (Englewood Cliffs, NJ: Prentice-Hall, 1984).

TABLE A-3

Properties of common liquids, solids, and foods

(a) Liquids

| Substance                           | Boiling data at 1 atm    |                                              | Freezing data      |                                        | Liquid properties |                                    |                               |
|-------------------------------------|--------------------------|----------------------------------------------|--------------------|----------------------------------------|-------------------|------------------------------------|-------------------------------|
|                                     | Normal boiling point, °C | Latent heat of vaporization $h_{fg}$ , kJ/kg | Freezing point, °C | Latent heat of fusion $h_{if}$ , kJ/kg | Temperature, °C   | Density $\rho$ , kg/m <sup>3</sup> | Specific heat $c_p$ , kJ/kg·K |
| Ammonia                             | -33.3                    | 1357                                         | -77.7              | 322.4                                  | -33.3             | 682                                | 4.43                          |
|                                     |                          |                                              |                    |                                        | -20               | 665                                | 4.52                          |
|                                     |                          |                                              |                    |                                        | 0                 | 639                                | 4.60                          |
|                                     |                          |                                              |                    |                                        | 25                | 602                                | 4.80                          |
| Argon                               | -185.9                   | 161.6                                        | -189.3             | 28                                     | -185.6            | 1394                               | 1.14                          |
| Benzene                             | 80.2                     | 394                                          | 5.5                | 126                                    | 20                | 879                                | 1.72                          |
| Brine (20% sodium chloride by mass) | 103.9                    | —                                            | -17.4              | —                                      | 20                | 1150                               | 3.11                          |
| <i>n</i> -Butane                    | -0.5                     | 385.2                                        | -138.5             | 80.3                                   | -0.5              | 601                                | 2.31                          |
| Carbon dioxide                      | -78.4*                   | 230.5 (at 0°C)                               | -56.6              | —                                      | 0                 | 298                                | 0.59                          |
| Ethanol                             | 78.2                     | 838.3                                        | -114.2             | 109                                    | 25                | 783                                | 2.46                          |
| Ethyl alcohol                       | 78.6                     | 855                                          | -156               | 108                                    | 20                | 789                                | 2.84                          |
| Ethylene glycol                     | 198.1                    | 800.1                                        | -10.8              | 181.1                                  | 20                | 1109                               | 2.84                          |
| Glycerine                           | 179.9                    | 974                                          | 18.9               | 200.6                                  | 20                | 1261                               | 2.32                          |
| Helium                              | -268.9                   | 22.8                                         | —                  | —                                      | -268.9            | 146.2                              | 22.8                          |
| Hydrogen                            | -252.8                   | 445.7                                        | -259.2             | 59.5                                   | -252.8            | 70.7                               | 10.0                          |
| Isobutane                           | -11.7                    | 367.1                                        | -160               | 105.7                                  | -11.7             | 593.8                              | 2.28                          |
| Kerosene                            | 204–293                  | 251                                          | -24.9              | —                                      | 20                | 820                                | 2.00                          |
| Mercury                             | 356.7                    | 294.7                                        | -38.9              | 11.4                                   | 25                | 13,560                             | 0.139                         |
| Methane                             | -161.5                   | 510.4                                        | -182.2             | 58.4                                   | -161.5            | 423                                | 3.49                          |
|                                     |                          |                                              |                    |                                        | -100              | 301                                | 5.79                          |
| Methanol                            | 64.5                     | 1100                                         | -97.7              | 99.2                                   | 25                | 787                                | 2.55                          |
| Nitrogen                            | -195.8                   | 198.6                                        | -210               | 25.3                                   | -195.8            | 809                                | 2.06                          |
|                                     |                          |                                              |                    |                                        | -160              | 596                                | 2.97                          |
| Octane                              | 124.8                    | 306.3                                        | -57.5              | 180.7                                  | 20                | 703                                | 2.10                          |
| Oil (light)                         |                          |                                              |                    |                                        | 25                | 910                                | 1.80                          |
| Oxygen                              | -183                     | 212.7                                        | -218.8             | 13.7                                   | -183              | 1141                               | 1.71                          |
| Petroleum                           | —                        | 230–384                                      |                    |                                        | 20                | 640                                | 2.0                           |
| Propane                             | -42.1                    | 427.8                                        | -187.7             | 80.0                                   | -42.1             | 581                                | 2.25                          |
|                                     |                          |                                              |                    |                                        | 0                 | 529                                | 2.53                          |
|                                     |                          |                                              |                    |                                        | 50                | 449                                | 3.13                          |
| Refrigerant-134a                    | -26.1                    | 217.0                                        | -96.6              | —                                      | -50               | 1443                               | 1.23                          |
|                                     |                          |                                              |                    |                                        | -26.1             | 1374                               | 1.27                          |
|                                     |                          |                                              |                    |                                        | 0                 | 1295                               | 1.34                          |
|                                     |                          |                                              |                    |                                        | 25                | 1207                               | 1.43                          |
| Water                               | 100                      | 2257                                         | 0.0                | 333.7                                  | 0                 | 1000                               | 4.22                          |
|                                     |                          |                                              |                    |                                        | 25                | 997                                | 4.18                          |
|                                     |                          |                                              |                    |                                        | 50                | 988                                | 4.18                          |
|                                     |                          |                                              |                    |                                        | 75                | 975                                | 4.19                          |
|                                     |                          |                                              |                    |                                        | 100               | 958                                | 4.22                          |

\* Sublimation temperature. (At pressures below the triple-point pressure of 518 kPa, carbon dioxide exists as a solid or gas. Also, the freezing-point temperature of carbon dioxide is the triple-point temperature of -56.5°C.)

TABLE A-3

Properties of common liquids, solids, and foods (*Concluded*)

(b) Solids (values are for room temperature unless indicated otherwise)

| Substance                         | Density,<br>$\rho$ kg/m <sup>3</sup> | Specific heat,<br>$c_p$ kJ/kg·K | Substance                      | Density,<br>$\rho$ kg/m <sup>3</sup> | Specific heat,<br>$c_p$ kJ/kg·K |
|-----------------------------------|--------------------------------------|---------------------------------|--------------------------------|--------------------------------------|---------------------------------|
| Metals                            |                                      |                                 | Nonmetals                      |                                      |                                 |
| Aluminum                          |                                      |                                 | Asphalt                        | 2110                                 | 0.920                           |
| 200 K                             |                                      | 0.797                           | Brick, common                  | 1922                                 | 0.79                            |
| 250 K                             |                                      | 0.859                           | Brick, fireclay (500°C)        | 2300                                 | 0.960                           |
| 300 K                             | 2,700                                | 0.902                           | Concrete                       | 2300                                 | 0.653                           |
| 350 K                             |                                      | 0.929                           | Clay                           | 1000                                 | 0.920                           |
| 400 K                             |                                      | 0.949                           | Diamond                        | 2420                                 | 0.616                           |
| 450 K                             |                                      | 0.973                           | Glass, window                  | 2700                                 | 0.800                           |
| 500 K                             |                                      | 0.997                           | Glass, pyrex                   | 2230                                 | 0.840                           |
| Bronze (76% Cu, 2% Zn,<br>2% Al)  | 8,280                                | 0.400                           | Graphite                       | 2500                                 | 0.711                           |
| Brass, yellow (65% Cu,<br>35% Zn) | 8,310                                | 0.400                           | Granite                        | 2700                                 | 1.017                           |
| Copper                            |                                      |                                 | Gypsum or plaster board        | 800                                  | 1.09                            |
| −173°C                            |                                      | 0.254                           | Ice                            |                                      |                                 |
| −100°C                            |                                      | 0.342                           | 200 K                          |                                      | 1.56                            |
| −50°C                             |                                      | 0.367                           | 220 K                          |                                      | 1.71                            |
| 0°C                               |                                      | 0.381                           | 240 K                          |                                      | 1.86                            |
| 27°C                              | 8,900                                | 0.386                           | 260 K                          |                                      | 2.01                            |
| 100°C                             |                                      | 0.393                           | 273 K                          | 921                                  | 2.11                            |
| 200°C                             |                                      | 0.403                           | Limestone                      | 1650                                 | 0.909                           |
| Iron                              | 7,840                                | 0.45                            | Marble                         | 2600                                 | 0.880                           |
| Lead                              | 11,310                               | 0.128                           | Plywood (Douglas Fir)          | 545                                  | 1.21                            |
| Magnesium                         | 1,730                                | 1.000                           | Rubber (soft)                  | 1100                                 | 1.840                           |
| Nickel                            | 8,890                                | 0.440                           | Rubber (hard)                  | 1150                                 | 2.009                           |
| Silver                            | 10,470                               | 0.235                           | Sand                           | 1520                                 | 0.800                           |
| Steel, mild                       | 7,830                                | 0.500                           | Stone                          | 1500                                 | 0.800                           |
| Tungsten                          | 19,400                               | 0.130                           | Woods, hard (maple, oak, etc.) | 721                                  | 1.26                            |
|                                   |                                      |                                 | Woods, soft (fir, pine, etc.)  | 513                                  | 1.38                            |

(c) Foods

| Food          | Water<br>content,<br>% (mass) | Freezing<br>point, °C | Specific heat,<br>kJ/kg·K |                   | Latent<br>heat of<br>fusion,<br>kJ/kg | Food           | Water<br>content,<br>% (mass) | Freezing<br>point, °C | Specific heat,<br>kJ/kg·K |                   | Latent<br>heat of<br>fusion,<br>kJ/kg |
|---------------|-------------------------------|-----------------------|---------------------------|-------------------|---------------------------------------|----------------|-------------------------------|-----------------------|---------------------------|-------------------|---------------------------------------|
|               |                               |                       | Above<br>freezing         | Below<br>freezing |                                       |                |                               |                       | Above<br>freezing         | Below<br>freezing |                                       |
| Apples        | 84                            | −1.1                  | 3.65                      | 1.90              | 281                                   | Lettuce        | 95                            | −0.2                  | 4.02                      | 2.04              | 317                                   |
| Bananas       | 75                            | −0.8                  | 3.35                      | 1.78              | 251                                   | Milk, whole    | 88                            | −0.6                  | 3.79                      | 1.95              | 294                                   |
| Beef round    | 67                            | —                     | 3.08                      | 1.68              | 224                                   | Oranges        | 87                            | −0.8                  | 3.75                      | 1.94              | 291                                   |
| Broccoli      | 90                            | −0.6                  | 3.86                      | 1.97              | 301                                   | Potatoes       | 78                            | −0.6                  | 3.45                      | 1.82              | 261                                   |
| Butter        | 16                            | —                     | —                         | 1.04              | 53                                    | Salmon fish    | 64                            | −2.2                  | 2.98                      | 1.65              | 214                                   |
| Cheese, swiss | 39                            | −10.0                 | 2.15                      | 1.33              | 130                                   | Shrimp         | 83                            | −2.2                  | 3.62                      | 1.89              | 277                                   |
| Cherries      | 80                            | −1.8                  | 3.52                      | 1.85              | 267                                   | Spinach        | 93                            | −0.3                  | 3.96                      | 2.01              | 311                                   |
| Chicken       | 74                            | −2.8                  | 3.32                      | 1.77              | 247                                   | Strawberries   | 90                            | −0.8                  | 3.86                      | 1.97              | 301                                   |
| Corn, sweet   | 74                            | −0.6                  | 3.32                      | 1.77              | 247                                   | Tomatoes, ripe | 94                            | −0.5                  | 3.99                      | 2.02              | 314                                   |
| Eggs, whole   | 74                            | −0.6                  | 3.32                      | 1.77              | 247                                   | Turkey         | 64                            | —                     | 2.98                      | 1.65              | 214                                   |
| Ice cream     | 63                            | −5.6                  | 2.95                      | 1.63              | 210                                   | Watermelon     | 93                            | −0.4                  | 3.96                      | 2.01              | 311                                   |

Source of Data: Values are obtained from various handbooks and other sources or are calculated. Water content and freezing-point data of foods are from ASHRAE, *Handbook of Fundamentals*, SI version (Atlanta, GA: American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc., 1993), Chapter 30, Table 1. Freezing point is the temperature at which freezing starts for fruits and vegetables, and the average freezing temperature for other foods.

TABLE A-4

Saturated water—Temperature table

| Temp.,<br>$T$ °C | Sat.<br>press.,<br>$P_{\text{sat}}$ kPa | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>$\text{kJ/kg}$ |                    |                         | Enthalpy,<br>$\text{kJ/kg}$ |                    |                         | Entropy,<br>$\text{kJ/kg}\cdot\text{K}$ |                    |                         |
|------------------|-----------------------------------------|--------------------------------------------|-------------------------|------------------------------------|--------------------|-------------------------|-----------------------------|--------------------|-------------------------|-----------------------------------------|--------------------|-------------------------|
|                  |                                         | Sat.<br>liquid,<br>$v_f$                   | Sat.<br>vapor,<br>$v_g$ | Sat.<br>liquid,<br>$u_f$           | Evap.,<br>$u_{fg}$ | Sat.<br>vapor,<br>$u_g$ | Sat.<br>liquid,<br>$h_f$    | Evap.,<br>$h_{fg}$ | Sat.<br>vapor,<br>$h_g$ | Sat.<br>liquid,<br>$s_f$                | Evap.,<br>$s_{fg}$ | Sat.<br>vapor,<br>$s_g$ |
| 0.01             | 0.6117                                  | 0.001000                                   | 206.00                  | 0.000                              | 2374.9             | 2374.9                  | 0.001                       | 2500.9             | 2500.9                  | 0.0000                                  | 9.1556             | 9.1556                  |
| 5                | 0.8725                                  | 0.001000                                   | 147.03                  | 21.019                             | 2360.8             | 2381.8                  | 21.020                      | 2489.1             | 2510.1                  | 0.0763                                  | 8.9487             | 9.0249                  |
| 10               | 1.2281                                  | 0.001000                                   | 106.32                  | 42.020                             | 2346.6             | 2388.7                  | 42.022                      | 2477.2             | 2519.2                  | 0.1511                                  | 8.7488             | 8.8999                  |
| 15               | 1.7057                                  | 0.001001                                   | 77.885                  | 62.980                             | 2332.5             | 2395.5                  | 62.982                      | 2465.4             | 2528.3                  | 0.2245                                  | 8.5559             | 8.7803                  |
| 20               | 2.3392                                  | 0.001002                                   | 57.762                  | 83.913                             | 2318.4             | 2402.3                  | 83.915                      | 2453.5             | 2537.4                  | 0.2965                                  | 8.3696             | 8.6661                  |
| 25               | 3.1698                                  | 0.001003                                   | 43.340                  | 104.83                             | 2304.3             | 2409.1                  | 104.83                      | 2441.7             | 2546.5                  | 0.3672                                  | 8.1895             | 8.5567                  |
| 30               | 4.2469                                  | 0.001004                                   | 32.879                  | 125.73                             | 2290.2             | 2415.9                  | 125.74                      | 2429.8             | 2555.6                  | 0.4368                                  | 8.0152             | 8.4520                  |
| 35               | 5.6291                                  | 0.001006                                   | 25.205                  | 146.63                             | 2276.0             | 2422.7                  | 146.64                      | 2417.9             | 2564.6                  | 0.5051                                  | 7.8466             | 8.3517                  |
| 40               | 7.3851                                  | 0.001008                                   | 19.515                  | 167.53                             | 2261.9             | 2429.4                  | 167.53                      | 2406.0             | 2573.5                  | 0.5724                                  | 7.6832             | 8.2556                  |
| 45               | 9.5953                                  | 0.001010                                   | 15.251                  | 188.43                             | 2247.7             | 2436.1                  | 188.44                      | 2394.0             | 2582.4                  | 0.6386                                  | 7.5247             | 8.1633                  |
| 50               | 12.352                                  | 0.001012                                   | 12.026                  | 209.33                             | 2233.4             | 2442.7                  | 209.34                      | 2382.0             | 2591.3                  | 0.7038                                  | 7.3710             | 8.0748                  |
| 55               | 15.763                                  | 0.001015                                   | 9.5639                  | 230.24                             | 2219.1             | 2449.3                  | 230.26                      | 2369.8             | 2600.1                  | 0.7680                                  | 7.2218             | 7.9898                  |
| 60               | 19.947                                  | 0.001017                                   | 7.6670                  | 251.16                             | 2204.7             | 2455.9                  | 251.18                      | 2357.7             | 2608.8                  | 0.8313                                  | 7.0769             | 7.9082                  |
| 65               | 25.043                                  | 0.001020                                   | 6.1935                  | 272.09                             | 2190.3             | 2462.4                  | 272.12                      | 2345.4             | 2617.5                  | 0.8937                                  | 6.9360             | 7.8296                  |
| 70               | 31.202                                  | 0.001023                                   | 5.0396                  | 293.04                             | 2175.8             | 2468.9                  | 293.07                      | 2333.0             | 2626.1                  | 0.9551                                  | 6.7989             | 7.7540                  |
| 75               | 38.597                                  | 0.001026                                   | 4.1291                  | 313.99                             | 2161.3             | 2475.3                  | 314.03                      | 2320.6             | 2634.6                  | 1.0158                                  | 6.6655             | 7.6812                  |
| 80               | 47.416                                  | 0.001029                                   | 3.4053                  | 334.97                             | 2146.6             | 2481.6                  | 335.02                      | 2308.0             | 2643.0                  | 1.0756                                  | 6.5355             | 7.6111                  |
| 85               | 57.868                                  | 0.001032                                   | 2.8261                  | 355.96                             | 2131.9             | 2487.8                  | 356.02                      | 2295.3             | 2651.4                  | 1.1346                                  | 6.4089             | 7.5435                  |
| 90               | 70.183                                  | 0.001036                                   | 2.3593                  | 376.97                             | 2117.0             | 2494.0                  | 377.04                      | 2282.5             | 2659.6                  | 1.1929                                  | 6.2853             | 7.4782                  |
| 95               | 84.609                                  | 0.001040                                   | 1.9808                  | 398.00                             | 2102.0             | 2500.1                  | 398.09                      | 2269.6             | 2667.6                  | 1.2504                                  | 6.1647             | 7.4151                  |
| 100              | 101.42                                  | 0.001043                                   | 1.6720                  | 419.06                             | 2087.0             | 2506.0                  | 419.17                      | 2256.4             | 2675.6                  | 1.3072                                  | 6.0470             | 7.3542                  |
| 105              | 120.90                                  | 0.001047                                   | 1.4186                  | 440.15                             | 2071.8             | 2511.9                  | 440.28                      | 2243.1             | 2683.4                  | 1.3634                                  | 5.9319             | 7.2952                  |
| 110              | 143.38                                  | 0.001052                                   | 1.2094                  | 461.27                             | 2056.4             | 2517.7                  | 461.42                      | 2229.7             | 2691.1                  | 1.4188                                  | 5.8193             | 7.2382                  |
| 115              | 169.18                                  | 0.001056                                   | 1.0360                  | 482.42                             | 2040.9             | 2523.3                  | 482.59                      | 2216.0             | 2698.6                  | 1.4737                                  | 5.7092             | 7.1829                  |
| 120              | 198.67                                  | 0.001060                                   | 0.89133                 | 503.60                             | 2025.3             | 2528.9                  | 503.81                      | 2202.1             | 2706.0                  | 1.5279                                  | 5.6013             | 7.1292                  |
| 125              | 232.23                                  | 0.001065                                   | 0.77012                 | 524.83                             | 2009.5             | 2534.3                  | 525.07                      | 2188.1             | 2713.1                  | 1.5816                                  | 5.4956             | 7.0771                  |
| 130              | 270.28                                  | 0.001070                                   | 0.66808                 | 546.10                             | 1993.4             | 2539.5                  | 546.38                      | 2173.7             | 2720.1                  | 1.6346                                  | 5.3919             | 7.0265                  |
| 135              | 313.22                                  | 0.001075                                   | 0.58179                 | 567.41                             | 1977.3             | 2544.7                  | 567.75                      | 2159.1             | 2726.9                  | 1.6872                                  | 5.2901             | 6.9773                  |
| 140              | 361.53                                  | 0.001080                                   | 0.50850                 | 588.77                             | 1960.9             | 2549.6                  | 589.16                      | 2144.3             | 2733.5                  | 1.7392                                  | 5.1901             | 6.9294                  |
| 145              | 415.68                                  | 0.001085                                   | 0.44600                 | 610.19                             | 1944.2             | 2554.4                  | 610.64                      | 2129.2             | 2739.8                  | 1.7908                                  | 5.0919             | 6.8827                  |
| 150              | 476.16                                  | 0.001091                                   | 0.39248                 | 631.66                             | 1927.4             | 2559.1                  | 632.18                      | 2113.8             | 2745.9                  | 1.8418                                  | 4.9953             | 6.8371                  |
| 155              | 543.49                                  | 0.001096                                   | 0.34648                 | 653.19                             | 1910.3             | 2563.5                  | 653.79                      | 2098.0             | 2751.8                  | 1.8924                                  | 4.9002             | 6.7927                  |
| 160              | 618.23                                  | 0.001102                                   | 0.30680                 | 674.79                             | 1893.0             | 2567.8                  | 675.47                      | 2082.0             | 2757.5                  | 1.9426                                  | 4.8066             | 6.7492                  |
| 165              | 700.93                                  | 0.001108                                   | 0.27244                 | 696.46                             | 1875.4             | 2571.9                  | 697.24                      | 2065.6             | 2762.8                  | 1.9923                                  | 4.7143             | 6.7067                  |
| 170              | 792.18                                  | 0.001114                                   | 0.24260                 | 718.20                             | 1857.5             | 2575.7                  | 719.08                      | 2048.8             | 2767.9                  | 2.0417                                  | 4.6233             | 6.6650                  |
| 175              | 892.60                                  | 0.001121                                   | 0.21659                 | 740.02                             | 1839.4             | 2579.4                  | 741.02                      | 2031.7             | 2772.7                  | 2.0906                                  | 4.5335             | 6.6242                  |
| 180              | 1002.8                                  | 0.001127                                   | 0.19384                 | 761.92                             | 1820.9             | 2582.8                  | 763.05                      | 2014.2             | 2777.2                  | 2.1392                                  | 4.4448             | 6.5841                  |
| 185              | 1123.5                                  | 0.001134                                   | 0.17390                 | 783.91                             | 1802.1             | 2586.0                  | 785.19                      | 1996.2             | 2781.4                  | 2.1875                                  | 4.3572             | 6.5447                  |
| 190              | 1255.2                                  | 0.001141                                   | 0.15636                 | 806.00                             | 1783.0             | 2589.0                  | 807.43                      | 1977.9             | 2785.3                  | 2.2355                                  | 4.2705             | 6.5059                  |
| 195              | 1398.8                                  | 0.001149                                   | 0.14089                 | 828.18                             | 1763.6             | 2591.7                  | 829.78                      | 1959.0             | 2788.8                  | 2.2831                                  | 4.1847             | 6.4678                  |
| 200              | 1554.9                                  | 0.001157                                   | 0.12721                 | 850.46                             | 1743.7             | 2594.2                  | 852.26                      | 1939.8             | 2792.0                  | 2.3305                                  | 4.0997             | 6.4302                  |



TABLE A-4

Saturated water—Temperature table (*Concluded*)

| Temp.,<br>$T$ °C | Sat.<br>press.,<br>$P_{\text{sat}}$ kPa | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>$\text{kJ/kg}$ |                    |                         | Enthalpy,<br>$\text{kJ/kg}$ |                    |                         | Entropy,<br>$\text{kJ/kg}\cdot\text{K}$ |                    |                         |
|------------------|-----------------------------------------|--------------------------------------------|-------------------------|------------------------------------|--------------------|-------------------------|-----------------------------|--------------------|-------------------------|-----------------------------------------|--------------------|-------------------------|
|                  |                                         | Sat.<br>liquid,<br>$v_f$                   | Sat.<br>vapor,<br>$v_g$ | Sat.<br>liquid,<br>$u_f$           | Evap.,<br>$u_{fg}$ | Sat.<br>vapor,<br>$u_g$ | Sat.<br>liquid,<br>$h_f$    | Evap.,<br>$h_{fg}$ | Sat.<br>vapor,<br>$h_g$ | Sat.<br>liquid,<br>$s_f$                | Evap.,<br>$s_{fg}$ | Sat.<br>vapor,<br>$s_g$ |
| 205              | 1724.3                                  | 0.001164                                   | 0.11508                 | 872.86                             | 1723.5             | 2596.4                  | 874.87                      | 1920.0             | 2794.8                  | 2.3776                                  | 4.0154             | 6.3930                  |
| 210              | 1907.7                                  | 0.001173                                   | 0.10429                 | 895.38                             | 1702.9             | 2598.3                  | 897.61                      | 1899.7             | 2797.3                  | 2.4245                                  | 3.9318             | 6.3563                  |
| 215              | 2105.9                                  | 0.001181                                   | 0.094680                | 918.02                             | 1681.9             | 2599.9                  | 920.50                      | 1878.8             | 2799.3                  | 2.4712                                  | 3.8489             | 6.3200                  |
| 220              | 2319.6                                  | 0.001190                                   | 0.086094                | 940.79                             | 1660.5             | 2601.3                  | 943.55                      | 1857.4             | 2801.0                  | 2.5176                                  | 3.7664             | 6.2840                  |
| 225              | 2549.7                                  | 0.001199                                   | 0.078405                | 963.70                             | 1638.6             | 2602.3                  | 966.76                      | 1835.4             | 2802.2                  | 2.5639                                  | 3.6844             | 6.2483                  |
| 230              | 2797.1                                  | 0.001209                                   | 0.071505                | 986.76                             | 1616.1             | 2602.9                  | 990.14                      | 1812.8             | 2802.9                  | 2.6100                                  | 3.6028             | 6.2128                  |
| 235              | 3062.6                                  | 0.001219                                   | 0.065300                | 1010.0                             | 1593.2             | 2603.2                  | 1013.7                      | 1789.5             | 2803.2                  | 2.6560                                  | 3.5216             | 6.1775                  |
| 240              | 3347.0                                  | 0.001229                                   | 0.059707                | 1033.4                             | 1569.8             | 2603.1                  | 1037.5                      | 1765.5             | 2803.0                  | 2.7018                                  | 3.4405             | 6.1424                  |
| 245              | 3651.2                                  | 0.001240                                   | 0.054656                | 1056.9                             | 1545.7             | 2602.7                  | 1061.5                      | 1740.8             | 2802.2                  | 2.7476                                  | 3.3596             | 6.1072                  |
| 250              | 3976.2                                  | 0.001252                                   | 0.050085                | 1080.7                             | 1521.1             | 2601.8                  | 1085.7                      | 1715.3             | 2801.0                  | 2.7933                                  | 3.2788             | 6.0721                  |
| 255              | 4322.9                                  | 0.001263                                   | 0.045941                | 1104.7                             | 1495.8             | 2600.5                  | 1110.1                      | 1689.0             | 2799.1                  | 2.8390                                  | 3.1979             | 6.0369                  |
| 260              | 4692.3                                  | 0.001276                                   | 0.042175                | 1128.8                             | 1469.9             | 2598.7                  | 1134.8                      | 1661.8             | 2796.6                  | 2.8847                                  | 3.1169             | 6.0017                  |
| 265              | 5085.3                                  | 0.001289                                   | 0.038748                | 1153.3                             | 1443.2             | 2596.5                  | 1159.8                      | 1633.7             | 2793.5                  | 2.9304                                  | 3.0358             | 5.9662                  |
| 270              | 5503.0                                  | 0.001303                                   | 0.035622                | 1177.9                             | 1415.7             | 2593.7                  | 1185.1                      | 1604.6             | 2789.7                  | 2.9762                                  | 2.9542             | 5.9305                  |
| 275              | 5946.4                                  | 0.001317                                   | 0.032767                | 1202.9                             | 1387.4             | 2590.3                  | 1210.7                      | 1574.5             | 2785.2                  | 3.0221                                  | 2.8723             | 5.8944                  |
| 280              | 6416.6                                  | 0.001333                                   | 0.030153                | 1228.2                             | 1358.2             | 2586.4                  | 1236.7                      | 1543.2             | 2779.9                  | 3.0681                                  | 2.7898             | 5.8579                  |
| 285              | 6914.6                                  | 0.001349                                   | 0.027756                | 1253.7                             | 1328.1             | 2581.8                  | 1263.1                      | 1510.7             | 2773.7                  | 3.1144                                  | 2.7066             | 5.8210                  |
| 290              | 7441.8                                  | 0.001366                                   | 0.025554                | 1279.7                             | 1296.9             | 2576.5                  | 1289.8                      | 1476.9             | 2766.7                  | 3.1608                                  | 2.6225             | 5.7834                  |
| 295              | 7999.0                                  | 0.001384                                   | 0.023528                | 1306.0                             | 1264.5             | 2570.5                  | 1317.1                      | 1441.6             | 2758.7                  | 3.2076                                  | 2.5374             | 5.7450                  |
| 300              | 8587.9                                  | 0.001404                                   | 0.021659                | 1332.7                             | 1230.9             | 2563.6                  | 1344.8                      | 1404.8             | 2749.6                  | 3.2548                                  | 2.4511             | 5.7059                  |
| 305              | 9209.4                                  | 0.001425                                   | 0.019932                | 1360.0                             | 1195.9             | 2555.8                  | 1373.1                      | 1366.3             | 2739.4                  | 3.3024                                  | 2.3633             | 5.6657                  |
| 310              | 9865.0                                  | 0.001447                                   | 0.018333                | 1387.7                             | 1159.3             | 2547.1                  | 1402.0                      | 1325.9             | 2727.9                  | 3.3506                                  | 2.2737             | 5.6243                  |
| 315              | 10,556                                  | 0.001472                                   | 0.016849                | 1416.1                             | 1121.1             | 2537.2                  | 1431.6                      | 1283.4             | 2715.0                  | 3.3994                                  | 2.1821             | 5.5816                  |
| 320              | 11,284                                  | 0.001499                                   | 0.015470                | 1445.1                             | 1080.9             | 2526.0                  | 1462.0                      | 1238.5             | 2700.6                  | 3.4491                                  | 2.0881             | 5.5372                  |
| 325              | 12,051                                  | 0.001528                                   | 0.014183                | 1475.0                             | 1038.5             | 2513.4                  | 1493.4                      | 1191.0             | 2684.3                  | 3.4998                                  | 1.9911             | 5.4908                  |
| 330              | 12,858                                  | 0.001560                                   | 0.012979                | 1505.7                             | 993.5              | 2499.2                  | 1525.8                      | 1140.3             | 2666.0                  | 3.5516                                  | 1.8906             | 5.4422                  |
| 335              | 13,707                                  | 0.001597                                   | 0.011848                | 1537.5                             | 945.5              | 2483.0                  | 1559.4                      | 1086.0             | 2645.4                  | 3.6050                                  | 1.7857             | 5.3907                  |
| 340              | 14,601                                  | 0.001638                                   | 0.010783                | 1570.7                             | 893.8              | 2464.5                  | 1594.6                      | 1027.4             | 2622.0                  | 3.6602                                  | 1.6756             | 5.3358                  |
| 345              | 15,541                                  | 0.001685                                   | 0.009772                | 1605.5                             | 837.7              | 2443.2                  | 1631.7                      | 963.4              | 2595.1                  | 3.7179                                  | 1.5585             | 5.2765                  |
| 350              | 16,529                                  | 0.001741                                   | 0.008806                | 1642.4                             | 775.9              | 2418.3                  | 1671.2                      | 892.7              | 2563.9                  | 3.7788                                  | 1.4326             | 5.2114                  |
| 355              | 17,570                                  | 0.001808                                   | 0.007872                | 1682.2                             | 706.4              | 2388.6                  | 1714.0                      | 812.9              | 2526.9                  | 3.8442                                  | 1.2942             | 5.1384                  |
| 360              | 18,666                                  | 0.001895                                   | 0.006950                | 1726.2                             | 625.7              | 2351.9                  | 1761.5                      | 720.1              | 2481.6                  | 3.9165                                  | 1.1373             | 5.0537                  |
| 365              | 19,822                                  | 0.002015                                   | 0.006009                | 1777.2                             | 526.4              | 2303.6                  | 1817.2                      | 605.5              | 2422.7                  | 4.0004                                  | 0.9489             | 4.9493                  |
| 370              | 21,044                                  | 0.002217                                   | 0.004953                | 1844.5                             | 385.6              | 2230.1                  | 1891.2                      | 443.1              | 2334.3                  | 4.1119                                  | 0.6890             | 4.8009                  |
| 373.95           | 22,064                                  | 0.003106                                   | 0.003106                | 2015.7                             | 0                  | 2015.7                  | 2084.3                      | 0                  | 2084.3                  | 4.4070                                  | 0                  | 4.4070                  |

Source of Data: Tables A-4 through A-8 are generated using the Engineering Equation Solver (EES) software developed by S. A. Klein and F. L. Alvarado. The routine used in calculations is the highly accurate Steam\_IAPWS, which incorporates the 1995 Formulation for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use, issued by The International Association for the Properties of Water and Steam (IAPWS). This formulation replaces the 1984 formulation of Haar, Gallagher, and Kell (NBS/NRC Steam Tables, Hemisphere Publishing Co., 1984), which is also available in EES as the routine STEAM. The new formulation is based on the correlations of Saul and Wagner (J. Phys. Chem. Ref. Data, 16, 893, 1987) with modifications to adjust to the International Temperature Scale of 1990. The modifications are described by Wagner and Pruss (J. Phys. Chem. Ref. Data, 22, 783, 1993). The properties of ice are based on Hyland and Wexler, "Formulations for the Thermodynamic Properties of the Saturated Phases of  $\text{H}_2\text{O}$  from 173.15 K to 473.15 K," *ASHRAE Trans.*, Part 2A, Paper 2793, 1983.

TABLE A-5

Saturated water—Pressure table

| Press.,<br><i>P</i> kPa | Sat.<br>temp.,<br><i>T</i> <sub>sat</sub> °C | Specific volume,<br>m <sup>3</sup> /kg   |                                         | Internal energy,<br>kJ/kg                |                                  |                                         | Enthalpy,<br>kJ/kg                       |                                  |                                         | Entropy,<br>kJ/kg·K                      |                                  |                                         |
|-------------------------|----------------------------------------------|------------------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|
|                         |                                              | Sat.<br>liquid,<br><i>v</i> <sub>f</sub> | Sat.<br>vapor,<br><i>v</i> <sub>g</sub> | Sat.<br>liquid,<br><i>u</i> <sub>f</sub> | Evap.,<br><i>u</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>u</i> <sub>g</sub> | Sat.<br>liquid,<br><i>h</i> <sub>f</sub> | Evap.,<br><i>h</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>h</i> <sub>g</sub> | Sat.<br>liquid,<br><i>s</i> <sub>f</sub> | Evap.,<br><i>s</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>s</i> <sub>g</sub> |
| 1.0                     | 6.97                                         | 0.001000                                 | 129.19                                  | 29.302                                   | 2355.2                           | 2384.5                                  | 29.303                                   | 2484.4                           | 2513.7                                  | 0.1059                                   | 8.8690                           | 8.9749                                  |
| 1.5                     | 13.02                                        | 0.001001                                 | 87.964                                  | 54.686                                   | 2338.1                           | 2392.8                                  | 54.688                                   | 2470.1                           | 2524.7                                  | 0.1956                                   | 8.6314                           | 8.8270                                  |
| 2.0                     | 17.50                                        | 0.001001                                 | 66.990                                  | 73.431                                   | 2325.5                           | 2398.9                                  | 73.433                                   | 2459.5                           | 2532.9                                  | 0.2606                                   | 8.4621                           | 8.7227                                  |
| 2.5                     | 21.08                                        | 0.001002                                 | 54.242                                  | 88.422                                   | 2315.4                           | 2403.8                                  | 88.424                                   | 2451.0                           | 2539.4                                  | 0.3118                                   | 8.3302                           | 8.6421                                  |
| 3.0                     | 24.08                                        | 0.001003                                 | 45.654                                  | 100.98                                   | 2306.9                           | 2407.9                                  | 100.98                                   | 2443.9                           | 2544.8                                  | 0.3543                                   | 8.2222                           | 8.5765                                  |
| 4.0                     | 28.96                                        | 0.001004                                 | 34.791                                  | 121.39                                   | 2293.1                           | 2414.5                                  | 121.39                                   | 2432.3                           | 2553.7                                  | 0.4224                                   | 8.0510                           | 8.4734                                  |
| 5.0                     | 32.87                                        | 0.001005                                 | 28.185                                  | 137.75                                   | 2282.1                           | 2419.8                                  | 137.75                                   | 2423.0                           | 2560.7                                  | 0.4762                                   | 7.9176                           | 8.3938                                  |
| 7.5                     | 40.29                                        | 0.001008                                 | 19.233                                  | 168.74                                   | 2261.1                           | 2429.8                                  | 168.75                                   | 2405.3                           | 2574.0                                  | 0.5763                                   | 7.6738                           | 8.2501                                  |
| 10                      | 45.81                                        | 0.001010                                 | 14.670                                  | 191.79                                   | 2245.4                           | 2437.2                                  | 191.81                                   | 2392.1                           | 2583.9                                  | 0.6492                                   | 7.4996                           | 8.1488                                  |
| 15                      | 53.97                                        | 0.001014                                 | 10.020                                  | 225.93                                   | 2222.1                           | 2448.0                                  | 225.94                                   | 2372.3                           | 2598.3                                  | 0.7549                                   | 7.2522                           | 8.0071                                  |
| 20                      | 60.06                                        | 0.001017                                 | 7.6481                                  | 251.40                                   | 2204.6                           | 2456.0                                  | 251.42                                   | 2357.5                           | 2608.9                                  | 0.8320                                   | 7.0752                           | 7.9073                                  |
| 25                      | 64.96                                        | 0.001020                                 | 6.2034                                  | 271.93                                   | 2190.4                           | 2462.4                                  | 271.96                                   | 2345.5                           | 2617.5                                  | 0.8932                                   | 6.9370                           | 7.8302                                  |
| 30                      | 69.09                                        | 0.001022                                 | 5.2287                                  | 289.24                                   | 2178.5                           | 2467.7                                  | 289.27                                   | 2335.3                           | 2624.6                                  | 0.9441                                   | 6.8234                           | 7.7675                                  |
| 40                      | 75.86                                        | 0.001026                                 | 3.9933                                  | 317.58                                   | 2158.8                           | 2476.3                                  | 317.62                                   | 2318.4                           | 2636.1                                  | 1.0261                                   | 6.6430                           | 7.6691                                  |
| 50                      | 81.32                                        | 0.001030                                 | 3.2403                                  | 340.49                                   | 2142.7                           | 2483.2                                  | 340.54                                   | 2304.7                           | 2645.2                                  | 1.0912                                   | 6.5019                           | 7.5931                                  |
| 75                      | 91.76                                        | 0.001037                                 | 2.2172                                  | 384.36                                   | 2111.8                           | 2496.1                                  | 384.44                                   | 2278.0                           | 2662.4                                  | 1.2132                                   | 6.2426                           | 7.4558                                  |
| 100                     | 99.61                                        | 0.001043                                 | 1.6941                                  | 417.40                                   | 2088.2                           | 2505.6                                  | 417.51                                   | 2257.5                           | 2675.0                                  | 1.3028                                   | 6.0562                           | 7.3589                                  |
| 101.325                 | 99.97                                        | 0.001043                                 | 1.6734                                  | 418.95                                   | 2087.0                           | 2506.0                                  | 419.06                                   | 2256.5                           | 2675.6                                  | 1.3069                                   | 6.0476                           | 7.3545                                  |
| 125                     | 105.97                                       | 0.001048                                 | 1.3750                                  | 444.23                                   | 2068.8                           | 2513.0                                  | 444.36                                   | 2240.6                           | 2684.9                                  | 1.3741                                   | 5.9100                           | 7.2841                                  |
| 150                     | 111.35                                       | 0.001053                                 | 1.1594                                  | 466.97                                   | 2052.3                           | 2519.2                                  | 467.13                                   | 2226.0                           | 2693.1                                  | 1.4337                                   | 5.7894                           | 7.2231                                  |
| 175                     | 116.04                                       | 0.001057                                 | 1.0037                                  | 486.82                                   | 2037.7                           | 2524.5                                  | 487.01                                   | 2213.1                           | 2700.2                                  | 1.4850                                   | 5.6865                           | 7.1716                                  |
| 200                     | 120.21                                       | 0.001061                                 | 0.88578                                 | 504.50                                   | 2024.6                           | 2529.1                                  | 504.71                                   | 2201.6                           | 2706.3                                  | 1.5302                                   | 5.5968                           | 7.1270                                  |
| 225                     | 123.97                                       | 0.001064                                 | 0.79329                                 | 520.47                                   | 2012.7                           | 2533.2                                  | 520.71                                   | 2191.0                           | 2711.7                                  | 1.5706                                   | 5.5171                           | 7.0877                                  |
| 250                     | 127.41                                       | 0.001067                                 | 0.71873                                 | 535.08                                   | 2001.8                           | 2536.8                                  | 535.35                                   | 2181.2                           | 2716.5                                  | 1.6072                                   | 5.4453                           | 7.0525                                  |
| 275                     | 130.58                                       | 0.001070                                 | 0.65732                                 | 548.57                                   | 1991.6                           | 2540.1                                  | 548.86                                   | 2172.0                           | 2720.9                                  | 1.6408                                   | 5.3800                           | 7.0207                                  |
| 300                     | 133.52                                       | 0.001073                                 | 0.60582                                 | 561.11                                   | 1982.1                           | 2543.2                                  | 561.43                                   | 2163.5                           | 2724.9                                  | 1.6717                                   | 5.3200                           | 6.9917                                  |
| 325                     | 136.27                                       | 0.001076                                 | 0.56199                                 | 572.84                                   | 1973.1                           | 2545.9                                  | 573.19                                   | 2155.4                           | 2728.6                                  | 1.7005                                   | 5.2645                           | 6.9650                                  |
| 350                     | 138.86                                       | 0.001079                                 | 0.52422                                 | 583.89                                   | 1964.6                           | 2548.5                                  | 584.26                                   | 2147.7                           | 2732.0                                  | 1.7274                                   | 5.2128                           | 6.9402                                  |
| 375                     | 141.30                                       | 0.001081                                 | 0.49133                                 | 594.32                                   | 1956.6                           | 2550.9                                  | 594.73                                   | 2140.4                           | 2735.1                                  | 1.7526                                   | 5.1645                           | 6.9171                                  |
| 400                     | 143.61                                       | 0.001084                                 | 0.46242                                 | 604.22                                   | 1948.9                           | 2553.1                                  | 604.66                                   | 2133.4                           | 2738.1                                  | 1.7765                                   | 5.1191                           | 6.8955                                  |
| 450                     | 147.90                                       | 0.001088                                 | 0.41392                                 | 622.65                                   | 1934.5                           | 2557.1                                  | 623.14                                   | 2120.3                           | 2743.4                                  | 1.8205                                   | 5.0356                           | 6.8561                                  |
| 500                     | 151.83                                       | 0.001093                                 | 0.37483                                 | 639.54                                   | 1921.2                           | 2560.7                                  | 640.09                                   | 2108.0                           | 2748.1                                  | 1.8604                                   | 4.9603                           | 6.8207                                  |
| 550                     | 155.46                                       | 0.001097                                 | 0.34261                                 | 655.16                                   | 1908.8                           | 2563.9                                  | 655.77                                   | 2096.6                           | 2752.4                                  | 1.8970                                   | 4.8916                           | 6.7886                                  |
| 600                     | 158.83                                       | 0.001101                                 | 0.31560                                 | 669.72                                   | 1897.1                           | 2566.8                                  | 670.38                                   | 2085.8                           | 2756.2                                  | 1.9308                                   | 4.8285                           | 6.7593                                  |
| 650                     | 161.98                                       | 0.001104                                 | 0.29260                                 | 683.37                                   | 1886.1                           | 2569.4                                  | 684.08                                   | 2075.5                           | 2759.6                                  | 1.9623                                   | 4.7699                           | 6.7322                                  |
| 700                     | 164.95                                       | 0.001108                                 | 0.27278                                 | 696.23                                   | 1875.6                           | 2571.8                                  | 697.00                                   | 2065.8                           | 2762.8                                  | 1.9918                                   | 4.7153                           | 6.7071                                  |
| 750                     | 167.75                                       | 0.001111                                 | 0.25552                                 | 708.40                                   | 1865.6                           | 2574.0                                  | 709.24                                   | 2056.4                           | 2765.7                                  | 2.0195                                   | 4.6642                           | 6.6837                                  |

TABLE A-5

Saturated water—Pressure table (*Concluded*)

| Press.,<br><i>P</i> kPa | Sat.<br>temp.,<br><i>T</i> <sub>sat</sub> °C | Specific volume,<br>m <sup>3</sup> /kg   |                                         | Internal energy,<br>kJ/kg                |                                  |                                         | Enthalpy,<br>kJ/kg                       |                                  |                                         | Entropy,<br>kJ/kg·K                      |                                  |                                         |
|-------------------------|----------------------------------------------|------------------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|------------------------------------------|----------------------------------|-----------------------------------------|
|                         |                                              | Sat.<br>liquid,<br><i>v</i> <sub>f</sub> | Sat.<br>vapor,<br><i>v</i> <sub>g</sub> | Sat.<br>liquid,<br><i>u</i> <sub>f</sub> | Evap.,<br><i>u</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>u</i> <sub>g</sub> | Sat.<br>liquid,<br><i>h</i> <sub>f</sub> | Evap.,<br><i>h</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>h</i> <sub>g</sub> | Sat.<br>liquid,<br><i>s</i> <sub>f</sub> | Evap.,<br><i>s</i> <sub>fg</sub> | Sat.<br>vapor,<br><i>s</i> <sub>g</sub> |
| 800                     | 170.41                                       | 0.001115                                 | 0.24035                                 | 719.97                                   | 1856.1                           | 2576.0                                  | 720.87                                   | 2047.5                           | 2768.3                                  | 2.0457                                   | 4.6160                           | 6.6616                                  |
| 850                     | 172.94                                       | 0.001118                                 | 0.22690                                 | 731.00                                   | 1846.9                           | 2577.9                                  | 731.95                                   | 2038.8                           | 2770.8                                  | 2.0705                                   | 4.5705                           | 6.6409                                  |
| 900                     | 175.35                                       | 0.001121                                 | 0.21489                                 | 741.55                                   | 1838.1                           | 2579.6                                  | 742.56                                   | 2030.5                           | 2773.0                                  | 2.0941                                   | 4.5273                           | 6.6213                                  |
| 950                     | 177.66                                       | 0.001124                                 | 0.20411                                 | 751.67                                   | 1829.6                           | 2581.3                                  | 752.74                                   | 2022.4                           | 2775.2                                  | 2.1166                                   | 4.4862                           | 6.6027                                  |
| 1000                    | 179.88                                       | 0.001127                                 | 0.19436                                 | 761.39                                   | 1821.4                           | 2582.8                                  | 762.51                                   | 2014.6                           | 2777.1                                  | 2.1381                                   | 4.4470                           | 6.5850                                  |
| 1100                    | 184.06                                       | 0.001133                                 | 0.17745                                 | 779.78                                   | 1805.7                           | 2585.5                                  | 781.03                                   | 1999.6                           | 2780.7                                  | 2.1785                                   | 4.3735                           | 6.5520                                  |
| 1200                    | 187.96                                       | 0.001138                                 | 0.16326                                 | 796.96                                   | 1790.9                           | 2587.8                                  | 798.33                                   | 1985.4                           | 2783.8                                  | 2.2159                                   | 4.3058                           | 6.5217                                  |
| 1300                    | 191.60                                       | 0.001144                                 | 0.15119                                 | 813.10                                   | 1776.8                           | 2589.9                                  | 814.59                                   | 1971.9                           | 2786.5                                  | 2.2508                                   | 4.2428                           | 6.4936                                  |
| 1400                    | 195.04                                       | 0.001149                                 | 0.14078                                 | 828.35                                   | 1763.4                           | 2591.8                                  | 829.96                                   | 1958.9                           | 2788.9                                  | 2.2835                                   | 4.1840                           | 6.4675                                  |
| 1500                    | 198.29                                       | 0.001154                                 | 0.13171                                 | 842.82                                   | 1750.6                           | 2593.4                                  | 844.55                                   | 1946.4                           | 2791.0                                  | 2.3143                                   | 4.1287                           | 6.4430                                  |
| 1750                    | 205.72                                       | 0.001166                                 | 0.11344                                 | 876.12                                   | 1720.6                           | 2596.7                                  | 878.16                                   | 1917.1                           | 2795.2                                  | 2.3844                                   | 4.0033                           | 6.3877                                  |
| 2000                    | 212.38                                       | 0.001177                                 | 0.099587                                | 906.12                                   | 1693.0                           | 2599.1                                  | 908.47                                   | 1889.8                           | 2798.3                                  | 2.4467                                   | 3.8923                           | 6.3390                                  |
| 2250                    | 218.41                                       | 0.001187                                 | 0.088717                                | 933.54                                   | 1667.3                           | 2600.9                                  | 936.21                                   | 1864.3                           | 2800.5                                  | 2.5029                                   | 3.7926                           | 6.2954                                  |
| 2500                    | 223.95                                       | 0.001197                                 | 0.079952                                | 958.87                                   | 1643.2                           | 2602.1                                  | 961.87                                   | 1840.1                           | 2801.9                                  | 2.5542                                   | 3.7016                           | 6.2558                                  |
| 3000                    | 233.85                                       | 0.001217                                 | 0.066667                                | 1004.6                                   | 1598.5                           | 2603.2                                  | 1008.3                                   | 1794.9                           | 2803.2                                  | 2.6454                                   | 3.5402                           | 6.1856                                  |
| 3500                    | 242.56                                       | 0.001235                                 | 0.057061                                | 1045.4                                   | 1557.6                           | 2603.0                                  | 1049.7                                   | 1753.0                           | 2802.7                                  | 2.7253                                   | 3.3991                           | 6.1244                                  |
| 4000                    | 250.35                                       | 0.001252                                 | 0.049779                                | 1082.4                                   | 1519.3                           | 2601.7                                  | 1087.4                                   | 1713.5                           | 2800.8                                  | 2.7966                                   | 3.2731                           | 6.0696                                  |
| 5000                    | 263.94                                       | 0.001286                                 | 0.039448                                | 1148.1                                   | 1448.9                           | 2597.0                                  | 1154.5                                   | 1639.7                           | 2794.2                                  | 2.9207                                   | 3.0530                           | 5.9737                                  |
| 6000                    | 275.59                                       | 0.001319                                 | 0.032449                                | 1205.8                                   | 1384.1                           | 2589.9                                  | 1213.8                                   | 1570.9                           | 2784.6                                  | 3.0275                                   | 2.8627                           | 5.8902                                  |
| 7000                    | 285.83                                       | 0.001352                                 | 0.027378                                | 1258.0                                   | 1323.0                           | 2581.0                                  | 1267.5                                   | 1505.2                           | 2772.6                                  | 3.1220                                   | 2.6927                           | 5.8148                                  |
| 8000                    | 295.01                                       | 0.001384                                 | 0.023525                                | 1306.0                                   | 1264.5                           | 2570.5                                  | 1317.1                                   | 1441.6                           | 2758.7                                  | 3.2077                                   | 2.5373                           | 5.7450                                  |
| 9000                    | 303.35                                       | 0.001418                                 | 0.020489                                | 1350.9                                   | 1207.6                           | 2558.5                                  | 1363.7                                   | 1379.3                           | 2742.9                                  | 3.2866                                   | 2.3925                           | 5.6791                                  |
| 10,000                  | 311.00                                       | 0.001452                                 | 0.018028                                | 1393.3                                   | 1151.8                           | 2545.2                                  | 1407.8                                   | 1317.6                           | 2725.5                                  | 3.3603                                   | 2.2556                           | 5.6159                                  |
| 11,000                  | 318.08                                       | 0.001488                                 | 0.015988                                | 1433.9                                   | 1096.6                           | 2530.4                                  | 1450.2                                   | 1256.1                           | 2706.3                                  | 3.4299                                   | 2.1245                           | 5.5544                                  |
| 12,000                  | 324.68                                       | 0.001526                                 | 0.014264                                | 1473.0                                   | 1041.3                           | 2514.3                                  | 1491.3                                   | 1194.1                           | 2685.4                                  | 3.4964                                   | 1.9975                           | 5.4939                                  |
| 13,000                  | 330.85                                       | 0.001566                                 | 0.012781                                | 1511.0                                   | 985.5                            | 2496.6                                  | 1531.4                                   | 1131.3                           | 2662.7                                  | 3.5606                                   | 1.8730                           | 5.4336                                  |
| 14,000                  | 336.67                                       | 0.001610                                 | 0.011487                                | 1548.4                                   | 928.7                            | 2477.1                                  | 1571.0                                   | 1067.0                           | 2637.9                                  | 3.6232                                   | 1.7497                           | 5.3728                                  |
| 15,000                  | 342.16                                       | 0.001657                                 | 0.010341                                | 1585.5                                   | 870.3                            | 2455.7                                  | 1610.3                                   | 1000.5                           | 2610.8                                  | 3.6848                                   | 1.6261                           | 5.3108                                  |
| 16,000                  | 347.36                                       | 0.001710                                 | 0.009312                                | 1622.6                                   | 809.4                            | 2432.0                                  | 1649.9                                   | 931.1                            | 2581.0                                  | 3.7461                                   | 1.5005                           | 5.2466                                  |
| 17,000                  | 352.29                                       | 0.001770                                 | 0.008374                                | 1660.2                                   | 745.1                            | 2405.4                                  | 1690.3                                   | 857.4                            | 2547.7                                  | 3.8082                                   | 1.3709                           | 5.1791                                  |
| 18,000                  | 356.99                                       | 0.001840                                 | 0.007504                                | 1699.1                                   | 675.9                            | 2375.0                                  | 1732.2                                   | 777.8                            | 2510.0                                  | 3.8720                                   | 1.2343                           | 5.1064                                  |
| 19,000                  | 361.47                                       | 0.001926                                 | 0.006677                                | 1740.3                                   | 598.9                            | 2339.2                                  | 1776.8                                   | 689.2                            | 2466.0                                  | 3.9396                                   | 1.0860                           | 5.0256                                  |
| 20,000                  | 365.75                                       | 0.002038                                 | 0.005862                                | 1785.8                                   | 509.0                            | 2294.8                                  | 1826.6                                   | 585.5                            | 2412.1                                  | 4.0146                                   | 0.9164                           | 4.9310                                  |
| 21,000                  | 369.83                                       | 0.002207                                 | 0.004994                                | 1841.6                                   | 391.9                            | 2233.5                                  | 1888.0                                   | 450.4                            | 2338.4                                  | 4.1071                                   | 0.7005                           | 4.8076                                  |
| 22,000                  | 373.71                                       | 0.002703                                 | 0.003644                                | 1951.7                                   | 140.8                            | 2092.4                                  | 2011.1                                   | 161.5                            | 2172.6                                  | 4.2942                                   | 0.2496                           | 4.5439                                  |
| 22,064                  | 373.95                                       | 0.003106                                 | 0.003106                                | 2015.7                                   | 0                                | 2015.7                                  | 2084.3                                   | 0                                | 2084.3                                  | 4.4070                                   | 0                                | 4.4070                                  |

TABLE A-6

Superheated water

| $T$<br>°C                         | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg         | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg         | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|-----------------------------------|---------------------------|--------------|--------------|----------------|-----------------------------------|--------------|--------------|----------------|-----------------------------------|--------------|--------------|----------------|
| $P = 0.01 \text{ MPa (45.81°C)*}$ |                           |              |              |                | $P = 0.05 \text{ MPa (81.32°C)}$  |              |              |                | $P = 0.10 \text{ MPa (99.61°C)}$  |              |              |                |
| Sat. <sup>†</sup>                 | 14.670                    | 2437.2       | 2583.9       | 8.1488         | 3.2403                            | 2483.2       | 2645.2       | 7.5931         | 1.6941                            | 2505.6       | 2675.0       | 7.3589         |
| 50                                | 14.867                    | 2443.3       | 2592.0       | 8.1741         |                                   |              |              |                |                                   |              |              |                |
| 100                               | 17.196                    | 2515.5       | 2687.5       | 8.4489         | 3.4187                            | 2511.5       | 2682.4       | 7.6953         | 1.6959                            | 2506.2       | 2675.8       | 7.3611         |
| 150                               | 19.513                    | 2587.9       | 2783.0       | 8.6893         | 3.8897                            | 2585.7       | 2780.2       | 7.9413         | 1.9367                            | 2582.9       | 2776.6       | 7.6148         |
| 200                               | 21.826                    | 2661.4       | 2879.6       | 8.9049         | 4.3562                            | 2660.0       | 2877.8       | 8.1592         | 2.1724                            | 2658.2       | 2875.5       | 7.8356         |
| 250                               | 24.136                    | 2736.1       | 2977.5       | 9.1015         | 4.8206                            | 2735.1       | 2976.2       | 8.3568         | 2.4062                            | 2733.9       | 2974.5       | 8.0346         |
| 300                               | 26.446                    | 2812.3       | 3076.7       | 9.2827         | 5.2841                            | 2811.6       | 3075.8       | 8.5387         | 2.6389                            | 2810.7       | 3074.5       | 8.2172         |
| 400                               | 31.063                    | 2969.3       | 3280.0       | 9.6094         | 6.2094                            | 2968.9       | 3279.3       | 8.8659         | 3.1027                            | 2968.3       | 3278.6       | 8.5452         |
| 500                               | 35.680                    | 3132.9       | 3489.7       | 9.8998         | 7.1338                            | 3132.6       | 3489.3       | 9.1566         | 3.5655                            | 3132.2       | 3488.7       | 8.8362         |
| 600                               | 40.296                    | 3303.3       | 3706.3       | 10.1631        | 8.0577                            | 3303.1       | 3706.0       | 9.4201         | 4.0279                            | 3302.8       | 3705.6       | 9.0999         |
| 700                               | 44.911                    | 3480.8       | 3929.9       | 10.4056        | 8.9813                            | 3480.6       | 3929.7       | 9.6626         | 4.4900                            | 3480.4       | 3929.4       | 9.3424         |
| 800                               | 49.527                    | 3665.4       | 4160.6       | 10.6312        | 9.9047                            | 3665.2       | 4160.4       | 9.8883         | 4.9519                            | 3665.0       | 4160.2       | 9.5682         |
| 900                               | 54.143                    | 3856.9       | 4398.3       | 10.8429        | 10.8280                           | 3856.8       | 4398.2       | 10.1000        | 5.4137                            | 3856.7       | 4398.0       | 9.7800         |
| 1000                              | 58.758                    | 4055.3       | 4642.8       | 11.0429        | 11.7513                           | 4055.2       | 4642.7       | 10.3000        | 5.8755                            | 4055.0       | 4642.6       | 9.9800         |
| 1100                              | 63.373                    | 4260.0       | 4893.8       | 11.2326        | 12.6745                           | 4259.9       | 4893.7       | 10.4897        | 6.3372                            | 4259.8       | 4893.6       | 10.1698        |
| 1200                              | 67.989                    | 4470.9       | 5150.8       | 11.4132        | 13.5977                           | 4470.8       | 5150.7       | 10.6704        | 6.7988                            | 4470.7       | 5150.6       | 10.3504        |
| 1300                              | 72.604                    | 4687.4       | 5413.4       | 11.5857        | 14.5209                           | 4687.3       | 5413.3       | 10.8429        | 7.2605                            | 4687.2       | 5413.3       | 10.5229        |
| $P = 0.20 \text{ MPa (120.21°C)}$ |                           |              |              |                | $P = 0.30 \text{ MPa (133.52°C)}$ |              |              |                | $P = 0.40 \text{ MPa (143.61°C)}$ |              |              |                |
| Sat.                              | 0.88578                   | 2529.1       | 2706.3       | 7.1270         | 0.60582                           | 2543.2       | 2724.9       | 6.9917         | 0.46242                           | 2553.1       | 2738.1       | 6.8955         |
| 150                               | 0.95986                   | 2577.1       | 2769.1       | 7.2810         | 0.63402                           | 2571.0       | 2761.2       | 7.0792         | 0.47088                           | 2564.4       | 2752.8       | 6.9306         |
| 200                               | 1.08049                   | 2654.6       | 2870.7       | 7.5081         | 0.71643                           | 2651.0       | 2865.9       | 7.3132         | 0.53434                           | 2647.2       | 2860.9       | 7.1723         |
| 250                               | 1.19890                   | 2731.4       | 2971.2       | 7.7100         | 0.79645                           | 2728.9       | 2967.9       | 7.5180         | 0.59520                           | 2726.4       | 2964.5       | 7.3804         |
| 300                               | 1.31623                   | 2808.8       | 3072.1       | 7.8941         | 0.87535                           | 2807.0       | 3069.6       | 7.7037         | 0.65489                           | 2805.1       | 3067.1       | 7.5677         |
| 400                               | 1.54934                   | 2967.2       | 3277.0       | 8.2236         | 1.03155                           | 2966.0       | 3275.5       | 8.0347         | 0.77265                           | 2964.9       | 3273.9       | 7.9003         |
| 500                               | 1.78142                   | 3131.4       | 3487.7       | 8.5153         | 1.18672                           | 3130.6       | 3486.6       | 8.3271         | 0.88936                           | 3129.8       | 3485.5       | 8.1933         |
| 600                               | 2.01302                   | 3302.2       | 3704.8       | 8.7793         | 1.34139                           | 3301.6       | 3704.0       | 8.5915         | 1.00558                           | 3301.0       | 3703.3       | 8.4580         |
| 700                               | 2.24434                   | 3479.9       | 3928.8       | 9.0221         | 1.49580                           | 3479.5       | 3928.2       | 8.8345         | 1.12152                           | 3479.0       | 3927.6       | 8.7012         |
| 800                               | 2.47550                   | 3664.7       | 4159.8       | 9.2479         | 1.65004                           | 3664.3       | 4159.3       | 9.0605         | 1.23730                           | 3663.9       | 4158.9       | 8.9274         |
| 900                               | 2.70656                   | 3856.3       | 4397.7       | 9.4598         | 1.80417                           | 3856.0       | 4397.3       | 9.2725         | 1.35298                           | 3855.7       | 4396.9       | 9.1394         |
| 1000                              | 2.93755                   | 4054.8       | 4642.3       | 9.6599         | 1.95824                           | 4054.5       | 4642.0       | 9.4726         | 1.46859                           | 4054.3       | 4641.7       | 9.3396         |
| 1100                              | 3.16848                   | 4259.6       | 4893.3       | 9.8497         | 2.11226                           | 4259.4       | 4893.1       | 9.6624         | 1.58414                           | 4259.2       | 4892.9       | 9.5295         |
| 1200                              | 3.39938                   | 4470.5       | 5150.4       | 10.0304        | 2.26624                           | 4470.3       | 5150.2       | 9.8431         | 1.69966                           | 4470.2       | 5150.0       | 9.7102         |
| 1300                              | 3.63026                   | 4687.1       | 5413.1       | 10.2029        | 2.42019                           | 4686.9       | 5413.0       | 10.0157        | 1.81516                           | 4686.7       | 5412.8       | 9.8828         |
| $P = 0.50 \text{ MPa (151.83°C)}$ |                           |              |              |                | $P = 0.60 \text{ MPa (158.83°C)}$ |              |              |                | $P = 0.80 \text{ MPa (170.41°C)}$ |              |              |                |
| Sat.                              | 0.37483                   | 2560.7       | 2748.1       | 6.8207         | 0.31560                           | 2566.8       | 2756.2       | 6.7593         | 0.24035                           | 2576.0       | 2768.3       | 6.6616         |
| 200                               | 0.42503                   | 2643.3       | 2855.8       | 7.0610         | 0.35212                           | 2639.4       | 2850.6       | 6.9683         | 0.26088                           | 2631.1       | 2839.8       | 6.8177         |
| 250                               | 0.47443                   | 2723.8       | 2961.0       | 7.2725         | 0.39390                           | 2721.2       | 2957.6       | 7.1833         | 0.29321                           | 2715.9       | 2950.4       | 7.0402         |
| 300                               | 0.52261                   | 2803.3       | 3064.6       | 7.4614         | 0.43442                           | 2801.4       | 3062.0       | 7.3740         | 0.32416                           | 2797.5       | 3056.9       | 7.2345         |
| 350                               | 0.57015                   | 2883.0       | 3168.1       | 7.6346         | 0.47428                           | 2881.6       | 3166.1       | 7.5481         | 0.35442                           | 2878.6       | 3162.2       | 7.4107         |
| 400                               | 0.61731                   | 2963.7       | 3272.4       | 7.7956         | 0.51374                           | 2962.5       | 3270.8       | 7.7097         | 0.38429                           | 2960.2       | 3267.7       | 7.5735         |
| 500                               | 0.71095                   | 3129.0       | 3484.5       | 8.0893         | 0.59200                           | 3128.2       | 3483.4       | 8.0041         | 0.44332                           | 3126.6       | 3481.3       | 7.8692         |
| 600                               | 0.80409                   | 3300.4       | 3702.5       | 8.3544         | 0.66976                           | 3299.8       | 3701.7       | 8.2695         | 0.50186                           | 3298.7       | 3700.1       | 8.1354         |
| 700                               | 0.89696                   | 3478.6       | 3927.0       | 8.5978         | 0.74725                           | 3478.1       | 3926.4       | 8.5132         | 0.56011                           | 3477.2       | 3925.3       | 8.3794         |
| 800                               | 0.98966                   | 3663.6       | 4158.4       | 8.8240         | 0.82457                           | 3663.2       | 4157.9       | 8.7395         | 0.61820                           | 3662.5       | 4157.0       | 8.6061         |
| 900                               | 1.08227                   | 3855.4       | 4396.6       | 9.0362         | 0.90179                           | 3855.1       | 4396.2       | 8.9518         | 0.67619                           | 3854.5       | 4395.5       | 8.8185         |
| 1000                              | 1.17480                   | 4054.0       | 4641.4       | 9.2364         | 0.97893                           | 4053.8       | 4641.1       | 9.1521         | 0.73411                           | 4053.3       | 4640.5       | 9.0189         |
| 1100                              | 1.26728                   | 4259.0       | 4892.6       | 9.4263         | 1.05603                           | 4258.8       | 4892.4       | 9.3420         | 0.79197                           | 4258.3       | 4891.9       | 9.2090         |
| 1200                              | 1.35972                   | 4470.0       | 5149.8       | 9.6071         | 1.13309                           | 4469.8       | 5149.6       | 9.5229         | 0.84980                           | 4469.4       | 5149.3       | 9.3898         |
| 1300                              | 1.45214                   | 4686.6       | 5412.6       | 9.7797         | 1.21012                           | 4686.4       | 5412.5       | 9.6955         | 0.90761                           | 4686.1       | 5412.2       | 9.5625         |

\*The temperature in parentheses is the saturation temperature at the specified pressure.

† Properties of saturated vapor at the specified pressure.

TABLE A-6

Superheated water (*Concluded*)

| $T$<br>°C                         | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg         | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg         | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|-----------------------------------|---------------------------|--------------|--------------|----------------|-----------------------------------|--------------|--------------|----------------|-----------------------------------|--------------|--------------|----------------|
| $P = 1.00 \text{ MPa (179.88°C)}$ |                           |              |              |                | $P = 1.20 \text{ MPa (187.96°C)}$ |              |              |                | $P = 1.40 \text{ MPa (195.04°C)}$ |              |              |                |
| Sat.                              | 0.19437                   | 2582.8       | 2777.1       | 6.5850         | 0.16326                           | 2587.8       | 2783.8       | 6.5217         | 0.14078                           | 2591.8       | 2788.9       | 6.4675         |
| 200                               | 0.20602                   | 2622.3       | 2828.3       | 6.6956         | 0.16934                           | 2612.9       | 2816.1       | 6.5909         | 0.14303                           | 2602.7       | 2803.0       | 6.4975         |
| 250                               | 0.23275                   | 2710.4       | 2943.1       | 6.9265         | 0.19241                           | 2704.7       | 2935.6       | 6.8313         | 0.16356                           | 2698.9       | 2927.9       | 6.7488         |
| 300                               | 0.25799                   | 2793.7       | 3051.6       | 7.1246         | 0.21386                           | 2789.7       | 3046.3       | 7.0335         | 0.18233                           | 2785.7       | 3040.9       | 6.9553         |
| 350                               | 0.28250                   | 2875.7       | 3158.2       | 7.3029         | 0.23455                           | 2872.7       | 3154.2       | 7.2139         | 0.20029                           | 2869.7       | 3150.1       | 7.1379         |
| 400                               | 0.30661                   | 2957.9       | 3264.5       | 7.4670         | 0.25482                           | 2955.5       | 3261.3       | 7.3793         | 0.21782                           | 2953.1       | 3258.1       | 7.3046         |
| 500                               | 0.35411                   | 3125.0       | 3479.1       | 7.7642         | 0.29464                           | 3123.4       | 3477.0       | 7.6779         | 0.25216                           | 3121.8       | 3474.8       | 7.6047         |
| 600                               | 0.40111                   | 3297.5       | 3698.6       | 8.0311         | 0.33395                           | 3296.3       | 3697.0       | 7.9456         | 0.28597                           | 3295.1       | 3695.5       | 7.8730         |
| 700                               | 0.44783                   | 3476.3       | 3924.1       | 8.2755         | 0.37297                           | 3475.3       | 3922.9       | 8.1904         | 0.31951                           | 3474.4       | 3921.7       | 8.1183         |
| 800                               | 0.49438                   | 3661.7       | 4156.1       | 8.5024         | 0.41184                           | 3661.0       | 4155.2       | 8.4176         | 0.35288                           | 3660.3       | 4154.3       | 8.3458         |
| 900                               | 0.54083                   | 3853.9       | 4394.8       | 8.7150         | 0.45059                           | 3853.3       | 4394.0       | 8.6303         | 0.38614                           | 3852.7       | 4393.3       | 8.5587         |
| 1000                              | 0.58721                   | 4052.7       | 4640.0       | 8.9155         | 0.48928                           | 4052.2       | 4639.4       | 8.8310         | 0.41933                           | 4051.7       | 4638.8       | 8.7595         |
| 1100                              | 0.63354                   | 4257.9       | 4891.4       | 9.1057         | 0.52792                           | 4257.5       | 4891.0       | 9.0212         | 0.45247                           | 4257.0       | 4890.5       | 8.9497         |
| 1200                              | 0.67983                   | 4469.0       | 5148.9       | 9.2866         | 0.56652                           | 4468.7       | 5148.5       | 9.2022         | 0.48558                           | 4468.3       | 5148.1       | 9.1308         |
| 1300                              | 0.72610                   | 4685.8       | 5411.9       | 9.4593         | 0.60509                           | 4685.5       | 5411.6       | 9.3750         | 0.51866                           | 4685.1       | 5411.3       | 9.3036         |
| $P = 1.60 \text{ MPa (201.37°C)}$ |                           |              |              |                | $P = 1.80 \text{ MPa (207.11°C)}$ |              |              |                | $P = 2.00 \text{ MPa (212.38°C)}$ |              |              |                |
| Sat.                              | 0.12374                   | 2594.8       | 2792.8       | 6.4200         | 0.11037                           | 2597.3       | 2795.9       | 6.3775         | 0.09959                           | 2599.1       | 2798.3       | 6.3390         |
| 225                               | 0.13293                   | 2645.1       | 2857.8       | 6.5537         | 0.11678                           | 2637.0       | 2847.2       | 6.4825         | 0.10381                           | 2628.5       | 2836.1       | 6.4160         |
| 250                               | 0.14190                   | 2692.9       | 2919.9       | 6.6753         | 0.12502                           | 2686.7       | 2911.7       | 6.6088         | 0.11150                           | 2680.3       | 2903.3       | 6.5475         |
| 300                               | 0.15866                   | 2781.6       | 3035.4       | 6.8864         | 0.14025                           | 2777.4       | 3029.9       | 6.8246         | 0.12551                           | 2773.2       | 3024.2       | 6.7684         |
| 350                               | 0.17459                   | 2866.6       | 3146.0       | 7.0713         | 0.15460                           | 2863.6       | 3141.9       | 7.0120         | 0.13860                           | 2860.5       | 3137.7       | 6.9583         |
| 400                               | 0.19007                   | 2950.8       | 3254.9       | 7.2394         | 0.16849                           | 2948.3       | 3251.6       | 7.1814         | 0.15122                           | 2945.9       | 3248.4       | 7.1292         |
| 500                               | 0.22029                   | 3120.1       | 3472.6       | 7.5410         | 0.19551                           | 3118.5       | 3470.4       | 7.4845         | 0.17568                           | 3116.9       | 3468.3       | 7.4337         |
| 600                               | 0.24999                   | 3293.9       | 3693.9       | 7.8101         | 0.22200                           | 3292.7       | 3692.3       | 7.7543         | 0.19962                           | 3291.5       | 3690.7       | 7.7043         |
| 700                               | 0.27941                   | 3473.5       | 3920.5       | 8.0558         | 0.24822                           | 3472.6       | 3919.4       | 8.0005         | 0.22326                           | 3471.7       | 3918.2       | 7.9509         |
| 800                               | 0.30865                   | 3659.5       | 4153.4       | 8.2834         | 0.27426                           | 3658.8       | 4152.4       | 8.2284         | 0.24674                           | 3658.0       | 4151.5       | 8.1791         |
| 900                               | 0.33780                   | 3852.1       | 4392.6       | 8.4965         | 0.30020                           | 3851.5       | 4391.9       | 8.4417         | 0.27012                           | 3850.9       | 4391.1       | 8.3925         |
| 1000                              | 0.36687                   | 4051.2       | 4638.2       | 8.6974         | 0.32606                           | 4050.7       | 4637.6       | 8.6427         | 0.29342                           | 4050.2       | 4637.1       | 8.5936         |
| 1100                              | 0.39589                   | 4256.6       | 4890.0       | 8.8878         | 0.35188                           | 4256.2       | 4889.6       | 8.8331         | 0.31667                           | 4255.7       | 4889.1       | 8.7842         |
| 1200                              | 0.42488                   | 4467.9       | 5147.7       | 9.0689         | 0.37766                           | 4467.6       | 5147.3       | 9.0143         | 0.33989                           | 4467.2       | 5147.0       | 8.9654         |
| 1300                              | 0.45383                   | 4684.8       | 5410.9       | 9.2418         | 0.40341                           | 4684.5       | 5410.6       | 9.1872         | 0.36308                           | 4684.2       | 5410.3       | 9.1384         |
| $P = 2.50 \text{ MPa (223.95°C)}$ |                           |              |              |                | $P = 3.00 \text{ MPa (233.85°C)}$ |              |              |                | $P = 3.50 \text{ MPa (242.56°C)}$ |              |              |                |
| Sat.                              | 0.07995                   | 2602.1       | 2801.9       | 6.2558         | 0.06667                           | 2603.2       | 2803.2       | 6.1856         | 0.05706                           | 2603.0       | 2802.7       | 6.1244         |
| 225                               | 0.08026                   | 2604.8       | 2805.5       | 6.2629         |                                   |              |              |                |                                   |              |              |                |
| 250                               | 0.08705                   | 2663.3       | 2880.9       | 6.4107         | 0.07063                           | 2644.7       | 2856.5       | 6.2893         | 0.05876                           | 2624.0       | 2829.7       | 6.1764         |
| 300                               | 0.09894                   | 2762.2       | 3009.6       | 6.6459         | 0.08118                           | 2750.8       | 2994.3       | 6.5412         | 0.06845                           | 2738.8       | 2978.4       | 6.4484         |
| 350                               | 0.10979                   | 2852.5       | 3127.0       | 6.8424         | 0.09056                           | 2844.4       | 3116.1       | 6.7450         | 0.07680                           | 2836.0       | 3104.9       | 6.6601         |
| 400                               | 0.12012                   | 2939.8       | 3240.1       | 7.0170         | 0.09938                           | 2933.6       | 3231.7       | 6.9235         | 0.08456                           | 2927.2       | 3223.2       | 6.8428         |
| 450                               | 0.13015                   | 3026.2       | 3351.6       | 7.1768         | 0.10789                           | 3021.2       | 3344.9       | 7.0856         | 0.09198                           | 3016.1       | 3338.1       | 7.0074         |
| 500                               | 0.13999                   | 3112.8       | 3462.8       | 7.3254         | 0.11620                           | 3108.6       | 3457.2       | 7.2359         | 0.09919                           | 3104.5       | 3451.7       | 7.1593         |
| 600                               | 0.15931                   | 3288.5       | 3686.8       | 7.5979         | 0.13245                           | 3285.5       | 3682.8       | 7.5103         | 0.11325                           | 3282.5       | 3678.9       | 7.4357         |
| 700                               | 0.17835                   | 3469.3       | 3915.2       | 7.8455         | 0.14841                           | 3467.0       | 3912.2       | 7.7590         | 0.12702                           | 3464.7       | 3909.3       | 7.6855         |
| 800                               | 0.19722                   | 3656.2       | 4149.2       | 8.0744         | 0.16420                           | 3654.3       | 4146.9       | 7.9885         | 0.14061                           | 3652.5       | 4144.6       | 7.9156         |
| 900                               | 0.21597                   | 3849.4       | 4389.3       | 8.2882         | 0.17988                           | 3847.9       | 4387.5       | 8.2028         | 0.15410                           | 3846.4       | 4385.7       | 8.1304         |
| 1000                              | 0.23466                   | 4049.0       | 4635.6       | 8.4897         | 0.19549                           | 4047.7       | 4634.2       | 8.4045         | 0.16751                           | 4046.4       | 4632.7       | 8.3324         |
| 1100                              | 0.25330                   | 4254.7       | 4887.9       | 8.6804         | 0.21105                           | 4253.6       | 4886.7       | 8.5955         | 0.18087                           | 4252.5       | 4885.6       | 8.5236         |
| 1200                              | 0.27190                   | 4466.3       | 5146.0       | 8.8618         | 0.22658                           | 4465.3       | 5145.1       | 8.7771         | 0.19420                           | 4464.4       | 5144.1       | 8.7053         |
| 1300                              | 0.29048                   | 4683.4       | 5409.5       | 9.0349         | 0.24207                           | 4682.6       | 5408.8       | 8.9502         | 0.20750                           | 4681.8       | 5408.0       | 8.8786         |

TABLE A-6

Superheated water (*Continued*)

| <i>T</i><br>°C                | <i>v</i><br>m <sup>3</sup> /kg | <i>u</i><br>kJ/kg | <i>h</i><br>kJ/kg | <i>s</i><br>kJ/kg·K | <i>v</i><br>m <sup>3</sup> /kg | <i>u</i><br>kJ/kg | <i>h</i><br>kJ/kg | <i>s</i><br>kJ/kg·K | <i>v</i><br>m <sup>3</sup> /kg | <i>u</i><br>kJ/kg | <i>h</i><br>kJ/kg | <i>s</i><br>kJ/kg·K |
|-------------------------------|--------------------------------|-------------------|-------------------|---------------------|--------------------------------|-------------------|-------------------|---------------------|--------------------------------|-------------------|-------------------|---------------------|
| <i>P</i> = 4.0 MPa (250.35°C) |                                |                   |                   |                     | <i>P</i> = 4.5 MPa (257.44°C)  |                   |                   |                     | <i>P</i> = 5.0 MPa (263.94°C)  |                   |                   |                     |
| Sat.                          | 0.04978                        | 2601.7            | 2800.8            | 6.0696              | 0.04406                        | 2599.7            | 2798.0            | 6.0198              | 0.03945                        | 2597.0            | 2794.2            | 5.9737              |
| 275                           | 0.05461                        | 2668.9            | 2887.3            | 6.2312              | 0.04733                        | 2651.4            | 2864.4            | 6.1429              | 0.04144                        | 2632.3            | 2839.5            | 6.0571              |
| 300                           | 0.05887                        | 2726.2            | 2961.7            | 6.3639              | 0.05138                        | 2713.0            | 2944.2            | 6.2854              | 0.04535                        | 2699.0            | 2925.7            | 6.2111              |
| 350                           | 0.06647                        | 2827.4            | 3093.3            | 6.5843              | 0.05842                        | 2818.6            | 3081.5            | 6.5153              | 0.05197                        | 2809.5            | 3069.3            | 6.4516              |
| 400                           | 0.07343                        | 2920.8            | 3214.5            | 6.7714              | 0.06477                        | 2914.2            | 3205.7            | 6.7071              | 0.05784                        | 2907.5            | 3196.7            | 6.6483              |
| 450                           | 0.08004                        | 3011.0            | 3331.2            | 6.9386              | 0.07076                        | 3005.8            | 3324.2            | 6.8770              | 0.06332                        | 3000.6            | 3317.2            | 6.8210              |
| 500                           | 0.08644                        | 3100.3            | 3446.0            | 7.0922              | 0.07652                        | 3096.0            | 3440.4            | 7.0323              | 0.06858                        | 3091.8            | 3434.7            | 6.9781              |
| 600                           | 0.09886                        | 3279.4            | 3674.9            | 7.3706              | 0.08766                        | 3276.4            | 3670.9            | 7.3127              | 0.07870                        | 3273.3            | 3666.9            | 7.2605              |
| 700                           | 0.11098                        | 3462.4            | 3906.3            | 7.6214              | 0.09850                        | 3460.0            | 3903.3            | 7.5647              | 0.08852                        | 3457.7            | 3900.3            | 7.5136              |
| 800                           | 0.12292                        | 3650.6            | 4142.3            | 7.8523              | 0.10916                        | 3648.8            | 4140.0            | 7.7962              | 0.09816                        | 3646.9            | 4137.7            | 7.7458              |
| 900                           | 0.13476                        | 3844.8            | 4383.9            | 8.0675              | 0.11972                        | 3843.3            | 4382.1            | 8.0118              | 0.10769                        | 3841.8            | 4380.2            | 7.9619              |
| 1000                          | 0.14653                        | 4045.1            | 4631.2            | 8.2698              | 0.13020                        | 4043.9            | 4629.8            | 8.2144              | 0.11715                        | 4042.6            | 4628.3            | 8.1648              |
| 1100                          | 0.15824                        | 4251.4            | 4884.4            | 8.4612              | 0.14064                        | 4250.4            | 4883.2            | 8.4060              | 0.12655                        | 4249.3            | 4882.1            | 8.3566              |
| 1200                          | 0.16992                        | 4463.5            | 5143.2            | 8.6430              | 0.15103                        | 4462.6            | 5142.2            | 8.5880              | 0.13592                        | 4461.6            | 5141.3            | 8.5388              |
| 1300                          | 0.18157                        | 4680.9            | 5407.2            | 8.8164              | 0.16140                        | 4680.1            | 5406.5            | 8.7616              | 0.14527                        | 4679.3            | 5405.7            | 8.7124              |
| <i>P</i> = 6.0 MPa (275.59°C) |                                |                   |                   |                     | <i>P</i> = 7.0 MPa (285.83°C)  |                   |                   |                     | <i>P</i> = 8.0 MPa (295.01°C)  |                   |                   |                     |
| Sat.                          | 0.03245                        | 2589.9            | 2784.6            | 5.8902              | 0.027378                       | 2581.0            | 2772.6            | 5.8148              | 0.023525                       | 2570.5            | 2758.7            | 5.7450              |
| 300                           | 0.03619                        | 2668.4            | 2885.6            | 6.0703              | 0.029492                       | 2633.5            | 2839.9            | 5.9337              | 0.024279                       | 2592.3            | 2786.5            | 5.7937              |
| 350                           | 0.04225                        | 2790.4            | 3043.9            | 6.3357              | 0.035262                       | 2770.1            | 3016.9            | 6.2305              | 0.029975                       | 2748.3            | 2988.1            | 6.1321              |
| 400                           | 0.04742                        | 2893.7            | 3178.3            | 6.5432              | 0.039958                       | 2879.5            | 3159.2            | 6.4502              | 0.034344                       | 2864.6            | 3139.4            | 6.3658              |
| 450                           | 0.05217                        | 2989.9            | 3302.9            | 6.7219              | 0.044187                       | 2979.0            | 3288.3            | 6.6353              | 0.038194                       | 2967.8            | 3273.3            | 6.5579              |
| 500                           | 0.05667                        | 3083.1            | 3423.1            | 6.8826              | 0.048157                       | 3074.3            | 3411.4            | 6.8000              | 0.041767                       | 3065.4            | 3399.5            | 6.7266              |
| 550                           | 0.06102                        | 3175.2            | 3541.3            | 7.0308              | 0.051966                       | 3167.9            | 3531.6            | 6.9507              | 0.045172                       | 3160.5            | 3521.8            | 6.8800              |
| 600                           | 0.06527                        | 3267.2            | 3658.8            | 7.1693              | 0.055665                       | 3261.0            | 3650.6            | 7.0910              | 0.048463                       | 3254.7            | 3642.4            | 7.0221              |
| 700                           | 0.07355                        | 3453.0            | 3894.3            | 7.4247              | 0.062850                       | 3448.3            | 3888.3            | 7.3487              | 0.054829                       | 3443.6            | 3882.2            | 7.2822              |
| 800                           | 0.08165                        | 3643.2            | 4133.1            | 7.6582              | 0.069856                       | 3639.5            | 4128.5            | 7.5836              | 0.061011                       | 3635.7            | 4123.8            | 7.5185              |
| 900                           | 0.08964                        | 3838.8            | 4376.6            | 7.8751              | 0.076750                       | 3835.7            | 4373.0            | 7.8014              | 0.067082                       | 3832.7            | 4369.3            | 7.7372              |
| 1000                          | 0.09756                        | 4040.1            | 4625.4            | 8.0786              | 0.083571                       | 4037.5            | 4622.5            | 8.0055              | 0.073079                       | 4035.0            | 4619.6            | 7.9419              |
| 1100                          | 0.10543                        | 4247.1            | 4879.7            | 8.2709              | 0.090341                       | 4245.0            | 4877.4            | 8.1982              | 0.079025                       | 4242.8            | 4875.0            | 8.1350              |
| 1200                          | 0.11326                        | 4459.8            | 5139.4            | 8.4534              | 0.097075                       | 4457.9            | 5137.4            | 8.3810              | 0.084934                       | 4456.1            | 5135.5            | 8.3181              |
| 1300                          | 0.12107                        | 4677.7            | 5404.1            | 8.6273              | 0.103781                       | 4676.1            | 5402.6            | 8.5551              | 0.090817                       | 4674.5            | 5401.0            | 8.4925              |
| <i>P</i> = 9.0 MPa (303.35°C) |                                |                   |                   |                     | <i>P</i> = 10.0 MPa (311.00°C) |                   |                   |                     | <i>P</i> = 12.5 MPa (327.81°C) |                   |                   |                     |
| Sat.                          | 0.020489                       | 2558.5            | 2742.9            | 5.6791              | 0.018028                       | 2545.2            | 2725.5            | 5.6159              | 0.013496                       | 2505.6            | 2674.3            | 5.4638              |
| 325                           | 0.023284                       | 2647.6            | 2857.1            | 5.8738              | 0.019877                       | 2611.6            | 2810.3            | 5.7596              |                                |                   |                   |                     |
| 350                           | 0.025816                       | 2725.0            | 2957.3            | 6.0380              | 0.022440                       | 2699.6            | 2924.0            | 5.9460              | 0.016138                       | 2624.9            | 2826.6            | 5.7130              |
| 400                           | 0.029960                       | 2849.2            | 3118.8            | 6.2876              | 0.026436                       | 2833.1            | 3097.5            | 6.2141              | 0.020030                       | 2789.6            | 3040.0            | 6.0433              |
| 450                           | 0.033524                       | 2956.3            | 3258.0            | 6.4872              | 0.029782                       | 2944.5            | 3242.4            | 6.4219              | 0.023019                       | 2913.7            | 3201.5            | 6.2749              |
| 500                           | 0.036793                       | 3056.3            | 3387.4            | 6.6603              | 0.032811                       | 3047.0            | 3375.1            | 6.5995              | 0.025630                       | 3023.2            | 3343.6            | 6.4651              |
| 550                           | 0.039885                       | 3153.0            | 3512.0            | 6.8164              | 0.035655                       | 3145.4            | 3502.0            | 6.7585              | 0.028033                       | 3126.1            | 3476.5            | 6.6317              |
| 600                           | 0.042861                       | 3248.4            | 3634.1            | 6.9605              | 0.038378                       | 3242.0            | 3625.8            | 6.9045              | 0.030306                       | 3225.8            | 3604.6            | 6.7828              |
| 650                           | 0.045755                       | 3343.4            | 3755.2            | 7.0954              | 0.041018                       | 3338.0            | 3748.1            | 7.0408              | 0.032491                       | 3324.1            | 3730.2            | 6.9227              |
| 700                           | 0.048589                       | 3438.8            | 3876.1            | 7.2229              | 0.043597                       | 3434.0            | 3870.0            | 7.1693              | 0.034612                       | 3422.0            | 3854.6            | 7.0540              |
| 800                           | 0.054132                       | 3632.0            | 4119.2            | 7.4606              | 0.048629                       | 3628.2            | 4114.5            | 7.4085              | 0.038724                       | 3618.8            | 4102.8            | 7.2967              |
| 900                           | 0.059562                       | 3829.6            | 4365.7            | 7.6802              | 0.053547                       | 3826.5            | 4362.0            | 7.6290              | 0.042720                       | 3818.9            | 4352.9            | 7.5195              |
| 1000                          | 0.064919                       | 4032.4            | 4616.7            | 7.8855              | 0.058391                       | 4029.9            | 4613.8            | 7.8349              | 0.046641                       | 4023.5            | 4606.5            | 7.7269              |
| 1100                          | 0.070224                       | 4240.7            | 4872.7            | 8.0791              | 0.063183                       | 4238.5            | 4870.3            | 8.0289              | 0.050510                       | 4233.1            | 4864.5            | 7.9220              |
| 1200                          | 0.075492                       | 4454.2            | 5133.6            | 8.2625              | 0.067938                       | 4452.4            | 5131.7            | 8.2126              | 0.054342                       | 4447.7            | 5127.0            | 8.1065              |
| 1300                          | 0.080733                       | 4672.9            | 5399.5            | 8.4371              | 0.072667                       | 4671.3            | 5398.0            | 8.3874              | 0.058147                       | 4667.3            | 5394.1            | 8.2819              |

TABLE A-6

Superheated water (*Concluded*)

| $T$<br>°C                         | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg         | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg         | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|-----------------------------------|---------------------------|--------------|--------------|----------------|-----------------------------------|--------------|--------------|----------------|-----------------------------------|--------------|--------------|----------------|
| $P = 15.0 \text{ MPa (342.16°C)}$ |                           |              |              |                | $P = 17.5 \text{ MPa (354.67°C)}$ |              |              |                | $P = 20.0 \text{ MPa (365.75°C)}$ |              |              |                |
| Sat.                              | 0.010341                  | 2455.7       | 2610.8       | 5.3108         | 0.007932                          | 2390.7       | 2529.5       | 5.1435         | 0.005862                          | 2294.8       | 2412.1       | 4.9310         |
| 350                               | 0.011481                  | 2520.9       | 2693.1       | 5.4438         |                                   |              |              |                |                                   |              |              |                |
| 400                               | 0.015671                  | 2740.6       | 2975.7       | 5.8819         | 0.012463                          | 2684.3       | 2902.4       | 5.7211         | 0.009950                          | 2617.9       | 2816.9       | 5.5526         |
| 450                               | 0.018477                  | 2880.8       | 3157.9       | 6.1434         | 0.015204                          | 2845.4       | 3111.4       | 6.0212         | 0.012721                          | 2807.3       | 3061.7       | 5.9043         |
| 500                               | 0.020828                  | 2998.4       | 3310.8       | 6.3480         | 0.017385                          | 2972.4       | 3276.7       | 6.2424         | 0.014793                          | 2945.3       | 3241.2       | 6.1446         |
| 550                               | 0.022945                  | 3106.2       | 3450.4       | 6.5230         | 0.019305                          | 3085.8       | 3423.6       | 6.4266         | 0.016571                          | 3064.7       | 3396.2       | 6.3390         |
| 600                               | 0.024921                  | 3209.3       | 3583.1       | 6.6796         | 0.021073                          | 3192.5       | 3561.3       | 6.5890         | 0.018185                          | 3175.3       | 3539.0       | 6.5075         |
| 650                               | 0.026804                  | 3310.1       | 3712.1       | 6.8233         | 0.022742                          | 3295.8       | 3693.8       | 6.7366         | 0.019695                          | 3281.4       | 3675.3       | 6.6593         |
| 700                               | 0.028621                  | 3409.8       | 3839.1       | 6.9573         | 0.024342                          | 3397.5       | 3823.5       | 6.8735         | 0.021134                          | 3385.1       | 3807.8       | 6.7991         |
| 800                               | 0.032121                  | 3609.3       | 4091.1       | 7.2037         | 0.027405                          | 3599.7       | 4079.3       | 7.1237         | 0.023870                          | 3590.1       | 4067.5       | 7.0531         |
| 900                               | 0.035503                  | 3811.2       | 4343.7       | 7.4288         | 0.030348                          | 3803.5       | 4334.6       | 7.3511         | 0.026484                          | 3795.7       | 4325.4       | 7.2829         |
| 1000                              | 0.038808                  | 4017.1       | 4599.2       | 7.6378         | 0.033215                          | 4010.7       | 4592.0       | 7.5616         | 0.029020                          | 4004.3       | 4584.7       | 7.4950         |
| 1100                              | 0.042062                  | 4227.7       | 4858.6       | 7.8339         | 0.036029                          | 4222.3       | 4852.8       | 7.7588         | 0.031504                          | 4216.9       | 4847.0       | 7.6933         |
| 1200                              | 0.045279                  | 4443.1       | 5122.3       | 8.0192         | 0.038806                          | 4438.5       | 5117.6       | 7.9449         | 0.033952                          | 4433.8       | 5112.9       | 7.8802         |
| 1300                              | 0.048469                  | 4663.3       | 5390.3       | 8.1952         | 0.041556                          | 4659.2       | 5386.5       | 8.1215         | 0.036371                          | 4655.2       | 5382.7       | 8.0574         |
| $P = 25.0 \text{ MPa}$            |                           |              |              |                | $P = 30.0 \text{ MPa}$            |              |              |                | $P = 35.0 \text{ MPa}$            |              |              |                |
| 375                               | 0.001978                  | 1799.9       | 1849.4       | 4.0345         | 0.001792                          | 1738.1       | 1791.9       | 3.9313         | 0.001701                          | 1702.8       | 1762.4       | 3.8724         |
| 400                               | 0.006005                  | 2428.5       | 2578.7       | 5.1400         | 0.002798                          | 2068.9       | 2152.8       | 4.4758         | 0.002105                          | 1914.9       | 1988.6       | 4.2144         |
| 425                               | 0.007886                  | 2607.8       | 2805.0       | 5.4708         | 0.005299                          | 2452.9       | 2611.8       | 5.1473         | 0.003434                          | 2253.3       | 2373.5       | 4.7751         |
| 450                               | 0.009176                  | 2721.2       | 2950.6       | 5.6759         | 0.006737                          | 2618.9       | 2821.0       | 5.4422         | 0.004957                          | 2497.5       | 2671.0       | 5.1946         |
| 500                               | 0.011143                  | 2887.3       | 3165.9       | 5.9643         | 0.008691                          | 2824.0       | 3084.8       | 5.7956         | 0.006933                          | 2755.3       | 2997.9       | 5.6331         |
| 550                               | 0.012736                  | 3020.8       | 3339.2       | 6.1816         | 0.010175                          | 2974.5       | 3279.7       | 6.0403         | 0.008348                          | 2925.8       | 3218.0       | 5.9093         |
| 600                               | 0.014140                  | 3140.0       | 3493.5       | 6.3637         | 0.011445                          | 3103.4       | 3446.8       | 6.2373         | 0.009523                          | 3065.6       | 3399.0       | 6.1229         |
| 650                               | 0.015430                  | 3251.9       | 3637.7       | 6.5243         | 0.012590                          | 3221.7       | 3599.4       | 6.4074         | 0.010565                          | 3190.9       | 3560.7       | 6.3030         |
| 700                               | 0.016643                  | 3359.9       | 3776.0       | 6.6702         | 0.013654                          | 3334.3       | 3743.9       | 6.5599         | 0.011523                          | 3308.3       | 3711.6       | 6.4623         |
| 800                               | 0.018922                  | 3570.7       | 4043.8       | 6.9322         | 0.015628                          | 3551.2       | 4020.0       | 6.8301         | 0.013278                          | 3531.6       | 3996.3       | 6.7409         |
| 900                               | 0.021075                  | 3780.2       | 4307.1       | 7.1668         | 0.017473                          | 3764.6       | 4288.8       | 7.0695         | 0.014904                          | 3749.0       | 4270.6       | 6.9853         |
| 1000                              | 0.023150                  | 3991.5       | 4570.2       | 7.3821         | 0.019240                          | 3978.6       | 4555.8       | 7.2880         | 0.016450                          | 3965.8       | 4541.5       | 7.2069         |
| 1100                              | 0.025172                  | 4206.1       | 4835.4       | 7.5825         | 0.020954                          | 4195.2       | 4823.9       | 7.4906         | 0.017942                          | 4184.4       | 4812.4       | 7.4118         |
| 1200                              | 0.027157                  | 4424.6       | 5103.5       | 7.7710         | 0.022630                          | 4415.3       | 5094.2       | 7.6807         | 0.019398                          | 4406.1       | 5085.0       | 7.6034         |
| 1300                              | 0.029115                  | 4647.2       | 5375.1       | 7.9494         | 0.024279                          | 4639.2       | 5367.6       | 7.8602         | 0.020827                          | 4631.2       | 5360.2       | 7.7841         |
| $P = 40.0 \text{ MPa}$            |                           |              |              |                | $P = 50.0 \text{ MPa}$            |              |              |                | $P = 60.0 \text{ MPa}$            |              |              |                |
| 375                               | 0.001641                  | 1677.0       | 1742.6       | 3.8290         | 0.001560                          | 1638.6       | 1716.6       | 3.7642         | 0.001503                          | 1609.7       | 1699.9       | 3.7149         |
| 400                               | 0.001911                  | 1855.0       | 1931.4       | 4.1145         | 0.001731                          | 1787.8       | 1874.4       | 4.0029         | 0.001633                          | 1745.2       | 1843.2       | 3.9317         |
| 425                               | 0.002538                  | 2097.5       | 2199.0       | 4.5044         | 0.002009                          | 1960.3       | 2060.7       | 4.2746         | 0.001816                          | 1892.9       | 2001.8       | 4.1630         |
| 450                               | 0.003692                  | 2364.2       | 2511.8       | 4.9449         | 0.002487                          | 2160.3       | 2284.7       | 4.5896         | 0.002086                          | 2055.1       | 2180.2       | 4.4140         |
| 500                               | 0.005623                  | 2681.6       | 2906.5       | 5.4744         | 0.003890                          | 2528.1       | 2722.6       | 5.1762         | 0.002952                          | 2393.2       | 2570.3       | 4.9356         |
| 550                               | 0.006985                  | 2875.1       | 3154.4       | 5.7857         | 0.005118                          | 2769.5       | 3025.4       | 5.5563         | 0.003955                          | 2664.6       | 2901.9       | 5.3517         |
| 600                               | 0.008089                  | 3026.8       | 3350.4       | 6.0170         | 0.006108                          | 2947.1       | 3252.6       | 5.8245         | 0.004833                          | 2866.8       | 3156.8       | 5.6527         |
| 650                               | 0.009053                  | 3159.5       | 3521.6       | 6.2078         | 0.006957                          | 3095.6       | 3443.5       | 6.0373         | 0.005591                          | 3031.3       | 3366.8       | 5.8867         |
| 700                               | 0.009930                  | 3282.0       | 3679.2       | 6.3740         | 0.007717                          | 3228.7       | 3614.6       | 6.2179         | 0.006265                          | 3175.4       | 3551.3       | 6.0814         |
| 800                               | 0.011521                  | 3511.8       | 3972.6       | 6.6613         | 0.009073                          | 3472.2       | 3925.8       | 6.5225         | 0.007456                          | 3432.6       | 3880.0       | 6.4033         |
| 900                               | 0.012980                  | 3733.3       | 4252.5       | 6.9107         | 0.010296                          | 3702.0       | 4216.8       | 6.7819         | 0.008519                          | 3670.9       | 4182.1       | 6.6725         |
| 1000                              | 0.014360                  | 3952.9       | 4527.3       | 7.1355         | 0.011441                          | 3927.4       | 4499.4       | 7.0131         | 0.009504                          | 3902.0       | 4472.2       | 6.9099         |
| 1100                              | 0.015686                  | 4173.7       | 4801.1       | 7.3425         | 0.012534                          | 4152.2       | 4778.9       | 7.2244         | 0.010439                          | 4130.9       | 4757.3       | 7.1255         |
| 1200                              | 0.016976                  | 4396.9       | 5075.9       | 7.5357         | 0.013590                          | 4378.6       | 5058.1       | 7.4207         | 0.011339                          | 4360.5       | 5040.8       | 7.3248         |
| 1300                              | 0.018239                  | 4623.3       | 5352.8       | 7.7175         | 0.014620                          | 4607.5       | 5338.5       | 7.6048         | 0.012213                          | 4591.8       | 5324.5       | 7.5111         |



TABLE A-7

Compressed liquid water

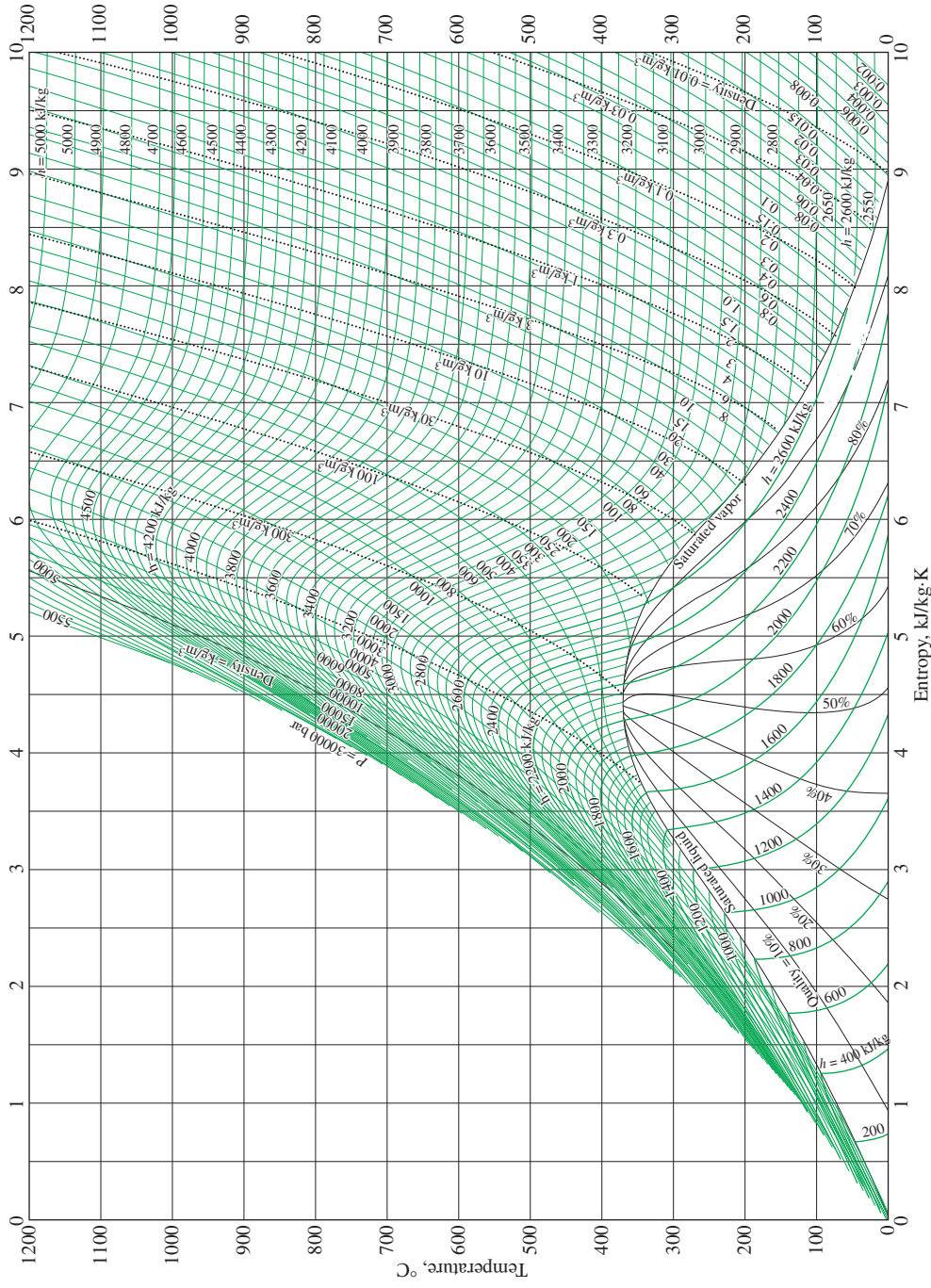
| $T$<br>°C                       | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg       | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg       | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|---------------------------------|---------------------------|--------------|--------------|----------------|---------------------------------|--------------|--------------|----------------|---------------------------------|--------------|--------------|----------------|
| $P = 5 \text{ MPa (263.94°C)}$  |                           |              |              |                | $P = 10 \text{ MPa (311.00°C)}$ |              |              |                | $P = 15 \text{ MPa (342.16°C)}$ |              |              |                |
| Sat.                            | 0.0012862                 | 1148.1       | 1154.5       | 2.9207         | 0.0014522                       | 1393.3       | 1407.9       | 3.3603         | 0.0016572                       | 1585.5       | 1610.3       | 3.6848         |
| 0                               | 0.0009977                 | 0.04         | 5.03         | 0.0001         | 0.0009952                       | 0.12         | 10.07        | 0.0003         | 0.0009928                       | 0.18         | 15.07        | 0.0004         |
| 20                              | 0.0009996                 | 83.61        | 88.61        | 0.2954         | 0.0009973                       | 83.31        | 93.28        | 0.2943         | 0.0009951                       | 83.01        | 97.93        | 0.2932         |
| 40                              | 0.0010057                 | 166.92       | 171.95       | 0.5705         | 0.0010035                       | 166.33       | 176.37       | 0.5685         | 0.0010013                       | 165.75       | 180.77       | 0.5666         |
| 60                              | 0.0010149                 | 250.29       | 255.36       | 0.8287         | 0.0010127                       | 249.43       | 259.55       | 0.8260         | 0.0010105                       | 248.58       | 263.74       | 0.8234         |
| 80                              | 0.0010267                 | 333.82       | 338.96       | 1.0723         | 0.0010244                       | 332.69       | 342.94       | 1.0691         | 0.0010221                       | 331.59       | 346.92       | 1.0659         |
| 100                             | 0.0010410                 | 417.65       | 422.85       | 1.3034         | 0.0010385                       | 416.23       | 426.62       | 1.2996         | 0.0010361                       | 414.85       | 430.39       | 1.2958         |
| 120                             | 0.0010576                 | 501.91       | 507.19       | 1.5236         | 0.0010549                       | 500.18       | 510.73       | 1.5191         | 0.0010522                       | 498.50       | 514.28       | 1.5148         |
| 140                             | 0.0010769                 | 586.80       | 592.18       | 1.7344         | 0.0010738                       | 584.72       | 595.45       | 1.7293         | 0.0010708                       | 582.69       | 598.75       | 1.7243         |
| 160                             | 0.0010988                 | 672.55       | 678.04       | 1.9374         | 0.0010954                       | 670.06       | 681.01       | 1.9316         | 0.0010920                       | 667.63       | 684.01       | 1.9259         |
| 180                             | 0.0011240                 | 759.47       | 765.09       | 2.1338         | 0.0011200                       | 756.48       | 767.68       | 2.1271         | 0.0011160                       | 753.58       | 770.32       | 2.1206         |
| 200                             | 0.0011531                 | 847.92       | 853.68       | 2.3251         | 0.0011482                       | 844.32       | 855.80       | 2.3174         | 0.0011435                       | 840.84       | 858.00       | 2.3100         |
| 220                             | 0.0011868                 | 938.39       | 944.32       | 2.5127         | 0.0011809                       | 934.01       | 945.82       | 2.5037         | 0.0011752                       | 929.81       | 947.43       | 2.4951         |
| 240                             | 0.0012268                 | 1031.6       | 1037.7       | 2.6983         | 0.0012192                       | 1026.2       | 1038.3       | 2.6876         | 0.0012121                       | 1021.0       | 1039.2       | 2.6774         |
| 260                             | 0.0012755                 | 1128.5       | 1134.9       | 2.8841         | 0.0012653                       | 1121.6       | 1134.3       | 2.8710         | 0.0012560                       | 1115.1       | 1134.0       | 2.8586         |
| 280                             |                           |              |              |                | 0.0013226                       | 1221.8       | 1235.0       | 3.0565         | 0.0013096                       | 1213.4       | 1233.0       | 3.0410         |
| 300                             |                           |              |              |                | 0.0013980                       | 1329.4       | 1343.3       | 3.2488         | 0.0013783                       | 1317.6       | 1338.3       | 3.2279         |
| 320                             |                           |              |              |                |                                 |              |              |                | 0.0014733                       | 1431.9       | 1454.0       | 3.4263         |
| 340                             |                           |              |              |                |                                 |              |              |                | 0.0016311                       | 1567.9       | 1592.4       | 3.6555         |
| $P = 20 \text{ MPa (365.75°C)}$ |                           |              |              |                | $P = 30 \text{ MPa}$            |              |              |                | $P = 50 \text{ MPa}$            |              |              |                |
| Sat.                            | 0.0020378                 | 1785.8       | 1826.6       | 4.0146         | 0.0009857                       | 0.29         | 29.86        | 0.0003         | 0.0009767                       | 0.29         | 49.13        | -0.0010        |
| 0                               | 0.0009904                 | 0.23         | 20.03        | 0.0005         | 0.0009886                       | 82.11        | 111.77       | 0.2897         | 0.0009805                       | 80.93        | 129.95       | 0.2845         |
| 20                              | 0.0009929                 | 82.71        | 102.57       | 0.2921         | 0.0009886                       | 82.11        | 111.77       | 0.2897         | 0.0009805                       | 80.93        | 129.95       | 0.2845         |
| 40                              | 0.0009992                 | 165.17       | 185.16       | 0.5646         | 0.0009951                       | 164.05       | 193.90       | 0.5607         | 0.0009872                       | 161.90       | 211.25       | 0.5528         |
| 60                              | 0.0010084                 | 247.75       | 267.92       | 0.8208         | 0.0010042                       | 246.14       | 276.26       | 0.8156         | 0.0009962                       | 243.08       | 292.88       | 0.8055         |
| 80                              | 0.0010199                 | 330.50       | 350.90       | 1.0627         | 0.0010155                       | 328.40       | 358.86       | 1.0564         | 0.0010072                       | 324.42       | 374.78       | 1.0442         |
| 100                             | 0.0010337                 | 413.50       | 434.17       | 1.2920         | 0.0010290                       | 410.87       | 441.74       | 1.2847         | 0.0010201                       | 405.94       | 456.94       | 1.2705         |
| 120                             | 0.0010496                 | 496.85       | 517.84       | 1.5105         | 0.0010445                       | 493.66       | 525.00       | 1.5020         | 0.0010349                       | 487.69       | 539.43       | 1.4859         |
| 140                             | 0.0010679                 | 580.71       | 602.07       | 1.7194         | 0.0010623                       | 576.90       | 608.76       | 1.7098         | 0.0010517                       | 569.77       | 622.36       | 1.6916         |
| 160                             | 0.0010886                 | 665.28       | 687.05       | 1.9203         | 0.0010823                       | 660.74       | 693.21       | 1.9094         | 0.0010704                       | 652.33       | 705.85       | 1.8889         |
| 180                             | 0.0011122                 | 750.78       | 773.02       | 2.1143         | 0.0011049                       | 745.40       | 778.55       | 2.1020         | 0.0010914                       | 735.49       | 790.06       | 2.0790         |
| 200                             | 0.0011390                 | 837.49       | 860.27       | 2.3027         | 0.0011304                       | 831.11       | 865.02       | 2.2888         | 0.0011149                       | 819.45       | 875.19       | 2.2628         |
| 220                             | 0.0011697                 | 925.77       | 949.16       | 2.4867         | 0.0011595                       | 918.15       | 952.93       | 2.4707         | 0.0011412                       | 904.39       | 961.45       | 2.4414         |
| 240                             | 0.0012053                 | 1016.1       | 1040.2       | 2.6676         | 0.0011927                       | 1006.9       | 1042.7       | 2.6491         | 0.0011708                       | 990.55       | 1049.1       | 2.6156         |
| 260                             | 0.0012472                 | 1109.0       | 1134.0       | 2.8469         | 0.0012314                       | 1097.8       | 1134.7       | 2.8250         | 0.0012044                       | 1078.2       | 1138.4       | 2.7864         |
| 280                             | 0.0012978                 | 1205.6       | 1231.5       | 3.0265         | 0.0012770                       | 1191.5       | 1229.8       | 3.0001         | 0.0012430                       | 1167.7       | 1229.9       | 2.9547         |
| 300                             | 0.0013611                 | 1307.2       | 1334.4       | 3.2091         | 0.0013322                       | 1288.9       | 1328.9       | 3.1761         | 0.0012879                       | 1259.6       | 1324.0       | 3.1218         |
| 320                             | 0.0014450                 | 1416.6       | 1445.5       | 3.3996         | 0.0014014                       | 1391.7       | 1433.7       | 3.3558         | 0.0013409                       | 1354.3       | 1421.4       | 3.2888         |
| 340                             | 0.0015693                 | 1540.2       | 1571.6       | 3.6086         | 0.0014932                       | 1502.4       | 1547.1       | 3.5438         | 0.0014049                       | 1452.9       | 1523.1       | 3.4575         |
| 360                             | 0.0018248                 | 1703.6       | 1740.1       | 3.8787         | 0.0016276                       | 1626.8       | 1675.6       | 3.7499         | 0.0014848                       | 1556.5       | 1630.7       | 3.6301         |
| 380                             |                           |              |              |                | 0.0018729                       | 1782.0       | 1838.2       | 4.0026         | 0.0015884                       | 1667.1       | 1746.5       | 3.8102         |



TABLE A-8

Saturated ice–water vapor

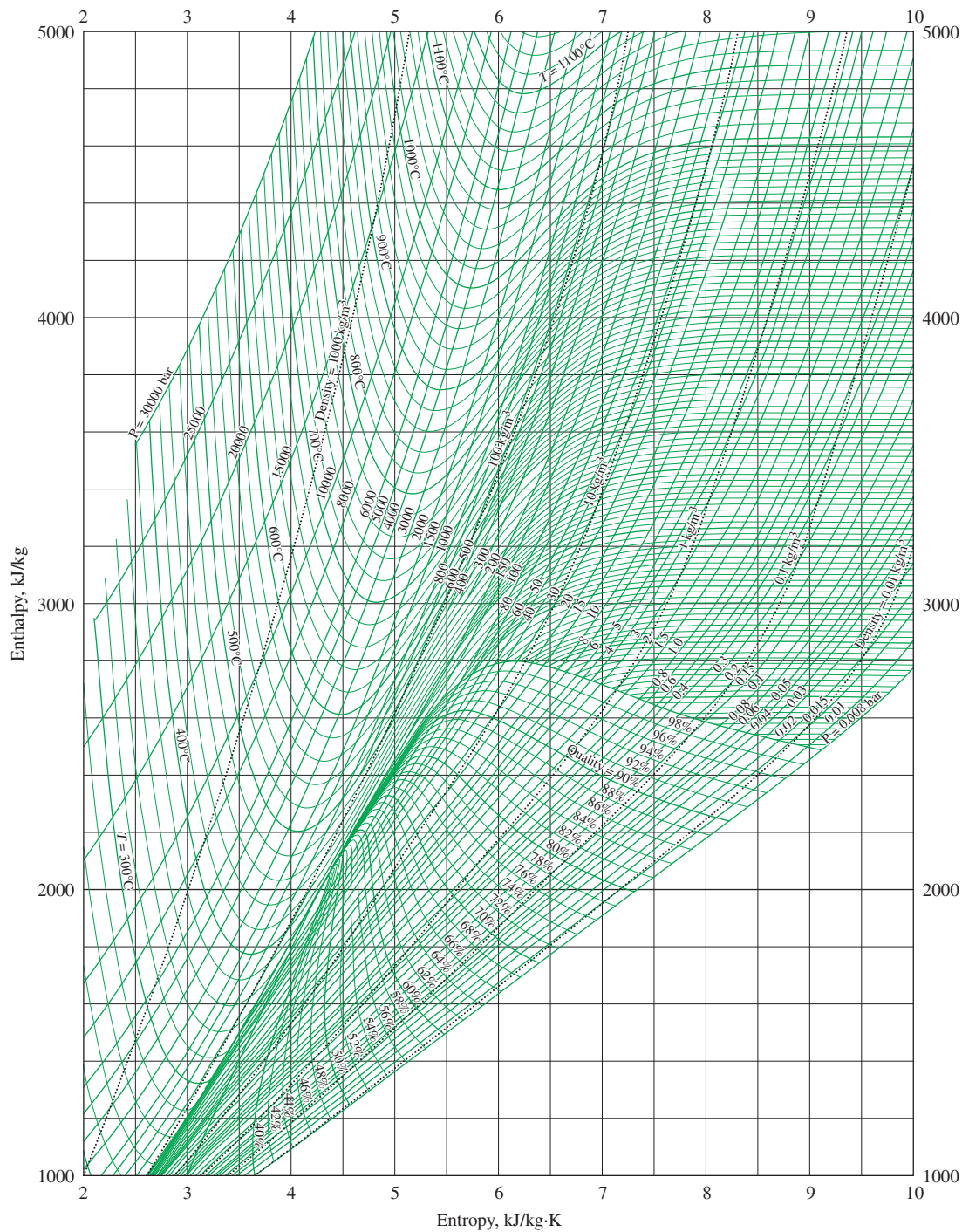
| Temp.,<br>$T$ °C | Sat.<br>press.,<br>$P_{\text{sat}}$ kPa | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>$\text{kJ/kg}$ |                    |                         | Enthalpy,<br>$\text{kJ/kg}$ |                    |                         | Entropy,<br>$\text{kJ/kg}\cdot\text{K}$ |                    |                         |
|------------------|-----------------------------------------|--------------------------------------------|-------------------------|------------------------------------|--------------------|-------------------------|-----------------------------|--------------------|-------------------------|-----------------------------------------|--------------------|-------------------------|
|                  |                                         | Sat.<br>ice,<br>$v_i$                      | Sat.<br>vapor,<br>$v_g$ | Sat.<br>ice,<br>$u_i$              | Subl.,<br>$u_{ig}$ | Sat.<br>vapor,<br>$u_g$ | Sat.<br>ice,<br>$h_i$       | Subl.,<br>$h_{ig}$ | Sat.<br>vapor,<br>$h_g$ | Sat.<br>ice,<br>$s_i$                   | Subl.,<br>$s_{ig}$ | Sat.<br>vapor,<br>$s_g$ |
| 0.01             | 0.61169                                 | 0.001091                                   | 205.99                  | −333.40                            | 2707.9             | 2374.5                  | −333.40                     | 2833.9             | 2500.5                  | −1.2202                                 | 10.374             | 9.154                   |
| 0                | 0.61115                                 | 0.001091                                   | 206.17                  | −333.43                            | 2707.9             | 2374.5                  | −333.43                     | 2833.9             | 2500.5                  | −1.2204                                 | 10.375             | 9.154                   |
| −2               | 0.51772                                 | 0.001091                                   | 241.62                  | −337.63                            | 2709.4             | 2371.8                  | −337.63                     | 2834.5             | 2496.8                  | −1.2358                                 | 10.453             | 9.218                   |
| −4               | 0.43748                                 | 0.001090                                   | 283.84                  | −341.80                            | 2710.8             | 2369.0                  | −341.80                     | 2835.0             | 2493.2                  | −1.2513                                 | 10.533             | 9.282                   |
| −6               | 0.36873                                 | 0.001090                                   | 334.27                  | −345.94                            | 2712.2             | 2366.2                  | −345.93                     | 2835.4             | 2489.5                  | −1.2667                                 | 10.613             | 9.347                   |
| −8               | 0.30998                                 | 0.001090                                   | 394.66                  | −350.04                            | 2713.5             | 2363.5                  | −350.04                     | 2835.8             | 2485.8                  | −1.2821                                 | 10.695             | 9.413                   |
| −10              | 0.25990                                 | 0.001089                                   | 467.17                  | −354.12                            | 2714.8             | 2360.7                  | −354.12                     | 2836.2             | 2482.1                  | −1.2976                                 | 10.778             | 9.480                   |
| −12              | 0.21732                                 | 0.001089                                   | 554.47                  | −358.17                            | 2716.1             | 2357.9                  | −358.17                     | 2836.6             | 2478.4                  | −1.3130                                 | 10.862             | 9.549                   |
| −14              | 0.18121                                 | 0.001088                                   | 659.88                  | −362.18                            | 2717.3             | 2355.2                  | −362.18                     | 2836.9             | 2474.7                  | −1.3284                                 | 10.947             | 9.618                   |
| −16              | 0.15068                                 | 0.001088                                   | 787.51                  | −366.17                            | 2718.6             | 2352.4                  | −366.17                     | 2837.2             | 2471.0                  | −1.3439                                 | 11.033             | 9.689                   |
| −18              | 0.12492                                 | 0.001088                                   | 942.51                  | −370.13                            | 2719.7             | 2349.6                  | −370.13                     | 2837.5             | 2467.3                  | −1.3593                                 | 11.121             | 9.761                   |
| −20              | 0.10326                                 | 0.001087                                   | 1131.3                  | −374.06                            | 2720.9             | 2346.8                  | −374.06                     | 2837.7             | 2463.6                  | −1.3748                                 | 11.209             | 9.835                   |
| −22              | 0.08510                                 | 0.001087                                   | 1362.0                  | −377.95                            | 2722.0             | 2344.1                  | −377.95                     | 2837.9             | 2459.9                  | −1.3903                                 | 11.300             | 9.909                   |
| −24              | 0.06991                                 | 0.001087                                   | 1644.7                  | −381.82                            | 2723.1             | 2341.3                  | −381.82                     | 2838.1             | 2456.2                  | −1.4057                                 | 11.391             | 9.985                   |
| −26              | 0.05725                                 | 0.001087                                   | 1992.2                  | −385.66                            | 2724.2             | 2338.5                  | −385.66                     | 2838.2             | 2452.5                  | −1.4212                                 | 11.484             | 10.063                  |
| −28              | 0.04673                                 | 0.001086                                   | 2421.0                  | −389.47                            | 2725.2             | 2335.7                  | −389.47                     | 2838.3             | 2448.8                  | −1.4367                                 | 11.578             | 10.141                  |
| −30              | 0.03802                                 | 0.001086                                   | 2951.7                  | −393.25                            | 2726.2             | 2332.9                  | −393.25                     | 2838.4             | 2445.1                  | −1.4521                                 | 11.673             | 10.221                  |
| −32              | 0.03082                                 | 0.001086                                   | 3610.9                  | −397.00                            | 2727.2             | 2330.2                  | −397.00                     | 2838.4             | 2441.4                  | −1.4676                                 | 11.770             | 10.303                  |
| −34              | 0.02490                                 | 0.001085                                   | 4432.4                  | −400.72                            | 2728.1             | 2327.4                  | −400.72                     | 2838.5             | 2437.7                  | −1.4831                                 | 11.869             | 10.386                  |
| −36              | 0.02004                                 | 0.001085                                   | 5460.1                  | −404.40                            | 2729.0             | 2324.6                  | −404.40                     | 2838.4             | 2434.0                  | −1.4986                                 | 11.969             | 10.470                  |
| −38              | 0.01608                                 | 0.001085                                   | 6750.5                  | −408.07                            | 2729.9             | 2321.8                  | −408.07                     | 2838.4             | 2430.3                  | −1.5141                                 | 12.071             | 10.557                  |
| −40              | 0.01285                                 | 0.001084                                   | 8376.7                  | −411.70                            | 2730.7             | 2319.0                  | −411.70                     | 2838.3             | 2426.6                  | −1.5296                                 | 12.174             | 10.644                  |



**FIGURE A-9**

T-s diagram for water.

Source of Data: From NBS/NRC Steam Tables/1 by Lester Haar, John S. Gallagher, and George S. Kell. Routledge/Taylor & Francis Books, Inc., 1984.



**FIGURE A-10**  
Mollier diagram for water.

Source of Data: From NBS/NRC Steam Tables/1 by Lester Haar, John S. Gallagher, and George S. Kell. Routledge/Taylor & Francis Books, Inc., 1984.

TABLE A-11

Saturated refrigerant-134a—Temperature table

| Temp.,<br>$T$ °C | Sat.<br>press.,<br>$P_{\text{sat}}$ kPa | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>kJ/kg |                    |                         | Enthalpy,<br>kJ/kg       |                    |                         | Entropy,<br>kJ/kg·K      |                    |                         |
|------------------|-----------------------------------------|--------------------------------------------|-------------------------|---------------------------|--------------------|-------------------------|--------------------------|--------------------|-------------------------|--------------------------|--------------------|-------------------------|
|                  |                                         | Sat.<br>liquid,<br>$v_f$                   | Sat.<br>vapor,<br>$v_g$ | Sat.<br>liquid,<br>$u_f$  | Evap.,<br>$u_{fg}$ | Sat.<br>vapor,<br>$u_g$ | Sat.<br>liquid,<br>$h_f$ | Evap.,<br>$h_{fg}$ | Sat.<br>vapor,<br>$h_g$ | Sat.<br>liquid,<br>$s_f$ | Evap.,<br>$s_{fg}$ | Sat.<br>vapor,<br>$s_g$ |
| −40              | 51.25                                   | 0.0007053                                  | 0.36064                 | −0.036                    | 207.42             | 207.38                  | 0.00                     | 225.86             | 225.86                  | 0.00000                  | 0.96869            | 0.96869                 |
| −38              | 56.86                                   | 0.0007082                                  | 0.32718                 | 2.472                     | 206.06             | 208.53                  | 2.512                    | 224.62             | 227.13                  | 0.01071                  | 0.95516            | 0.96588                 |
| −36              | 62.95                                   | 0.0007111                                  | 0.29740                 | 4.987                     | 204.69             | 209.68                  | 5.032                    | 223.37             | 228.40                  | 0.02137                  | 0.94182            | 0.96319                 |
| −34              | 69.56                                   | 0.0007141                                  | 0.27082                 | 7.509                     | 203.32             | 210.83                  | 7.559                    | 222.10             | 229.66                  | 0.03196                  | 0.92867            | 0.96063                 |
| −32              | 76.71                                   | 0.0007171                                  | 0.24706                 | 10.04                     | 201.94             | 211.97                  | 10.09                    | 220.83             | 230.93                  | 0.04249                  | 0.91569            | 0.95819                 |
| −30              | 84.43                                   | 0.0007201                                  | 0.22577                 | 12.58                     | 200.55             | 213.12                  | 12.64                    | 219.55             | 232.19                  | 0.05297                  | 0.90289            | 0.95586                 |
| −28              | 92.76                                   | 0.0007232                                  | 0.20666                 | 15.12                     | 199.15             | 214.27                  | 15.19                    | 218.25             | 233.44                  | 0.06339                  | 0.89024            | 0.95364                 |
| −26              | 101.73                                  | 0.0007264                                  | 0.18947                 | 17.67                     | 197.75             | 215.42                  | 17.75                    | 216.95             | 234.70                  | 0.07376                  | 0.87776            | 0.95152                 |
| −24              | 111.37                                  | 0.0007296                                  | 0.17398                 | 20.23                     | 196.34             | 216.57                  | 20.31                    | 215.63             | 235.94                  | 0.08408                  | 0.86542            | 0.94950                 |
| −22              | 121.72                                  | 0.0007328                                  | 0.15999                 | 22.80                     | 194.92             | 217.71                  | 22.89                    | 214.30             | 237.19                  | 0.09435                  | 0.85323            | 0.94758                 |
| −20              | 132.82                                  | 0.0007361                                  | 0.14735                 | 25.37                     | 193.49             | 218.86                  | 25.47                    | 212.96             | 238.43                  | 0.10456                  | 0.84119            | 0.94575                 |
| −18              | 144.69                                  | 0.0007394                                  | 0.13589                 | 27.96                     | 192.05             | 220.00                  | 28.07                    | 211.60             | 239.67                  | 0.11473                  | 0.82927            | 0.94401                 |
| −16              | 157.38                                  | 0.0007428                                  | 0.12550                 | 30.55                     | 190.60             | 221.15                  | 30.67                    | 210.23             | 240.90                  | 0.12486                  | 0.81749            | 0.94234                 |
| −14              | 170.93                                  | 0.0007463                                  | 0.11605                 | 33.15                     | 189.14             | 222.29                  | 33.28                    | 208.84             | 242.12                  | 0.13493                  | 0.80583            | 0.94076                 |
| −12              | 185.37                                  | 0.0007498                                  | 0.10744                 | 35.76                     | 187.66             | 223.42                  | 35.90                    | 207.44             | 243.34                  | 0.14497                  | 0.79429            | 0.93925                 |
| −10              | 200.74                                  | 0.0007533                                  | 0.099600                | 38.38                     | 186.18             | 224.56                  | 38.53                    | 206.02             | 244.55                  | 0.15496                  | 0.78286            | 0.93782                 |
| −8               | 217.08                                  | 0.0007570                                  | 0.092438                | 41.01                     | 184.69             | 225.69                  | 41.17                    | 204.59             | 245.76                  | 0.16491                  | 0.77154            | 0.93645                 |
| −6               | 234.44                                  | 0.0007607                                  | 0.085888                | 43.64                     | 183.18             | 226.82                  | 43.82                    | 203.14             | 246.95                  | 0.17482                  | 0.76033            | 0.93514                 |
| −4               | 252.85                                  | 0.0007644                                  | 0.079889                | 46.29                     | 181.66             | 227.94                  | 46.48                    | 201.66             | 248.14                  | 0.18469                  | 0.74921            | 0.93390                 |
| −2               | 272.36                                  | 0.0007683                                  | 0.074388                | 48.94                     | 180.12             | 229.07                  | 49.15                    | 200.17             | 249.33                  | 0.19452                  | 0.73819            | 0.93271                 |
| 0                | 293.01                                  | 0.0007722                                  | 0.069335                | 51.61                     | 178.58             | 230.18                  | 51.83                    | 198.67             | 250.50                  | 0.20432                  | 0.72726            | 0.93158                 |
| 2                | 314.84                                  | 0.0007761                                  | 0.064690                | 54.28                     | 177.01             | 231.30                  | 54.53                    | 197.14             | 251.66                  | 0.21408                  | 0.71641            | 0.93050                 |
| 4                | 337.90                                  | 0.0007802                                  | 0.060412                | 56.97                     | 175.44             | 232.40                  | 57.23                    | 195.58             | 252.82                  | 0.22381                  | 0.70565            | 0.92946                 |
| 6                | 362.23                                  | 0.0007843                                  | 0.056469                | 59.66                     | 173.84             | 233.51                  | 59.95                    | 194.01             | 253.96                  | 0.23351                  | 0.69496            | 0.92847                 |
| 8                | 387.88                                  | 0.0007886                                  | 0.052829                | 62.37                     | 172.23             | 234.60                  | 62.68                    | 192.42             | 255.09                  | 0.24318                  | 0.68435            | 0.92752                 |
| 10               | 414.89                                  | 0.0007929                                  | 0.049466                | 65.09                     | 170.61             | 235.69                  | 65.42                    | 190.80             | 256.22                  | 0.25282                  | 0.67380            | 0.92661                 |
| 12               | 443.31                                  | 0.0007973                                  | 0.046354                | 67.82                     | 168.96             | 236.78                  | 68.17                    | 189.16             | 257.33                  | 0.26243                  | 0.66331            | 0.92574                 |
| 14               | 473.19                                  | 0.0008018                                  | 0.043471                | 70.56                     | 167.30             | 237.86                  | 70.94                    | 187.49             | 258.43                  | 0.27201                  | 0.65289            | 0.92490                 |
| 16               | 504.58                                  | 0.0008064                                  | 0.040798                | 73.31                     | 165.62             | 238.93                  | 73.72                    | 185.80             | 259.51                  | 0.28157                  | 0.64252            | 0.92409                 |
| 18               | 537.52                                  | 0.0008112                                  | 0.038317                | 76.07                     | 163.92             | 239.99                  | 76.51                    | 184.08             | 260.59                  | 0.29111                  | 0.63219            | 0.92330                 |
| 20               | 572.07                                  | 0.0008160                                  | 0.036012                | 78.85                     | 162.19             | 241.04                  | 79.32                    | 182.33             | 261.64                  | 0.30062                  | 0.62192            | 0.92254                 |
| 22               | 608.27                                  | 0.0008209                                  | 0.033867                | 81.64                     | 160.45             | 242.09                  | 82.14                    | 180.55             | 262.69                  | 0.31012                  | 0.61168            | 0.92180                 |
| 24               | 646.18                                  | 0.0008260                                  | 0.031869                | 84.44                     | 158.68             | 243.13                  | 84.98                    | 178.74             | 263.72                  | 0.31959                  | 0.60148            | 0.92107                 |
| 26               | 685.84                                  | 0.0008312                                  | 0.030008                | 87.26                     | 156.89             | 244.15                  | 87.83                    | 176.90             | 264.73                  | 0.32905                  | 0.59131            | 0.92036                 |
| 28               | 727.31                                  | 0.0008366                                  | 0.028271                | 90.09                     | 155.08             | 245.17                  | 90.70                    | 175.03             | 265.73                  | 0.33849                  | 0.58117            | 0.91967                 |
| 30               | 770.64                                  | 0.0008421                                  | 0.026648                | 92.93                     | 153.24             | 246.17                  | 93.58                    | 173.13             | 266.71                  | 0.34792                  | 0.57105            | 0.91897                 |
| 32               | 815.89                                  | 0.0008477                                  | 0.025131                | 95.79                     | 151.37             | 247.17                  | 96.49                    | 171.19             | 267.67                  | 0.35734                  | 0.56095            | 0.91829                 |
| 34               | 863.11                                  | 0.0008535                                  | 0.023712                | 98.67                     | 149.48             | 248.15                  | 99.41                    | 169.21             | 268.61                  | 0.36675                  | 0.55086            | 0.91760                 |
| 36               | 912.35                                  | 0.0008595                                  | 0.022383                | 101.56                    | 147.55             | 249.11                  | 102.34                   | 167.19             | 269.53                  | 0.37615                  | 0.54077            | 0.91692                 |
| 38               | 963.68                                  | 0.0008657                                  | 0.021137                | 104.47                    | 145.60             | 250.07                  | 105.30                   | 165.13             | 270.44                  | 0.38554                  | 0.53068            | 0.91622                 |
| 40               | 1017.1                                  | 0.0008720                                  | 0.019968                | 107.39                    | 143.61             | 251.00                  | 108.28                   | 163.03             | 271.31                  | 0.39493                  | 0.52059            | 0.91552                 |
| 42               | 1072.8                                  | 0.0008786                                  | 0.018870                | 110.34                    | 141.59             | 251.92                  | 111.28                   | 160.89             | 272.17                  | 0.40432                  | 0.51048            | 0.91480                 |
| 44               | 1130.7                                  | 0.0008854                                  | 0.017837                | 113.30                    | 139.53             | 252.83                  | 114.30                   | 158.70             | 273.00                  | 0.41371                  | 0.50036            | 0.91407                 |

TABLE A-11

Saturated refrigerant-134a—Temperature table (*Concluded*)

| Temp.,<br>$T$ °C | Sat.<br>press.,<br>$P_{\text{sat}}$ kPa | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>kJ/kg |                    |                         | Enthalpy,<br>kJ/kg       |                    |                         | Entropy,<br>kJ/kg·K      |                    |                         |
|------------------|-----------------------------------------|--------------------------------------------|-------------------------|---------------------------|--------------------|-------------------------|--------------------------|--------------------|-------------------------|--------------------------|--------------------|-------------------------|
|                  |                                         | Sat.<br>liquid,<br>$v_f$                   | Sat.<br>vapor,<br>$v_g$ | Sat.<br>liquid,<br>$u_f$  | Evap.,<br>$u_{fg}$ | Sat.<br>vapor,<br>$u_g$ | Sat.<br>liquid,<br>$h_f$ | Evap.,<br>$h_{fg}$ | Sat.<br>vapor,<br>$h_g$ | Sat.<br>liquid,<br>$s_f$ | Evap.,<br>$s_{fg}$ | Sat.<br>vapor,<br>$s_g$ |
| 46               | 1191.0                                  | 0.0008924                                  | 0.016866                | 116.28                    | 137.43             | 253.71                  | 117.34                   | 156.46             | 273.80                  | 0.42311                  | 0.49020            | 0.91331                 |
| 48               | 1253.6                                  | 0.0008997                                  | 0.015951                | 119.28                    | 135.30             | 254.58                  | 120.41                   | 154.17             | 274.57                  | 0.43251                  | 0.48001            | 0.91252                 |
| 52               | 1386.2                                  | 0.0009151                                  | 0.014276                | 125.35                    | 130.89             | 256.24                  | 126.62                   | 149.41             | 276.03                  | 0.45136                  | 0.45948            | 0.91084                 |
| 56               | 1529.1                                  | 0.0009317                                  | 0.012782                | 131.52                    | 126.29             | 257.81                  | 132.94                   | 144.41             | 277.35                  | 0.47028                  | 0.43870            | 0.90898                 |
| 60               | 1682.8                                  | 0.0009498                                  | 0.011434                | 137.79                    | 121.45             | 259.23                  | 139.38                   | 139.09             | 278.47                  | 0.48930                  | 0.41746            | 0.90676                 |
| 65               | 1891.0                                  | 0.0009751                                  | 0.009959                | 145.80                    | 115.06             | 260.86                  | 147.64                   | 132.05             | 279.69                  | 0.51330                  | 0.39048            | 0.90379                 |
| 70               | 2118.2                                  | 0.0010037                                  | 0.008650                | 154.03                    | 108.17             | 262.20                  | 156.15                   | 124.37             | 280.52                  | 0.53763                  | 0.36239            | 0.90002                 |
| 75               | 2365.8                                  | 0.0010373                                  | 0.007486                | 162.55                    | 100.62             | 263.17                  | 165.01                   | 115.87             | 280.88                  | 0.56252                  | 0.33279            | 0.89531                 |
| 80               | 2635.3                                  | 0.0010774                                  | 0.006439                | 171.43                    | 92.22              | 263.66                  | 174.27                   | 106.35             | 280.63                  | 0.58812                  | 0.30113            | 0.88925                 |
| 85               | 2928.2                                  | 0.0011273                                  | 0.005484                | 180.81                    | 82.64              | 263.45                  | 184.11                   | 95.39              | 279.51                  | 0.61487                  | 0.26632            | 0.88120                 |
| 90               | 3246.9                                  | 0.0011938                                  | 0.004591                | 190.94                    | 71.19              | 262.13                  | 194.82                   | 82.22              | 277.04                  | 0.64354                  | 0.22638            | 0.86991                 |
| 95               | 3594.1                                  | 0.0012945                                  | 0.003713                | 202.49                    | 56.25              | 258.73                  | 207.14                   | 64.94              | 272.08                  | 0.67605                  | 0.17638            | 0.85243                 |
| 100              | 3975.1                                  | 0.0015269                                  | 0.002657                | 218.73                    | 29.72              | 248.46                  | 224.80                   | 34.22              | 259.02                  | 0.72224                  | 0.09169            | 0.81393                 |

Source of Data: Tables A-11 through A-13 are generated using the Engineering Equation Solver (EES) software developed by S. A. Klein and F. L. Alvarado. The routine used in calculations is the R134a, which is based on the fundamental equation of state developed by R. Tillner-Roth and H.D. Baehr, "An International Standard Formulation for the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for temperatures from 170 K to 455 K and pressures up to 70 MPa," *J. Phys. Chem., Ref. Data*, Vol. 23, No. 5, 1994. The enthalpy and entropy values of saturated liquid are set to zero at  $-40^\circ\text{C}$  (and  $-40^\circ\text{F}$ ).

TABLE A-12

Saturated refrigerant-134a—Pressure table

| Press.,<br>$P$<br>kPa | Sat.<br>temp.,<br>$T_{\text{sat}}$ °C | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>kJ/kg |                    |                         | Enthalpy,<br>kJ/kg       |                    |                         | Entropy,<br>kJ/kg·K      |                    |                         |
|-----------------------|---------------------------------------|--------------------------------------------|-------------------------|---------------------------|--------------------|-------------------------|--------------------------|--------------------|-------------------------|--------------------------|--------------------|-------------------------|
|                       |                                       | Sat.<br>liquid,<br>$v_f$                   | Sat.<br>vapor,<br>$v_g$ | Sat.<br>liquid,<br>$u_f$  | Evap.,<br>$u_{fg}$ | Sat.<br>vapor,<br>$u_g$ | Sat.<br>liquid,<br>$h_f$ | Evap.,<br>$h_{fg}$ | Sat.<br>vapor,<br>$h_g$ | Sat.<br>liquid,<br>$s_f$ | Evap.,<br>$s_{fg}$ | Sat.<br>vapor,<br>$s_g$ |
| 60                    | −36.95                                | 0.0007097                                  | 0.31108                 | 3.795                     | 205.34             | 209.13                  | 3.837                    | 223.96             | 227.80                  | 0.01633                  | 0.94812            | 0.96445                 |
| 70                    | −33.87                                | 0.0007143                                  | 0.26921                 | 7.672                     | 203.23             | 210.90                  | 7.722                    | 222.02             | 229.74                  | 0.03264                  | 0.92783            | 0.96047                 |
| 80                    | −31.13                                | 0.0007184                                  | 0.23749                 | 11.14                     | 201.33             | 212.48                  | 11.20                    | 220.27             | 231.47                  | 0.04707                  | 0.91009            | 0.95716                 |
| 90                    | −28.65                                | 0.0007222                                  | 0.21261                 | 14.30                     | 199.60             | 213.90                  | 14.36                    | 218.67             | 233.04                  | 0.06003                  | 0.89431            | 0.95434                 |
| 100                   | −26.37                                | 0.0007258                                  | 0.19255                 | 17.19                     | 198.01             | 215.21                  | 17.27                    | 217.19             | 234.46                  | 0.07182                  | 0.88008            | 0.95191                 |
| 120                   | −22.32                                | 0.0007323                                  | 0.16216                 | 22.38                     | 195.15             | 217.53                  | 22.47                    | 214.52             | 236.99                  | 0.09269                  | 0.85520            | 0.94789                 |
| 140                   | −18.77                                | 0.0007381                                  | 0.14020                 | 26.96                     | 192.60             | 219.56                  | 27.06                    | 212.13             | 239.19                  | 0.11080                  | 0.83387            | 0.94467                 |
| 160                   | −15.60                                | 0.0007435                                  | 0.12355                 | 31.06                     | 190.31             | 221.37                  | 31.18                    | 209.96             | 241.14                  | 0.12686                  | 0.81517            | 0.94202                 |
| 180                   | −12.73                                | 0.0007485                                  | 0.11049                 | 34.81                     | 188.20             | 223.01                  | 34.94                    | 207.95             | 242.90                  | 0.14131                  | 0.79848            | 0.93979                 |
| 200                   | −10.09                                | 0.0007532                                  | 0.099951                | 38.26                     | 186.25             | 224.51                  | 38.41                    | 206.09             | 244.50                  | 0.15449                  | 0.78339            | 0.93788                 |
| 240                   | −5.38                                 | 0.0007618                                  | 0.083983                | 44.46                     | 182.71             | 227.17                  | 44.64                    | 202.68             | 247.32                  | 0.17786                  | 0.75689            | 0.93475                 |
| 280                   | −1.25                                 | 0.0007697                                  | 0.072434                | 49.95                     | 179.54             | 229.49                  | 50.16                    | 199.61             | 249.77                  | 0.19822                  | 0.73406            | 0.93228                 |
| 320                   | 2.46                                  | 0.0007771                                  | 0.063681                | 54.90                     | 176.65             | 231.55                  | 55.14                    | 196.78             | 251.93                  | 0.21631                  | 0.71395            | 0.93026                 |
| 360                   | 5.82                                  | 0.0007840                                  | 0.056809                | 59.42                     | 173.99             | 233.41                  | 59.70                    | 194.15             | 253.86                  | 0.23265                  | 0.69591            | 0.92856                 |
| 400                   | 8.91                                  | 0.0007905                                  | 0.051266                | 63.61                     | 171.49             | 235.10                  | 63.92                    | 191.68             | 255.61                  | 0.24757                  | 0.67954            | 0.92711                 |
| 450                   | 12.46                                 | 0.0007983                                  | 0.045677                | 68.44                     | 168.58             | 237.03                  | 68.80                    | 188.78             | 257.58                  | 0.26462                  | 0.66093            | 0.92555                 |
| 500                   | 15.71                                 | 0.0008058                                  | 0.041168                | 72.92                     | 165.86             | 238.77                  | 73.32                    | 186.04             | 259.36                  | 0.28021                  | 0.64399            | 0.92420                 |
| 550                   | 18.73                                 | 0.0008129                                  | 0.037452                | 77.09                     | 163.29             | 240.38                  | 77.54                    | 183.44             | 260.98                  | 0.29460                  | 0.62842            | 0.92302                 |
| 600                   | 21.55                                 | 0.0008198                                  | 0.034335                | 81.01                     | 160.84             | 241.86                  | 81.50                    | 180.95             | 262.46                  | 0.30799                  | 0.61398            | 0.92196                 |
| 650                   | 24.20                                 | 0.0008265                                  | 0.031680                | 84.72                     | 158.51             | 243.23                  | 85.26                    | 178.56             | 263.82                  | 0.32052                  | 0.60048            | 0.92100                 |
| 700                   | 26.69                                 | 0.0008331                                  | 0.029392                | 88.24                     | 156.27             | 244.51                  | 88.82                    | 176.26             | 265.08                  | 0.33232                  | 0.58780            | 0.92012                 |
| 750                   | 29.06                                 | 0.0008395                                  | 0.027398                | 91.59                     | 154.11             | 245.70                  | 92.22                    | 174.03             | 266.25                  | 0.34348                  | 0.57582            | 0.91930                 |
| 800                   | 31.31                                 | 0.0008457                                  | 0.025645                | 94.80                     | 152.02             | 246.82                  | 95.48                    | 171.86             | 267.34                  | 0.35408                  | 0.56445            | 0.91853                 |
| 850                   | 33.45                                 | 0.0008519                                  | 0.024091                | 97.88                     | 150.00             | 247.88                  | 98.61                    | 169.75             | 268.36                  | 0.36417                  | 0.55362            | 0.91779                 |
| 900                   | 35.51                                 | 0.0008580                                  | 0.022703                | 100.84                    | 148.03             | 248.88                  | 101.62                   | 167.69             | 269.31                  | 0.37383                  | 0.54326            | 0.91709                 |
| 950                   | 37.48                                 | 0.0008640                                  | 0.021456                | 103.70                    | 146.11             | 249.82                  | 104.52                   | 165.68             | 270.20                  | 0.38307                  | 0.53333            | 0.91641                 |
| 1000                  | 39.37                                 | 0.0008700                                  | 0.020329                | 106.47                    | 144.24             | 250.71                  | 107.34                   | 163.70             | 271.04                  | 0.39196                  | 0.52378            | 0.91574                 |
| 1200                  | 46.29                                 | 0.0008935                                  | 0.016728                | 116.72                    | 137.12             | 253.84                  | 117.79                   | 156.12             | 273.92                  | 0.42449                  | 0.48870            | 0.91320                 |
| 1400                  | 52.40                                 | 0.0009167                                  | 0.014119                | 125.96                    | 130.44             | 256.40                  | 127.25                   | 148.92             | 276.17                  | 0.45325                  | 0.45742            | 0.91067                 |
| 1600                  | 57.88                                 | 0.0009400                                  | 0.012134                | 134.45                    | 124.05             | 258.50                  | 135.96                   | 141.96             | 277.92                  | 0.47921                  | 0.42881            | 0.90802                 |
| 1800                  | 62.87                                 | 0.0009639                                  | 0.010568                | 142.36                    | 117.85             | 260.21                  | 144.09                   | 135.14             | 279.23                  | 0.50304                  | 0.40213            | 0.90517                 |
| 2000                  | 67.45                                 | 0.0009887                                  | 0.009297                | 149.81                    | 111.75             | 261.56                  | 151.78                   | 128.36             | 280.15                  | 0.52519                  | 0.37684            | 0.90204                 |
| 2500                  | 77.54                                 | 0.0010567                                  | 0.006941                | 167.02                    | 96.47              | 263.49                  | 169.66                   | 111.18             | 280.84                  | 0.57542                  | 0.31701            | 0.89243                 |
| 3000                  | 86.16                                 | 0.0011410                                  | 0.005272                | 183.09                    | 80.17              | 263.26                  | 186.51                   | 92.57              | 279.08                  | 0.62133                  | 0.25759            | 0.87893                 |



TABLE A-13

Superheated refrigerant-134a

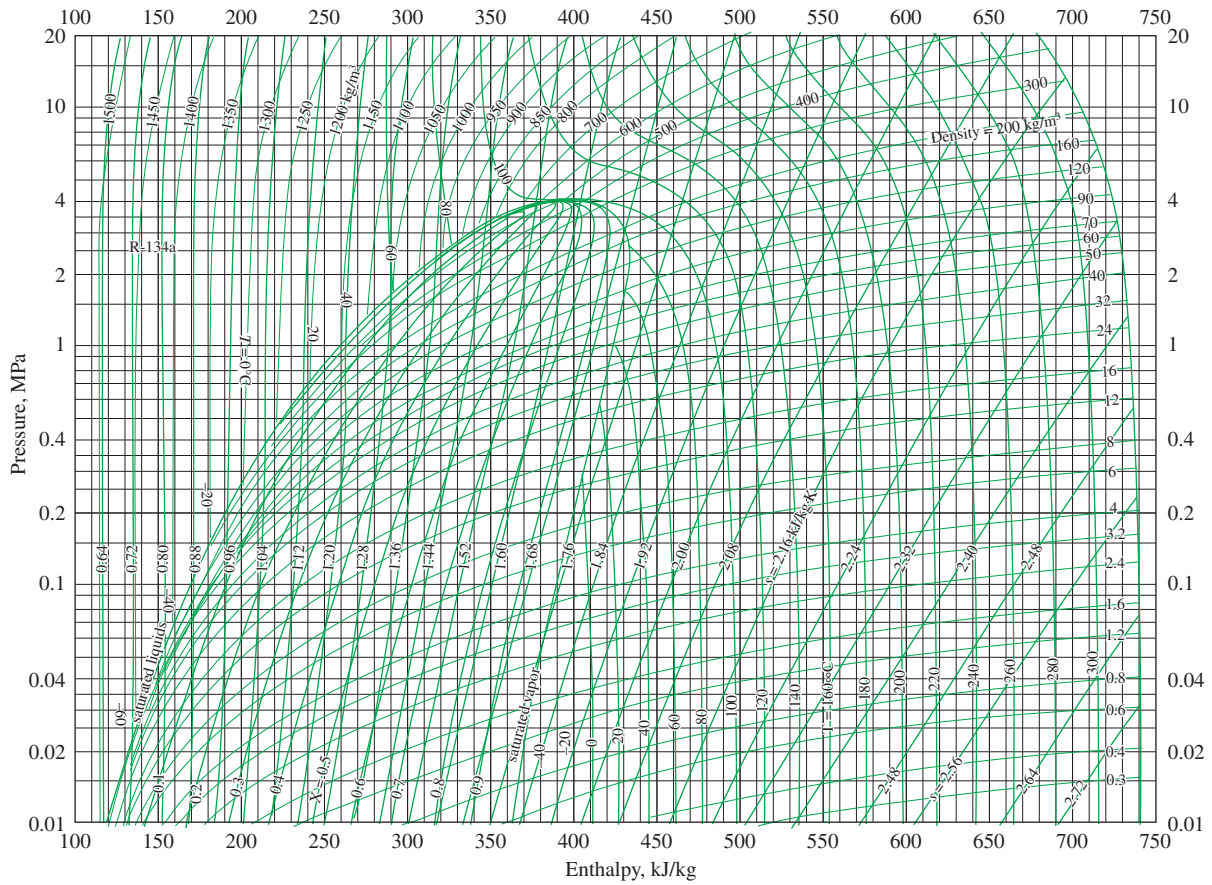
| $T$<br>°C                                                       | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg                                       | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg                                       | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|-----------------------------------------------------------------|---------------------------|--------------|--------------|----------------|-----------------------------------------------------------------|--------------|--------------|----------------|-----------------------------------------------------------------|--------------|--------------|----------------|
| $P = 0.06 \text{ MPa } (T_{\text{sat}} = -36.95^\circ\text{C})$ |                           |              |              |                | $P = 0.10 \text{ MPa } (T_{\text{sat}} = -26.37^\circ\text{C})$ |              |              |                | $P = 0.14 \text{ MPa } (T_{\text{sat}} = -18.77^\circ\text{C})$ |              |              |                |
| Sat.                                                            | 0.31108                   | 209.13       | 227.80       | 0.9645         | 0.19255                                                         | 215.21       | 234.46       | 0.9519         | 0.14020                                                         | 219.56       | 239.19       | 0.9447         |
| -20                                                             | 0.33608                   | 220.62       | 240.78       | 1.0175         | 0.19841                                                         | 219.68       | 239.52       | 0.9721         |                                                                 |              |              |                |
| -10                                                             | 0.35048                   | 227.57       | 248.60       | 1.0478         | 0.20743                                                         | 226.77       | 247.51       | 1.0031         | 0.14605                                                         | 225.93       | 246.37       | 0.9724         |
| 0                                                               | 0.36476                   | 234.67       | 256.56       | 1.0775         | 0.21630                                                         | 233.97       | 255.60       | 1.0333         | 0.15263                                                         | 233.25       | 254.61       | 1.0032         |
| 10                                                              | 0.37893                   | 241.94       | 264.68       | 1.1067         | 0.22506                                                         | 241.32       | 263.82       | 1.0628         | 0.15908                                                         | 240.68       | 262.95       | 1.0331         |
| 20                                                              | 0.39302                   | 249.37       | 272.95       | 1.1354         | 0.23373                                                         | 248.81       | 272.18       | 1.0919         | 0.16544                                                         | 248.24       | 271.40       | 1.0625         |
| 30                                                              | 0.40705                   | 256.97       | 281.39       | 1.1637         | 0.24233                                                         | 256.46       | 280.69       | 1.1204         | 0.17172                                                         | 255.95       | 279.99       | 1.0913         |
| 40                                                              | 0.42102                   | 264.73       | 289.99       | 1.1916         | 0.25088                                                         | 264.27       | 289.36       | 1.1485         | 0.17794                                                         | 263.80       | 288.72       | 1.1196         |
| 50                                                              | 0.43495                   | 272.66       | 298.75       | 1.2192         | 0.25937                                                         | 272.24       | 298.17       | 1.1762         | 0.18412                                                         | 271.81       | 297.59       | 1.1475         |
| 60                                                              | 0.44883                   | 280.75       | 307.68       | 1.2464         | 0.26783                                                         | 280.36       | 307.15       | 1.2036         | 0.19025                                                         | 279.97       | 306.61       | 1.1750         |
| 70                                                              | 0.46269                   | 289.01       | 316.77       | 1.2732         | 0.27626                                                         | 288.65       | 316.28       | 1.2306         | 0.19635                                                         | 288.29       | 315.78       | 1.2021         |
| 80                                                              | 0.47651                   | 297.43       | 326.02       | 1.2998         | 0.28465                                                         | 297.10       | 325.57       | 1.2573         | 0.20242                                                         | 296.77       | 325.11       | 1.2289         |
| 90                                                              | 0.49032                   | 306.02       | 335.43       | 1.3261         | 0.29303                                                         | 305.71       | 335.01       | 1.2836         | 0.20847                                                         | 305.40       | 334.59       | 1.2554         |
| 100                                                             | 0.50410                   | 314.76       | 345.01       | 1.3521         | 0.30138                                                         | 314.48       | 344.61       | 1.3097         | 0.21449                                                         | 314.19       | 344.22       | 1.2815         |
| $P = 0.18 \text{ MPa } (T_{\text{sat}} = -12.73^\circ\text{C})$ |                           |              |              |                | $P = 0.20 \text{ MPa } (T_{\text{sat}} = -10.09^\circ\text{C})$ |              |              |                | $P = 0.24 \text{ MPa } (T_{\text{sat}} = -5.38^\circ\text{C})$  |              |              |                |
| Sat.                                                            | 0.11049                   | 223.01       | 242.90       | 0.9398         | 0.09995                                                         | 224.51       | 244.50       | 0.9379         | 0.08398                                                         | 227.17       | 247.32       | 0.9348         |
| -10                                                             | 0.11189                   | 225.04       | 245.18       | 0.9485         | 0.09991                                                         | 224.57       | 244.56       | 0.9381         |                                                                 |              |              |                |
| 0                                                               | 0.11722                   | 232.49       | 253.59       | 0.9799         | 0.10481                                                         | 232.11       | 253.07       | 0.9699         | 0.08617                                                         | 231.30       | 251.98       | 0.9520         |
| 10                                                              | 0.12240                   | 240.02       | 262.05       | 1.0103         | 0.10955                                                         | 239.69       | 261.60       | 1.0005         | 0.09026                                                         | 239.00       | 260.66       | 0.9832         |
| 20                                                              | 0.12748                   | 247.66       | 270.60       | 1.0400         | 0.11418                                                         | 247.36       | 270.20       | 1.0304         | 0.09423                                                         | 246.76       | 269.38       | 1.0134         |
| 30                                                              | 0.13248                   | 255.43       | 279.27       | 1.0691         | 0.11874                                                         | 255.16       | 278.91       | 1.0596         | 0.09812                                                         | 254.63       | 278.17       | 1.0429         |
| 40                                                              | 0.13741                   | 263.33       | 288.07       | 1.0976         | 0.12322                                                         | 263.09       | 287.74       | 1.0882         | 0.10193                                                         | 262.61       | 287.07       | 1.0718         |
| 50                                                              | 0.14230                   | 271.38       | 297.00       | 1.1257         | 0.12766                                                         | 271.16       | 296.70       | 1.1164         | 0.10570                                                         | 270.73       | 296.09       | 1.1002         |
| 60                                                              | 0.14715                   | 279.58       | 306.07       | 1.1533         | 0.13206                                                         | 279.38       | 305.79       | 1.1441         | 0.10942                                                         | 278.98       | 305.24       | 1.1281         |
| 70                                                              | 0.15196                   | 287.93       | 315.28       | 1.1806         | 0.13641                                                         | 287.75       | 315.03       | 1.1714         | 0.11310                                                         | 287.38       | 314.53       | 1.1555         |
| 80                                                              | 0.15673                   | 296.43       | 324.65       | 1.2075         | 0.14074                                                         | 296.27       | 324.41       | 1.1984         | 0.11675                                                         | 295.93       | 323.95       | 1.1826         |
| 90                                                              | 0.16149                   | 305.09       | 334.16       | 1.2340         | 0.14504                                                         | 304.93       | 333.94       | 1.2250         | 0.12038                                                         | 304.62       | 333.51       | 1.2093         |
| 100                                                             | 0.16622                   | 313.90       | 343.82       | 1.2603         | 0.14933                                                         | 313.75       | 343.62       | 1.2513         | 0.12398                                                         | 313.46       | 343.22       | 1.2356         |
| $P = 0.28 \text{ MPa } (T_{\text{sat}} = -1.25^\circ\text{C})$  |                           |              |              |                | $P = 0.32 \text{ MPa } (T_{\text{sat}} = 2.46^\circ\text{C})$   |              |              |                | $P = 0.40 \text{ MPa } (T_{\text{sat}} = 8.91^\circ\text{C})$   |              |              |                |
| Sat.                                                            | 0.07243                   | 229.49       | 249.77       | 0.9323         | 0.06368                                                         | 231.55       | 251.93       | 0.9303         | 0.051266                                                        | 235.10       | 255.61       | 0.9271         |
| 0                                                               | 0.07282                   | 230.46       | 250.85       | 0.9362         |                                                                 |              |              |                |                                                                 |              |              |                |
| 10                                                              | 0.07646                   | 238.29       | 259.70       | 0.9681         | 0.06609                                                         | 237.56       | 258.70       | 0.9545         | 0.051506                                                        | 235.99       | 256.59       | 0.9306         |
| 20                                                              | 0.07997                   | 246.15       | 268.54       | 0.9987         | 0.06925                                                         | 245.51       | 267.67       | 0.9856         | 0.054213                                                        | 244.19       | 265.88       | 0.9628         |
| 30                                                              | 0.08338                   | 254.08       | 277.42       | 1.0285         | 0.07231                                                         | 253.52       | 276.66       | 1.0158         | 0.056796                                                        | 252.37       | 275.09       | 0.9937         |
| 40                                                              | 0.08672                   | 262.12       | 286.40       | 1.0577         | 0.07530                                                         | 261.62       | 285.72       | 1.0452         | 0.059292                                                        | 260.60       | 284.32       | 1.0237         |
| 50                                                              | 0.09000                   | 270.28       | 295.48       | 1.0862         | 0.07823                                                         | 269.83       | 294.87       | 1.0739         | 0.061724                                                        | 268.92       | 293.61       | 1.0529         |
| 60                                                              | 0.09324                   | 278.58       | 304.69       | 1.1143         | 0.08111                                                         | 278.17       | 304.12       | 1.1022         | 0.064104                                                        | 277.34       | 302.98       | 1.0814         |
| 70                                                              | 0.09644                   | 287.01       | 314.01       | 1.1419         | 0.08395                                                         | 286.64       | 313.50       | 1.1299         | 0.066443                                                        | 285.88       | 312.45       | 1.1095         |
| 80                                                              | 0.09961                   | 295.59       | 323.48       | 1.1690         | 0.08675                                                         | 295.24       | 323.00       | 1.1572         | 0.068747                                                        | 294.54       | 322.04       | 1.1370         |
| 90                                                              | 0.10275                   | 304.30       | 333.07       | 1.1958         | 0.08953                                                         | 303.99       | 332.64       | 1.1841         | 0.071023                                                        | 303.34       | 331.75       | 1.1641         |
| 100                                                             | 0.10587                   | 313.17       | 342.81       | 1.2223         | 0.09229                                                         | 312.87       | 342.41       | 1.2106         | 0.073274                                                        | 312.28       | 341.59       | 1.1908         |
| 110                                                             | 0.10897                   | 322.18       | 352.69       | 1.2484         | 0.09503                                                         | 321.91       | 352.31       | 1.2368         | 0.075504                                                        | 321.35       | 351.55       | 1.2172         |
| 120                                                             | 0.11205                   | 331.34       | 362.72       | 1.2742         | 0.09775                                                         | 331.08       | 362.36       | 1.2627         | 0.077717                                                        | 330.56       | 361.65       | 1.2432         |
| 130                                                             | 0.11512                   | 340.65       | 372.88       | 1.2998         | 0.10045                                                         | 340.41       | 372.55       | 1.2883         | 0.079913                                                        | 339.92       | 371.89       | 1.2689         |
| 140                                                             | 0.11818                   | 350.11       | 383.20       | 1.3251         | 0.10314                                                         | 349.88       | 382.89       | 1.3136         | 0.082096                                                        | 349.42       | 382.26       | 1.2943         |

TABLE A-13

Superheated refrigerant-134a (Concluded)

| $T$<br>°C                                                      | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg                                      | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K | $v$<br>m <sup>3</sup> /kg                                      | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|----------------------------------------------------------------|---------------------------|--------------|--------------|----------------|----------------------------------------------------------------|--------------|--------------|----------------|----------------------------------------------------------------|--------------|--------------|----------------|
| $P = 0.50 \text{ MPa } (T_{\text{sat}} = 15.71^\circ\text{C})$ |                           |              |              |                | $P = 0.60 \text{ MPa } (T_{\text{sat}} = 21.55^\circ\text{C})$ |              |              |                | $P = 0.70 \text{ MPa } (T_{\text{sat}} = 26.69^\circ\text{C})$ |              |              |                |
| Sat.                                                           | 0.041168                  | 238.77       | 259.36       | 0.9242         | 0.034335                                                       | 241.86       | 262.46       | 0.9220         | 0.029392                                                       | 244.51       | 265.08       | 0.9201         |
| 20                                                             | 0.042115                  | 242.42       | 263.48       | 0.9384         |                                                                |              |              |                |                                                                |              |              |                |
| 30                                                             | 0.044338                  | 250.86       | 273.03       | 0.9704         | 0.035984                                                       | 249.24       | 270.83       | 0.9500         | 0.029966                                                       | 247.49       | 268.47       | 0.9314         |
| 40                                                             | 0.046456                  | 259.27       | 282.50       | 1.0011         | 0.037865                                                       | 257.88       | 280.60       | 0.9817         | 0.031696                                                       | 256.41       | 278.59       | 0.9642         |
| 50                                                             | 0.048499                  | 267.73       | 291.98       | 1.0309         | 0.039659                                                       | 266.50       | 290.30       | 1.0122         | 0.033322                                                       | 265.22       | 288.54       | 0.9955         |
| 60                                                             | 0.050485                  | 276.27       | 301.51       | 1.0600         | 0.041389                                                       | 275.17       | 300.00       | 1.0417         | 0.034875                                                       | 274.03       | 298.44       | 1.0257         |
| 70                                                             | 0.052427                  | 284.91       | 311.12       | 1.0884         | 0.043069                                                       | 283.91       | 309.75       | 1.0706         | 0.036373                                                       | 282.88       | 308.34       | 1.0550         |
| 80                                                             | 0.054331                  | 293.65       | 320.82       | 1.1163         | 0.044710                                                       | 292.74       | 319.57       | 1.0988         | 0.037829                                                       | 291.81       | 318.29       | 1.0835         |
| 90                                                             | 0.056205                  | 302.52       | 330.63       | 1.1436         | 0.046318                                                       | 301.69       | 329.48       | 1.1265         | 0.039250                                                       | 300.84       | 328.31       | 1.1115         |
| 100                                                            | 0.058053                  | 311.52       | 340.55       | 1.1706         | 0.047900                                                       | 310.75       | 339.49       | 1.1536         | 0.040642                                                       | 309.96       | 338.41       | 1.1389         |
| 110                                                            | 0.059880                  | 320.65       | 350.59       | 1.1971         | 0.049458                                                       | 319.93       | 349.61       | 1.1804         | 0.042010                                                       | 319.21       | 348.61       | 1.1659         |
| 120                                                            | 0.061687                  | 329.91       | 360.75       | 1.2233         | 0.050997                                                       | 329.24       | 359.84       | 1.2068         | 0.043358                                                       | 328.57       | 358.92       | 1.1925         |
| 130                                                            | 0.063479                  | 339.31       | 371.05       | 1.2492         | 0.052519                                                       | 338.69       | 370.20       | 1.2328         | 0.044688                                                       | 338.06       | 369.34       | 1.2186         |
| 140                                                            | 0.065256                  | 348.85       | 381.47       | 1.2747         | 0.054027                                                       | 348.26       | 380.68       | 1.2585         | 0.046004                                                       | 347.67       | 379.88       | 1.2445         |
| 150                                                            | 0.067021                  | 358.52       | 392.04       | 1.3000         | 0.055522                                                       | 357.98       | 391.29       | 1.2838         | 0.047306                                                       | 357.42       | 390.54       | 1.2700         |
| 160                                                            | 0.068775                  | 368.34       | 402.73       | 1.3250         | 0.057006                                                       | 367.83       | 402.03       | 1.3089         | 0.048597                                                       | 367.31       | 401.32       | 1.2952         |
| $P = 0.80 \text{ MPa } (T_{\text{sat}} = 31.31^\circ\text{C})$ |                           |              |              |                | $P = 0.90 \text{ MPa } (T_{\text{sat}} = 35.51^\circ\text{C})$ |              |              |                | $P = 1.00 \text{ MPa } (T_{\text{sat}} = 39.37^\circ\text{C})$ |              |              |                |
| Sat.                                                           | 0.025645                  | 246.82       | 267.34       | 0.9185         | 0.022686                                                       | 248.82       | 269.25       | 0.9169         | 0.020319                                                       | 250.71       | 271.04       | 0.9157         |
| 40                                                             | 0.027035                  | 254.84       | 276.46       | 0.9481         | 0.023375                                                       | 253.15       | 274.19       | 0.9328         | 0.020406                                                       | 251.32       | 271.73       | 0.9180         |
| 50                                                             | 0.028547                  | 263.87       | 286.71       | 0.9803         | 0.024809                                                       | 262.46       | 284.79       | 0.9661         | 0.021796                                                       | 260.96       | 282.76       | 0.9526         |
| 60                                                             | 0.029973                  | 272.85       | 296.82       | 1.0111         | 0.026146                                                       | 271.62       | 295.15       | 0.9977         | 0.023068                                                       | 270.33       | 293.40       | 0.9851         |
| 70                                                             | 0.031340                  | 281.83       | 306.90       | 1.0409         | 0.027413                                                       | 280.74       | 305.41       | 1.0280         | 0.024261                                                       | 279.61       | 303.87       | 1.0160         |
| 80                                                             | 0.032659                  | 290.86       | 316.99       | 1.0699         | 0.028630                                                       | 289.88       | 315.65       | 1.0574         | 0.025398                                                       | 288.87       | 314.27       | 1.0459         |
| 90                                                             | 0.033941                  | 299.97       | 327.12       | 1.0982         | 0.029806                                                       | 299.08       | 325.90       | 1.0861         | 0.026492                                                       | 298.17       | 324.66       | 1.0749         |
| 100                                                            | 0.035193                  | 309.17       | 337.32       | 1.1259         | 0.030951                                                       | 308.35       | 336.21       | 1.1141         | 0.027552                                                       | 307.52       | 335.08       | 1.1032         |
| 110                                                            | 0.036420                  | 318.47       | 347.61       | 1.1531         | 0.032068                                                       | 317.72       | 346.58       | 1.1415         | 0.028584                                                       | 316.96       | 345.54       | 1.1309         |
| 120                                                            | 0.037625                  | 327.89       | 357.99       | 1.1798         | 0.033164                                                       | 327.19       | 357.04       | 1.1684         | 0.029592                                                       | 326.49       | 356.08       | 1.1580         |
| 130                                                            | 0.038813                  | 337.42       | 368.47       | 1.2062         | 0.034241                                                       | 336.78       | 367.59       | 1.1949         | 0.030581                                                       | 336.12       | 366.70       | 1.1847         |
| 140                                                            | 0.039985                  | 347.08       | 379.07       | 1.2321         | 0.035302                                                       | 346.48       | 378.25       | 1.2211         | 0.031554                                                       | 345.87       | 377.42       | 1.2110         |
| 150                                                            | 0.041143                  | 356.86       | 389.78       | 1.2577         | 0.036349                                                       | 356.30       | 389.01       | 1.2468         | 0.032512                                                       | 355.73       | 388.24       | 1.2369         |
| 160                                                            | 0.042290                  | 366.78       | 400.61       | 1.2830         | 0.037384                                                       | 366.25       | 399.89       | 1.2722         | 0.033457                                                       | 365.71       | 399.17       | 1.2624         |
| 170                                                            | 0.043427                  | 376.83       | 411.57       | 1.3081         | 0.038408                                                       | 376.33       | 410.89       | 1.2973         | 0.034392                                                       | 375.82       | 410.22       | 1.2876         |
| 180                                                            | 0.044554                  | 387.01       | 422.65       | 1.3328         | 0.039423                                                       | 386.54       | 422.02       | 1.3221         | 0.035317                                                       | 386.06       | 421.38       | 1.3125         |
| $P = 1.20 \text{ MPa } (T_{\text{sat}} = 46.29^\circ\text{C})$ |                           |              |              |                | $P = 1.40 \text{ MPa } (T_{\text{sat}} = 52.40^\circ\text{C})$ |              |              |                | $P = 1.60 \text{ MPa } (T_{\text{sat}} = 57.88^\circ\text{C})$ |              |              |                |
| Sat.                                                           | 0.016728                  | 253.84       | 273.92       | 0.9132         | 0.014119                                                       | 256.40       | 276.17       | 0.9107         | 0.012134                                                       | 258.50       | 277.92       | 0.9080         |
| 50                                                             | 0.017201                  | 257.64       | 278.28       | 0.9268         |                                                                |              |              |                |                                                                |              |              |                |
| 60                                                             | 0.018404                  | 267.57       | 289.66       | 0.9615         | 0.015005                                                       | 264.46       | 285.47       | 0.9389         | 0.012372                                                       | 260.91       | 280.71       | 0.9164         |
| 70                                                             | 0.019502                  | 277.23       | 300.63       | 0.9939         | 0.016060                                                       | 274.62       | 297.10       | 0.9733         | 0.013430                                                       | 271.78       | 293.27       | 0.9536         |
| 80                                                             | 0.020529                  | 286.77       | 311.40       | 1.0249         | 0.017023                                                       | 284.51       | 308.34       | 1.0056         | 0.014362                                                       | 282.11       | 305.09       | 0.9875         |
| 90                                                             | 0.021506                  | 296.28       | 322.09       | 1.0547         | 0.017923                                                       | 294.28       | 319.37       | 1.0364         | 0.015215                                                       | 292.19       | 316.53       | 1.0195         |
| 100                                                            | 0.022442                  | 305.81       | 332.74       | 1.0836         | 0.018778                                                       | 304.01       | 330.30       | 1.0661         | 0.016014                                                       | 302.16       | 327.78       | 1.0501         |
| 110                                                            | 0.023348                  | 315.40       | 343.41       | 1.1119         | 0.019597                                                       | 313.76       | 341.19       | 1.0949         | 0.016773                                                       | 312.09       | 338.93       | 1.0795         |
| 120                                                            | 0.024228                  | 325.05       | 354.12       | 1.1395         | 0.020388                                                       | 323.55       | 352.09       | 1.1230         | 0.017500                                                       | 322.03       | 350.03       | 1.1081         |
| 130                                                            | 0.025086                  | 334.79       | 364.90       | 1.1665         | 0.021155                                                       | 333.41       | 363.02       | 1.1504         | 0.018201                                                       | 332.02       | 361.14       | 1.1360         |
| 140                                                            | 0.025927                  | 344.63       | 375.74       | 1.1931         | 0.021904                                                       | 343.34       | 374.01       | 1.1773         | 0.018882                                                       | 342.06       | 372.27       | 1.1633         |
| 150                                                            | 0.026753                  | 354.57       | 386.68       | 1.2192         | 0.022636                                                       | 353.37       | 385.07       | 1.2038         | 0.019545                                                       | 352.19       | 383.46       | 1.1901         |
| 160                                                            | 0.027566                  | 364.63       | 397.71       | 1.2450         | 0.023355                                                       | 363.51       | 396.20       | 1.2298         | 0.020194                                                       | 362.40       | 394.71       | 1.2164         |
| 170                                                            | 0.028367                  | 374.80       | 408.84       | 1.2704         | 0.024061                                                       | 373.75       | 407.43       | 1.2554         | 0.020830                                                       | 372.71       | 406.04       | 1.2422         |
| 180                                                            | 0.029158                  | 385.10       | 420.09       | 1.2955         | 0.024757                                                       | 384.12       | 418.78       | 1.2808         | 0.021456                                                       | 383.13       | 417.46       | 1.2677         |





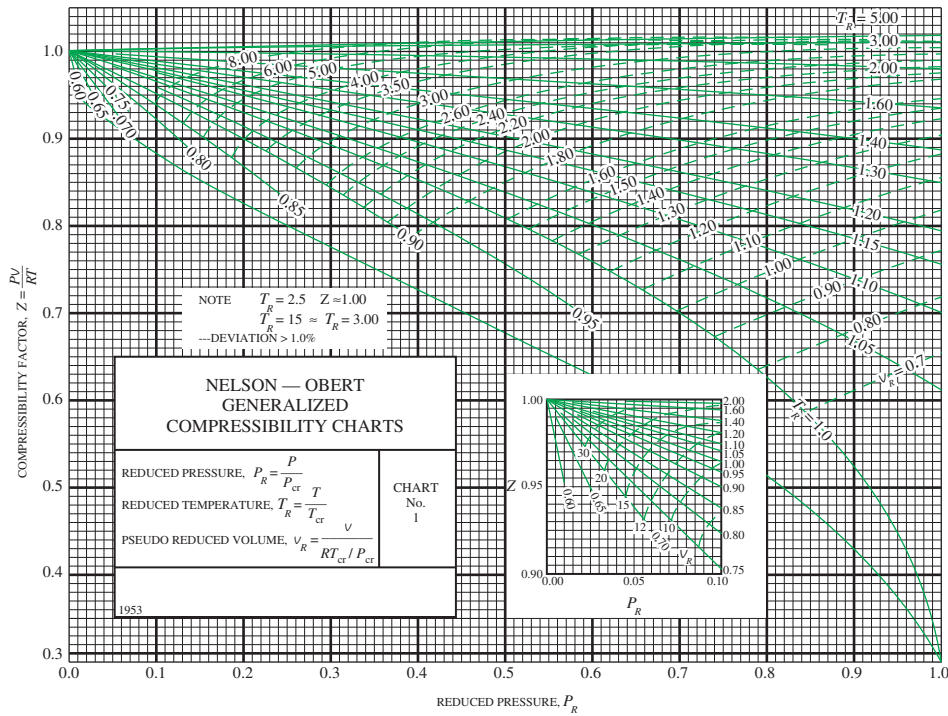
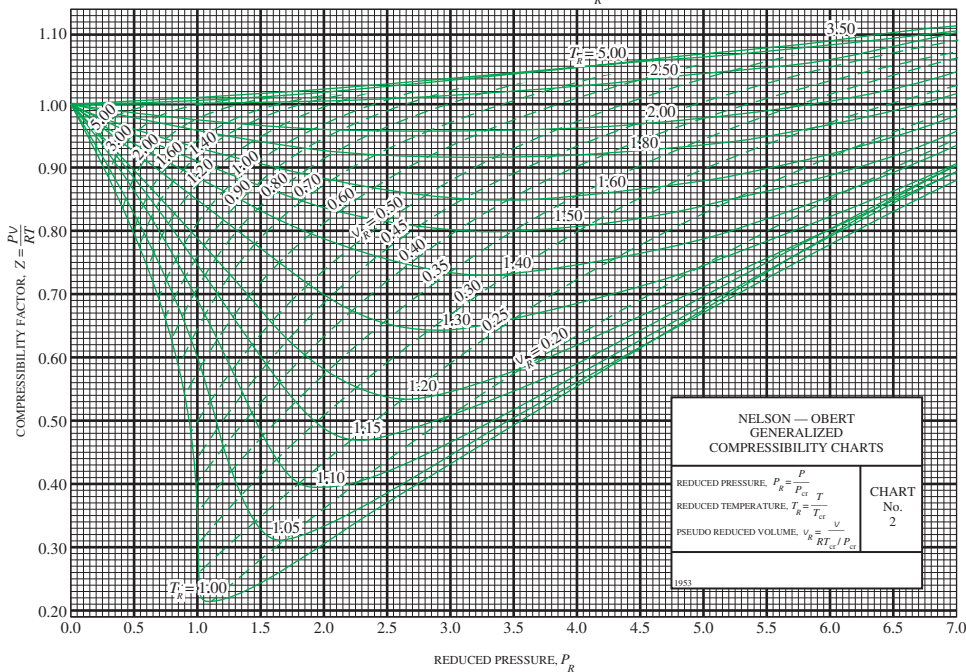
(a) Low pressures,  $0 < P_R < 1.0$ (b) Intermediate pressures,  $0 < P_R < 7$ 

FIGURE A-15

Nelson-Obert generalized compressibility chart.

Used with permission of Dr. Edward E. Obert, University of Wisconsin.

TABLE A-16

Properties of the atmosphere at high altitude

| Altitude,<br>m | Temperature,<br>°C | Pressure,<br>kPa | Gravity<br>$g$ , m/s <sup>2</sup> | Speed of<br>Sound,<br>m/s | Density,<br>kg/m <sup>3</sup> | Viscosity<br>$\mu$ , kg/m·s | Thermal<br>Conductivity,<br>W/m·K |
|----------------|--------------------|------------------|-----------------------------------|---------------------------|-------------------------------|-----------------------------|-----------------------------------|
| 0              | 15.00              | 101.33           | 9.807                             | 340.3                     | 1.225                         | $1.789 \times 10^{-5}$      | 0.0253                            |
| 200            | 13.70              | 98.95            | 9.806                             | 339.5                     | 1.202                         | $1.783 \times 10^{-5}$      | 0.0252                            |
| 400            | 12.40              | 96.61            | 9.805                             | 338.8                     | 1.179                         | $1.777 \times 10^{-5}$      | 0.0252                            |
| 600            | 11.10              | 94.32            | 9.805                             | 338.0                     | 1.156                         | $1.771 \times 10^{-5}$      | 0.0251                            |
| 800            | 9.80               | 92.08            | 9.804                             | 337.2                     | 1.134                         | $1.764 \times 10^{-5}$      | 0.0250                            |
| 1000           | 8.50               | 89.88            | 9.804                             | 336.4                     | 1.112                         | $1.758 \times 10^{-5}$      | 0.0249                            |
| 1200           | 7.20               | 87.72            | 9.803                             | 335.7                     | 1.090                         | $1.752 \times 10^{-5}$      | 0.0248                            |
| 1400           | 5.90               | 85.60            | 9.802                             | 334.9                     | 1.069                         | $1.745 \times 10^{-5}$      | 0.0247                            |
| 1600           | 4.60               | 83.53            | 9.802                             | 334.1                     | 1.048                         | $1.739 \times 10^{-5}$      | 0.0245                            |
| 1800           | 3.30               | 81.49            | 9.801                             | 333.3                     | 1.027                         | $1.732 \times 10^{-5}$      | 0.0244                            |
| 2000           | 2.00               | 79.50            | 9.800                             | 332.5                     | 1.007                         | $1.726 \times 10^{-5}$      | 0.0243                            |
| 2200           | 0.70               | 77.55            | 9.800                             | 331.7                     | 0.987                         | $1.720 \times 10^{-5}$      | 0.0242                            |
| 2400           | -0.59              | 75.63            | 9.799                             | 331.0                     | 0.967                         | $1.713 \times 10^{-5}$      | 0.0241                            |
| 2600           | -1.89              | 73.76            | 9.799                             | 330.2                     | 0.947                         | $1.707 \times 10^{-5}$      | 0.0240                            |
| 2800           | -3.19              | 71.92            | 9.798                             | 329.4                     | 0.928                         | $1.700 \times 10^{-5}$      | 0.0239                            |
| 3000           | -4.49              | 70.12            | 9.797                             | 328.6                     | 0.909                         | $1.694 \times 10^{-5}$      | 0.0238                            |
| 3200           | -5.79              | 68.36            | 9.797                             | 327.8                     | 0.891                         | $1.687 \times 10^{-5}$      | 0.0237                            |
| 3400           | -7.09              | 66.63            | 9.796                             | 327.0                     | 0.872                         | $1.681 \times 10^{-5}$      | 0.0236                            |
| 3600           | -8.39              | 64.94            | 9.796                             | 326.2                     | 0.854                         | $1.674 \times 10^{-5}$      | 0.0235                            |
| 3800           | -9.69              | 63.28            | 9.795                             | 325.4                     | 0.837                         | $1.668 \times 10^{-5}$      | 0.0234                            |
| 4000           | -10.98             | 61.66            | 9.794                             | 324.6                     | 0.819                         | $1.661 \times 10^{-5}$      | 0.0233                            |
| 4200           | -12.3              | 60.07            | 9.794                             | 323.8                     | 0.802                         | $1.655 \times 10^{-5}$      | 0.0232                            |
| 4400           | -13.6              | 58.52            | 9.793                             | 323.0                     | 0.785                         | $1.648 \times 10^{-5}$      | 0.0231                            |
| 4600           | -14.9              | 57.00            | 9.793                             | 322.2                     | 0.769                         | $1.642 \times 10^{-5}$      | 0.0230                            |
| 4800           | -16.2              | 55.51            | 9.792                             | 321.4                     | 0.752                         | $1.635 \times 10^{-5}$      | 0.0229                            |
| 5000           | -17.5              | 54.05            | 9.791                             | 320.5                     | 0.736                         | $1.628 \times 10^{-5}$      | 0.0228                            |
| 5200           | -18.8              | 52.62            | 9.791                             | 319.7                     | 0.721                         | $1.622 \times 10^{-5}$      | 0.0227                            |
| 5400           | -20.1              | 51.23            | 9.790                             | 318.9                     | 0.705                         | $1.615 \times 10^{-5}$      | 0.0226                            |
| 5600           | -21.4              | 49.86            | 9.789                             | 318.1                     | 0.690                         | $1.608 \times 10^{-5}$      | 0.0224                            |
| 5800           | -22.7              | 48.52            | 9.785                             | 317.3                     | 0.675                         | $1.602 \times 10^{-5}$      | 0.0223                            |
| 6000           | -24.0              | 47.22            | 9.788                             | 316.5                     | 0.660                         | $1.595 \times 10^{-5}$      | 0.0222                            |
| 6200           | -25.3              | 45.94            | 9.788                             | 315.6                     | 0.646                         | $1.588 \times 10^{-5}$      | 0.0221                            |
| 6400           | -26.6              | 44.69            | 9.787                             | 314.8                     | 0.631                         | $1.582 \times 10^{-5}$      | 0.0220                            |
| 6600           | -27.9              | 43.47            | 9.786                             | 314.0                     | 0.617                         | $1.575 \times 10^{-5}$      | 0.0219                            |
| 6800           | -29.2              | 42.27            | 9.785                             | 313.1                     | 0.604                         | $1.568 \times 10^{-5}$      | 0.0218                            |
| 7000           | -30.5              | 41.11            | 9.785                             | 312.3                     | 0.590                         | $1.561 \times 10^{-5}$      | 0.0217                            |
| 8000           | -36.9              | 35.65            | 9.782                             | 308.1                     | 0.526                         | $1.527 \times 10^{-5}$      | 0.0212                            |
| 9000           | -43.4              | 30.80            | 9.779                             | 303.8                     | 0.467                         | $1.493 \times 10^{-5}$      | 0.0206                            |
| 10,000         | -49.9              | 26.50            | 9.776                             | 299.5                     | 0.414                         | $1.458 \times 10^{-5}$      | 0.0201                            |
| 12,000         | -56.5              | 19.40            | 9.770                             | 295.1                     | 0.312                         | $1.422 \times 10^{-5}$      | 0.0195                            |
| 14,000         | -56.5              | 14.17            | 9.764                             | 295.1                     | 0.228                         | $1.422 \times 10^{-5}$      | 0.0195                            |
| 16,000         | -56.5              | 10.53            | 9.758                             | 295.1                     | 0.166                         | $1.422 \times 10^{-5}$      | 0.0195                            |
| 18,000         | -56.5              | 7.57             | 9.751                             | 295.1                     | 0.122                         | $1.422 \times 10^{-5}$      | 0.0195                            |

Source of Data: U.S. Standard Atmosphere Supplements, U.S. Government Printing Office, 1966. Based on year-round mean conditions at 45° latitude and varies with the time of the year and the weather patterns. The conditions at sea level ( $z = 0$ ) are taken to be  $P = 101.325$  kPa,  $T = 15^\circ\text{C}$ ,  $\rho = 1.2250$  kg/m<sup>3</sup>,  $g = 9.80665$  m/s<sup>2</sup>.

TABLE A-17

Ideal-gas properties of air

| $T$<br>K | $h$<br>kJ/kg | $P_r$  | $u$<br>kJ/kg | $v_r$  | $s^\circ$<br>kJ/kg·K | $T$<br>K | $h$<br>kJ/kg | $P_r$ | $u$<br>kJ/kg | $v_r$  | $s^\circ$<br>kJ/kg·K |
|----------|--------------|--------|--------------|--------|----------------------|----------|--------------|-------|--------------|--------|----------------------|
| 200      | 199.97       | 0.3363 | 142.56       | 1707.0 | 1.29559              | 580      | 586.04       | 14.38 | 419.55       | 115.7  | 2.37348              |
| 210      | 209.97       | 0.3987 | 149.69       | 1512.0 | 1.34444              | 590      | 596.52       | 15.31 | 427.15       | 110.6  | 2.39140              |
| 220      | 219.97       | 0.4690 | 156.82       | 1346.0 | 1.39105              | 600      | 607.02       | 16.28 | 434.78       | 105.8  | 2.40902              |
| 230      | 230.02       | 0.5477 | 164.00       | 1205.0 | 1.43557              | 610      | 617.53       | 17.30 | 442.42       | 101.2  | 2.42644              |
| 240      | 240.02       | 0.6355 | 171.13       | 1084.0 | 1.47824              | 620      | 628.07       | 18.36 | 450.09       | 96.92  | 2.44356              |
| 250      | 250.05       | 0.7329 | 178.28       | 979.0  | 1.51917              | 630      | 638.63       | 19.84 | 457.78       | 92.84  | 2.46048              |
| 260      | 260.09       | 0.8405 | 185.45       | 887.8  | 1.55848              | 640      | 649.22       | 20.64 | 465.50       | 88.99  | 2.47716              |
| 270      | 270.11       | 0.9590 | 192.60       | 808.0  | 1.59634              | 650      | 659.84       | 21.86 | 473.25       | 85.34  | 2.49364              |
| 280      | 280.13       | 1.0889 | 199.75       | 738.0  | 1.63279              | 660      | 670.47       | 23.13 | 481.01       | 81.89  | 2.50985              |
| 285      | 285.14       | 1.1584 | 203.33       | 706.1  | 1.65055              | 670      | 681.14       | 24.46 | 488.81       | 78.61  | 2.52589              |
| 290      | 290.16       | 1.2311 | 206.91       | 676.1  | 1.66802              | 680      | 691.82       | 25.85 | 496.62       | 75.50  | 2.54175              |
| 295      | 295.17       | 1.3068 | 210.49       | 647.9  | 1.68515              | 690      | 702.52       | 27.29 | 504.45       | 72.56  | 2.55731              |
| 298      | 298.18       | 1.3543 | 212.64       | 631.9  | 1.69528              | 700      | 713.27       | 28.80 | 512.33       | 69.76  | 2.57277              |
| 300      | 300.19       | 1.3860 | 214.07       | 621.2  | 1.70203              | 710      | 724.04       | 30.38 | 520.23       | 67.07  | 2.58810              |
| 305      | 305.22       | 1.4686 | 217.67       | 596.0  | 1.71865              | 720      | 734.82       | 32.02 | 528.14       | 64.53  | 2.60319              |
| 310      | 310.24       | 1.5546 | 221.25       | 572.3  | 1.73498              | 730      | 745.62       | 33.72 | 536.07       | 62.13  | 2.61803              |
| 315      | 315.27       | 1.6442 | 224.85       | 549.8  | 1.75106              | 740      | 756.44       | 35.50 | 544.02       | 59.82  | 2.63280              |
| 320      | 320.29       | 1.7375 | 228.42       | 528.6  | 1.76690              | 750      | 767.29       | 37.35 | 551.99       | 57.63  | 2.64737              |
| 325      | 325.31       | 1.8345 | 232.02       | 508.4  | 1.78249              | 760      | 778.18       | 39.27 | 560.01       | 55.54  | 2.66176              |
| 330      | 330.34       | 1.9352 | 235.61       | 489.4  | 1.79783              | 780      | 800.03       | 43.35 | 576.12       | 51.64  | 2.69013              |
| 340      | 340.42       | 2.149  | 242.82       | 454.1  | 1.82790              | 800      | 821.95       | 47.75 | 592.30       | 48.08  | 2.71787              |
| 350      | 350.49       | 2.379  | 250.02       | 422.2  | 1.85708              | 820      | 843.98       | 52.59 | 608.59       | 44.84  | 2.74504              |
| 360      | 360.58       | 2.626  | 257.24       | 393.4  | 1.88543              | 840      | 866.08       | 57.60 | 624.95       | 41.85  | 2.77170              |
| 370      | 370.67       | 2.892  | 264.46       | 367.2  | 1.91313              | 860      | 888.27       | 63.09 | 641.40       | 39.12  | 2.79783              |
| 380      | 380.77       | 3.176  | 271.69       | 343.4  | 1.94001              | 880      | 910.56       | 68.98 | 657.95       | 36.61  | 2.82344              |
| 390      | 390.88       | 3.481  | 278.93       | 321.5  | 1.96633              | 900      | 932.93       | 75.29 | 674.58       | 34.31  | 2.84856              |
| 400      | 400.98       | 3.806  | 286.16       | 301.6  | 1.99194              | 920      | 955.38       | 82.05 | 691.28       | 32.18  | 2.87324              |
| 410      | 411.12       | 4.153  | 293.43       | 283.3  | 2.01699              | 940      | 977.92       | 89.28 | 708.08       | 30.22  | 2.89748              |
| 420      | 421.26       | 4.522  | 300.69       | 266.6  | 2.04142              | 960      | 1000.55      | 97.00 | 725.02       | 28.40  | 2.92128              |
| 430      | 431.43       | 4.915  | 307.99       | 251.1  | 2.06533              | 980      | 1023.25      | 105.2 | 741.98       | 26.73  | 2.94468              |
| 440      | 441.61       | 5.332  | 315.30       | 236.8  | 2.08870              | 1000     | 1046.04      | 114.0 | 758.94       | 25.17  | 2.96770              |
| 450      | 451.80       | 5.775  | 322.62       | 223.6  | 2.11161              | 1020     | 1068.89      | 123.4 | 776.10       | 23.72  | 2.99034              |
| 460      | 462.02       | 6.245  | 329.97       | 211.4  | 2.13407              | 1040     | 1091.85      | 133.3 | 793.36       | 23.29  | 3.01260              |
| 470      | 472.24       | 6.742  | 337.32       | 200.1  | 2.15604              | 1060     | 1114.86      | 143.9 | 810.62       | 21.14  | 3.03449              |
| 480      | 482.49       | 7.268  | 344.70       | 189.5  | 2.17760              | 1080     | 1137.89      | 155.2 | 827.88       | 19.98  | 3.05608              |
| 490      | 492.74       | 7.824  | 352.08       | 179.7  | 2.19876              | 1100     | 1161.07      | 167.1 | 845.33       | 18.896 | 3.07732              |
| 500      | 503.02       | 8.411  | 359.49       | 170.6  | 2.21952              | 1120     | 1184.28      | 179.7 | 862.79       | 17.886 | 3.09825              |
| 510      | 513.32       | 9.031  | 366.92       | 162.1  | 2.23993              | 1140     | 1207.57      | 193.1 | 880.35       | 16.946 | 3.11883              |
| 520      | 523.63       | 9.684  | 374.36       | 154.1  | 2.25997              | 1160     | 1230.92      | 207.2 | 897.91       | 16.064 | 3.13916              |
| 530      | 533.98       | 10.37  | 381.84       | 146.7  | 2.27967              | 1180     | 1254.34      | 222.2 | 915.57       | 15.241 | 3.15916              |
| 540      | 544.35       | 11.10  | 389.34       | 139.7  | 2.29906              | 1200     | 1277.79      | 238.0 | 933.33       | 14.470 | 3.17888              |
| 550      | 555.74       | 11.86  | 396.86       | 133.1  | 2.31809              | 1220     | 1301.31      | 254.7 | 951.09       | 13.747 | 3.19834              |
| 560      | 565.17       | 12.66  | 404.42       | 127.0  | 2.33685              | 1240     | 1324.93      | 272.3 | 968.95       | 13.069 | 3.21751              |
| 570      | 575.59       | 13.50  | 411.97       | 121.2  | 2.35531              |          |              |       |              |        |                      |

TABLE A-17

Ideal-gas properties of air (*Concluded*)

| $T$<br>K | $h$<br>kJ/kg | $P_r$ | $u$<br>kJ/kg | $v_r$  | $s^\circ$<br>kJ/kg·K | $T$<br>K | $h$<br>kJ/kg | $P_r$ | $u$<br>kJ/kg | $v_r$ | $s^\circ$<br>kJ/kg·K |
|----------|--------------|-------|--------------|--------|----------------------|----------|--------------|-------|--------------|-------|----------------------|
| 1260     | 1348.55      | 290.8 | 986.90       | 12.435 | 3.23638              | 1600     | 1757.57      | 791.2 | 1298.30      | 5.804 | 3.52364              |
| 1280     | 1372.24      | 310.4 | 1004.76      | 11.835 | 3.25510              | 1620     | 1782.00      | 834.1 | 1316.96      | 5.574 | 3.53879              |
| 1300     | 1395.97      | 330.9 | 1022.82      | 11.275 | 3.27345              | 1640     | 1806.46      | 878.9 | 1335.72      | 5.355 | 3.55381              |
| 1320     | 1419.76      | 352.5 | 1040.88      | 10.747 | 3.29160              | 1660     | 1830.96      | 925.6 | 1354.48      | 5.147 | 3.56867              |
| 1340     | 1443.60      | 375.3 | 1058.94      | 10.247 | 3.30959              | 1680     | 1855.50      | 974.2 | 1373.24      | 4.949 | 3.58335              |
| 1360     | 1467.49      | 399.1 | 1077.10      | 9.780  | 3.32724              | 1700     | 1880.1       | 1025  | 1392.7       | 4.761 | 3.5979               |
| 1380     | 1491.44      | 424.2 | 1095.26      | 9.337  | 3.34474              | 1750     | 1941.6       | 1161  | 1439.8       | 4.328 | 3.6336               |
| 1400     | 1515.42      | 450.5 | 1113.52      | 8.919  | 3.36200              | 1800     | 2003.3       | 1310  | 1487.2       | 3.994 | 3.6684               |
| 1420     | 1539.44      | 478.0 | 1131.77      | 8.526  | 3.37901              | 1850     | 2065.3       | 1475  | 1534.9       | 3.601 | 3.7023               |
| 1440     | 1563.51      | 506.9 | 1150.13      | 8.153  | 3.39586              | 1900     | 2127.4       | 1655  | 1582.6       | 3.295 | 3.7354               |
| 1460     | 1587.63      | 537.1 | 1168.49      | 7.801  | 3.41247              | 1950     | 2189.7       | 1852  | 1630.6       | 3.022 | 3.7677               |
| 1480     | 1611.79      | 568.8 | 1186.95      | 7.468  | 3.42892              | 2000     | 2252.1       | 2068  | 1678.7       | 2.776 | 3.7994               |
| 1500     | 1635.97      | 601.9 | 1205.41      | 7.152  | 3.44516              | 2050     | 2314.6       | 2303  | 1726.8       | 2.555 | 3.8303               |
| 1520     | 1660.23      | 636.5 | 1223.87      | 6.854  | 3.46120              | 2100     | 2377.7       | 2559  | 1775.3       | 2.356 | 3.8605               |
| 1540     | 1684.51      | 672.8 | 1242.43      | 6.569  | 3.47712              | 2150     | 2440.3       | 2837  | 1823.8       | 2.175 | 3.8901               |
| 1560     | 1708.82      | 710.5 | 1260.99      | 6.301  | 3.49276              | 2200     | 2503.2       | 3138  | 1872.4       | 2.012 | 3.9191               |
| 1580     | 1733.17      | 750.0 | 1279.65      | 6.046  | 3.50829              | 2250     | 2566.4       | 3464  | 1921.3       | 1.864 | 3.9474               |

*Note:* The properties  $P_r$  (relative pressure) and  $v_r$  (relative specific volume) are dimensionless quantities used in the analysis of isentropic processes, and should not be confused with the properties pressure and specific volume.

*Source of Data:* Kenneth Wark, *Thermodynamics*, 4th ed. (New York: McGraw-Hill, 1983), pp. 785–86, table A–5. Originally published in J. H. Keenan and J. Kaye, *Gas Tables* (New York: John Wiley & Sons, 1948).

TABLE A-18

Ideal-gas properties of nitrogen,  $N_2$ 

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 600      | 17,563               | 12,574               | 212.066                      |
| 220      | 6,391                | 4,562                | 182.639                      | 610      | 17,864               | 12,792               | 212.564                      |
| 230      | 6,683                | 4,770                | 183.938                      | 620      | 18,166               | 13,011               | 213.055                      |
| 240      | 6,975                | 4,979                | 185.180                      | 630      | 18,468               | 13,230               | 213.541                      |
| 250      | 7,266                | 5,188                | 186.370                      | 640      | 18,772               | 13,450               | 214.018                      |
| 260      | 7,558                | 5,396                | 187.514                      | 650      | 19,075               | 13,671               | 214.489                      |
| 270      | 7,849                | 5,604                | 188.614                      | 660      | 19,380               | 13,892               | 214.954                      |
| 280      | 8,141                | 5,813                | 189.673                      | 670      | 19,685               | 14,114               | 215.413                      |
| 290      | 8,432                | 6,021                | 190.695                      | 680      | 19,991               | 14,337               | 215.866                      |
| 298      | 8,669                | 6,190                | 191.502                      | 690      | 20,297               | 14,560               | 216.314                      |
| 300      | 8,723                | 6,229                | 191.682                      | 700      | 20,604               | 14,784               | 216.756                      |
| 310      | 9,014                | 6,437                | 192.638                      | 710      | 20,912               | 15,008               | 217.192                      |
| 320      | 9,306                | 6,645                | 193.562                      | 720      | 21,220               | 15,234               | 217.624                      |
| 330      | 9,597                | 6,853                | 194.459                      | 730      | 21,529               | 15,460               | 218.059                      |
| 340      | 9,888                | 7,061                | 195.328                      | 740      | 21,839               | 15,686               | 218.472                      |
| 350      | 10,180               | 7,270                | 196.173                      | 750      | 22,149               | 15,913               | 218.889                      |
| 360      | 10,471               | 7,478                | 196.995                      | 760      | 22,460               | 16,141               | 219.301                      |
| 370      | 10,763               | 7,687                | 197.794                      | 770      | 22,772               | 16,370               | 219.709                      |
| 380      | 11,055               | 7,895                | 198.572                      | 780      | 23,085               | 16,599               | 220.113                      |
| 390      | 11,347               | 8,104                | 199.331                      | 790      | 23,398               | 16,830               | 220.512                      |
| 400      | 11,640               | 8,314                | 200.071                      | 800      | 23,714               | 17,061               | 220.907                      |
| 410      | 11,932               | 8,523                | 200.794                      | 810      | 24,027               | 17,292               | 221.298                      |
| 420      | 12,225               | 8,733                | 201.499                      | 820      | 24,342               | 17,524               | 221.684                      |
| 430      | 12,518               | 8,943                | 202.189                      | 830      | 24,658               | 17,757               | 222.067                      |
| 440      | 12,811               | 9,153                | 202.863                      | 840      | 24,974               | 17,990               | 222.447                      |
| 450      | 13,105               | 9,363                | 203.523                      | 850      | 25,292               | 18,224               | 222.822                      |
| 460      | 13,399               | 9,574                | 204.170                      | 860      | 25,610               | 18,459               | 223.194                      |
| 470      | 13,693               | 9,786                | 204.803                      | 870      | 25,928               | 18,695               | 223.562                      |
| 480      | 13,988               | 9,997                | 205.424                      | 880      | 26,248               | 18,931               | 223.927                      |
| 490      | 14,285               | 10,210               | 206.033                      | 890      | 26,568               | 19,168               | 224.288                      |
| 500      | 14,581               | 10,423               | 206.630                      | 900      | 26,890               | 19,407               | 224.647                      |
| 510      | 14,876               | 10,635               | 207.216                      | 910      | 27,210               | 19,644               | 225.002                      |
| 520      | 15,172               | 10,848               | 207.792                      | 920      | 27,532               | 19,883               | 225.353                      |
| 530      | 15,469               | 11,062               | 208.358                      | 930      | 27,854               | 20,122               | 225.701                      |
| 540      | 15,766               | 11,277               | 208.914                      | 940      | 28,178               | 20,362               | 226.047                      |
| 550      | 16,064               | 11,492               | 209.461                      | 950      | 28,501               | 20,603               | 226.389                      |
| 560      | 16,363               | 11,707               | 209.999                      | 960      | 28,826               | 20,844               | 226.728                      |
| 570      | 16,662               | 11,923               | 210.528                      | 970      | 29,151               | 21,086               | 227.064                      |
| 580      | 16,962               | 12,139               | 211.049                      | 980      | 29,476               | 21,328               | 227.398                      |
| 590      | 17,262               | 12,356               | 211.562                      | 990      | 29,803               | 21,571               | 227.728                      |

TABLE A-18

Ideal-gas properties of nitrogen,  $N_2$  (Concluded)

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 1000     | 30,129               | 21,815               | 228.057                      | 1760     | 56,227               | 41,594               | 247.396                      |
| 1020     | 30,784               | 22,304               | 228.706                      | 1780     | 56,938               | 42,139               | 247.798                      |
| 1040     | 31,442               | 22,795               | 229.344                      | 1800     | 57,651               | 42,685               | 248.195                      |
| 1060     | 32,101               | 23,288               | 229.973                      | 1820     | 58,363               | 43,231               | 248.589                      |
| 1080     | 32,762               | 23,782               | 230.591                      | 1840     | 59,075               | 43,777               | 248.979                      |
| 1100     | 33,426               | 24,280               | 231.199                      | 1860     | 59,790               | 44,324               | 249.365                      |
| 1120     | 34,092               | 24,780               | 231.799                      | 1880     | 60,504               | 44,873               | 249.748                      |
| 1140     | 34,760               | 25,282               | 232.391                      | 1900     | 61,220               | 45,423               | 250.128                      |
| 1160     | 35,430               | 25,786               | 232.973                      | 1920     | 61,936               | 45,973               | 250.502                      |
| 1180     | 36,104               | 26,291               | 233.549                      | 1940     | 62,654               | 46,524               | 250.874                      |
| 1200     | 36,777               | 26,799               | 234.115                      | 1960     | 63,381               | 47,075               | 251.242                      |
| 1220     | 37,452               | 27,308               | 234.673                      | 1980     | 64,090               | 47,627               | 251.607                      |
| 1240     | 38,129               | 27,819               | 235.223                      | 2000     | 64,810               | 48,181               | 251.969                      |
| 1260     | 38,807               | 28,331               | 235.766                      | 2050     | 66,612               | 49,567               | 252.858                      |
| 1280     | 39,488               | 28,845               | 236.302                      | 2100     | 68,417               | 50,957               | 253.726                      |
| 1300     | 40,170               | 29,361               | 236.831                      | 2150     | 70,226               | 52,351               | 254.578                      |
| 1320     | 40,853               | 29,378               | 237.353                      | 2200     | 72,040               | 53,749               | 255.412                      |
| 1340     | 41,539               | 30,398               | 237.867                      | 2250     | 73,856               | 55,149               | 256.227                      |
| 1360     | 42,227               | 30,919               | 238.376                      | 2300     | 75,676               | 56,553               | 257.027                      |
| 1380     | 42,915               | 31,441               | 238.878                      | 2350     | 77,496               | 57,958               | 257.810                      |
| 1400     | 43,605               | 31,964               | 239.375                      | 2400     | 79,320               | 59,366               | 258.580                      |
| 1420     | 44,295               | 32,489               | 239.865                      | 2450     | 81,149               | 60,779               | 259.332                      |
| 1440     | 44,988               | 33,014               | 240.350                      | 2500     | 82,981               | 62,195               | 260.073                      |
| 1460     | 45,682               | 33,543               | 240.827                      | 2550     | 84,814               | 63,613               | 260.799                      |
| 1480     | 46,377               | 34,071               | 241.301                      | 2600     | 86,650               | 65,033               | 261.512                      |
| 1500     | 47,073               | 34,601               | 241.768                      | 2650     | 88,488               | 66,455               | 262.213                      |
| 1520     | 47,771               | 35,133               | 242.228                      | 2700     | 90,328               | 67,880               | 262.902                      |
| 1540     | 48,470               | 35,665               | 242.685                      | 2750     | 92,171               | 69,306               | 263.577                      |
| 1560     | 49,168               | 36,197               | 243.137                      | 2800     | 94,014               | 70,734               | 264.241                      |
| 1580     | 49,869               | 36,732               | 243.585                      | 2850     | 95,859               | 72,163               | 264.895                      |
| 1600     | 50,571               | 37,268               | 244.028                      | 2900     | 97,705               | 73,593               | 265.538                      |
| 1620     | 51,275               | 37,806               | 244.464                      | 2950     | 99,556               | 75,028               | 266.170                      |
| 1640     | 51,980               | 38,344               | 244.896                      | 3000     | 101,407              | 76,464               | 266.793                      |
| 1660     | 52,686               | 38,884               | 245.324                      | 3050     | 103,260              | 77,902               | 267.404                      |
| 1680     | 53,393               | 39,424               | 245.747                      | 3100     | 105,115              | 79,341               | 268.007                      |
| 1700     | 54,099               | 39,965               | 246.166                      | 3150     | 106,972              | 80,782               | 268.601                      |
| 1720     | 54,807               | 40,507               | 246.580                      | 3200     | 108,830              | 82,224               | 269.186                      |
| 1740     | 55,516               | 41,049               | 246.990                      | 3250     | 110,690              | 83,668               | 269.763                      |

Source of Data: Tables A-18 through A-25 are adapted from Kenneth Wark, *Thermodynamics*, 4th ed. (New York: McGraw-Hill, 1983), pp. 787-98. Originally published in JANAF, *Thermochemical Tables*, NSRDS-NBS-37, 1971.



TABLE A-19

Ideal-gas properties of oxygen,  $O_2$ 

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 600      | 17,929               | 12,940               | 226.346                      |
| 220      | 6,404                | 4,575                | 196.171                      | 610      | 18,250               | 13,178               | 226.877                      |
| 230      | 6,694                | 4,782                | 197.461                      | 620      | 18,572               | 13,417               | 227.400                      |
| 240      | 6,984                | 4,989                | 198.696                      | 630      | 18,895               | 13,657               | 227.918                      |
| 250      | 7,275                | 5,197                | 199.885                      | 640      | 19,219               | 13,898               | 228.429                      |
| 260      | 7,566                | 5,405                | 201.027                      | 650      | 19,544               | 14,140               | 228.932                      |
| 270      | 7,858                | 5,613                | 202.128                      | 660      | 19,870               | 14,383               | 229.430                      |
| 280      | 8,150                | 5,822                | 203.191                      | 670      | 20,197               | 14,626               | 229.920                      |
| 290      | 8,443                | 6,032                | 204.218                      | 680      | 20,524               | 14,871               | 230.405                      |
| 298      | 8,682                | 6,203                | 205.033                      | 690      | 20,854               | 15,116               | 230.885                      |
| 300      | 8,736                | 6,242                | 205.213                      | 700      | 21,184               | 15,364               | 231.358                      |
| 310      | 9,030                | 6,453                | 206.177                      | 710      | 21,514               | 15,611               | 231.827                      |
| 320      | 9,325                | 6,664                | 207.112                      | 720      | 21,845               | 15,859               | 232.291                      |
| 330      | 9,620                | 6,877                | 208.020                      | 730      | 22,177               | 16,107               | 232.748                      |
| 340      | 9,916                | 7,090                | 208.904                      | 740      | 22,510               | 16,357               | 233.201                      |
| 350      | 10,213               | 7,303                | 209.765                      | 750      | 22,844               | 16,607               | 233.649                      |
| 360      | 10,511               | 7,518                | 210.604                      | 760      | 23,178               | 16,859               | 234.091                      |
| 370      | 10,809               | 7,733                | 211.423                      | 770      | 23,513               | 17,111               | 234.528                      |
| 380      | 11,109               | 7,949                | 212.222                      | 780      | 23,850               | 17,364               | 234.960                      |
| 390      | 11,409               | 8,166                | 213.002                      | 790      | 24,186               | 17,618               | 235.387                      |
| 400      | 11,711               | 8,384                | 213.765                      | 800      | 24,523               | 17,872               | 235.810                      |
| 410      | 12,012               | 8,603                | 214.510                      | 810      | 24,861               | 18,126               | 236.230                      |
| 420      | 12,314               | 8,822                | 215.241                      | 820      | 25,199               | 18,382               | 236.644                      |
| 430      | 12,618               | 9,043                | 215.955                      | 830      | 25,537               | 18,637               | 237.055                      |
| 440      | 12,923               | 9,264                | 216.656                      | 840      | 25,877               | 18,893               | 237.462                      |
| 450      | 13,228               | 9,487                | 217.342                      | 850      | 26,218               | 19,150               | 237.864                      |
| 460      | 13,525               | 9,710                | 218.016                      | 860      | 26,559               | 19,408               | 238.264                      |
| 470      | 13,842               | 9,935                | 218.676                      | 870      | 26,899               | 19,666               | 238.660                      |
| 480      | 14,151               | 10,160               | 219.326                      | 880      | 27,242               | 19,925               | 239.051                      |
| 490      | 14,460               | 10,386               | 219.963                      | 890      | 27,584               | 20,185               | 239.439                      |
| 500      | 14,770               | 10,614               | 220.589                      | 900      | 27,928               | 20,445               | 239.823                      |
| 510      | 15,082               | 10,842               | 221.206                      | 910      | 28,272               | 20,706               | 240.203                      |
| 520      | 15,395               | 11,071               | 221.812                      | 920      | 28,616               | 20,967               | 240.580                      |
| 530      | 15,708               | 11,301               | 222.409                      | 930      | 28,960               | 21,228               | 240.953                      |
| 540      | 16,022               | 11,533               | 222.997                      | 940      | 29,306               | 21,491               | 241.323                      |
| 550      | 16,338               | 11,765               | 223.576                      | 950      | 29,652               | 21,754               | 241.689                      |
| 560      | 16,654               | 11,998               | 224.146                      | 960      | 29,999               | 22,017               | 242.052                      |
| 570      | 16,971               | 12,232               | 224.708                      | 970      | 30,345               | 22,280               | 242.411                      |
| 580      | 17,290               | 12,467               | 225.262                      | 980      | 30,692               | 22,544               | 242.768                      |
| 590      | 17,609               | 12,703               | 225.808                      | 990      | 31,041               | 22,809               | 242.120                      |

TABLE A-19

Ideal-gas properties of oxygen,  $O_2$  (Concluded)

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 1000     | 31,389               | 23,075               | 243.471                      | 1760     | 58,880               | 44,247               | 263.861                      |
| 1020     | 32,088               | 23,607               | 244.164                      | 1780     | 59,624               | 44,825               | 264.283                      |
| 1040     | 32,789               | 24,142               | 244.844                      | 1800     | 60,371               | 45,405               | 264.701                      |
| 1060     | 33,490               | 24,677               | 245.513                      | 1820     | 61,118               | 45,986               | 265.113                      |
| 1080     | 34,194               | 25,214               | 246.171                      | 1840     | 61,866               | 46,568               | 265.521                      |
| 1100     | 34,899               | 25,753               | 246.818                      | 1860     | 62,616               | 47,151               | 265.925                      |
| 1120     | 35,606               | 26,294               | 247.454                      | 1880     | 63,365               | 47,734               | 266.326                      |
| 1140     | 36,314               | 26,836               | 248.081                      | 1900     | 64,116               | 48,319               | 266.722                      |
| 1160     | 37,023               | 27,379               | 248.698                      | 1920     | 64,868               | 48,904               | 267.115                      |
| 1180     | 37,734               | 27,923               | 249.307                      | 1940     | 65,620               | 49,490               | 267.505                      |
| 1200     | 38,447               | 28,469               | 249.906                      | 1960     | 66,374               | 50,078               | 267.891                      |
| 1220     | 39,162               | 29,018               | 250.497                      | 1980     | 67,127               | 50,665               | 268.275                      |
| 1240     | 39,877               | 29,568               | 251.079                      | 2000     | 67,881               | 51,253               | 268.655                      |
| 1260     | 40,594               | 30,118               | 251.653                      | 2050     | 69,772               | 52,727               | 269.588                      |
| 1280     | 41,312               | 30,670               | 252.219                      | 2100     | 71,668               | 54,208               | 270.504                      |
| 1300     | 42,033               | 31,224               | 252.776                      | 2150     | 73,573               | 55,697               | 271.399                      |
| 1320     | 42,753               | 31,778               | 253.325                      | 2200     | 75,484               | 57,192               | 272.278                      |
| 1340     | 43,475               | 32,334               | 253.868                      | 2250     | 77,397               | 58,690               | 273.136                      |
| 1360     | 44,198               | 32,891               | 254.404                      | 2300     | 79,316               | 60,193               | 273.891                      |
| 1380     | 44,923               | 33,449               | 254.932                      | 2350     | 81,243               | 61,704               | 274.809                      |
| 1400     | 45,648               | 34,008               | 255.454                      | 2400     | 83,174               | 63,219               | 275.625                      |
| 1420     | 46,374               | 34,567               | 255.968                      | 2450     | 85,112               | 64,742               | 276.424                      |
| 1440     | 47,102               | 35,129               | 256.475                      | 2500     | 87,057               | 66,271               | 277.207                      |
| 1460     | 47,831               | 35,692               | 256.978                      | 2550     | 89,004               | 67,802               | 277.979                      |
| 1480     | 48,561               | 36,256               | 257.474                      | 2600     | 90,956               | 69,339               | 278.738                      |
| 1500     | 49,292               | 36,821               | 257.965                      | 2650     | 92,916               | 70,883               | 279.485                      |
| 1520     | 50,024               | 37,387               | 258.450                      | 2700     | 94,881               | 72,433               | 280.219                      |
| 1540     | 50,756               | 37,952               | 258.928                      | 2750     | 96,852               | 73,987               | 280.942                      |
| 1560     | 51,490               | 38,520               | 259.402                      | 2800     | 98,826               | 75,546               | 281.654                      |
| 1580     | 52,224               | 39,088               | 259.870                      | 2850     | 100,808              | 77,112               | 282.357                      |
| 1600     | 52,961               | 39,658               | 260.333                      | 2900     | 102,793              | 78,682               | 283.048                      |
| 1620     | 53,696               | 40,227               | 260.791                      | 2950     | 104,785              | 80,258               | 283.728                      |
| 1640     | 54,434               | 40,799               | 261.242                      | 3000     | 106,780              | 81,837               | 284.399                      |
| 1660     | 55,172               | 41,370               | 261.690                      | 3050     | 108,778              | 83,419               | 285.060                      |
| 1680     | 55,912               | 41,944               | 262.132                      | 3100     | 110,784              | 85,009               | 285.713                      |
| 1700     | 56,652               | 42,517               | 262.571                      | 3150     | 112,795              | 86,601               | 286.355                      |
| 1720     | 57,394               | 43,093               | 263.005                      | 3200     | 114,809              | 88,203               | 286.989                      |
| 1740     | 58,136               | 43,669               | 263.435                      | 3250     | 116,827              | 89,804               | 287.614                      |

TABLE A-20

Ideal-gas properties of carbon dioxide, CO<sub>2</sub>

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 600      | 22,280               | 17,291               | 243.199                      |
| 220      | 6,601                | 4,772                | 202.966                      | 610      | 22,754               | 17,683               | 243.983                      |
| 230      | 6,938                | 5,026                | 204.464                      | 620      | 23,231               | 18,076               | 244.758                      |
| 240      | 7,280                | 5,285                | 205.920                      | 630      | 23,709               | 18,471               | 245.524                      |
| 250      | 7,627                | 5,548                | 207.337                      | 640      | 24,190               | 18,869               | 246.282                      |
| 260      | 7,979                | 5,817                | 208.717                      | 650      | 24,674               | 19,270               | 247.032                      |
| 270      | 8,335                | 6,091                | 210.062                      | 660      | 25,160               | 19,672               | 247.773                      |
| 280      | 8,697                | 6,369                | 211.376                      | 670      | 25,648               | 20,078               | 248.507                      |
| 290      | 9,063                | 6,651                | 212.660                      | 680      | 26,138               | 20,484               | 249.233                      |
| 298      | 9,364                | 6,885                | 213.685                      | 690      | 26,631               | 20,894               | 249.952                      |
| 300      | 9,431                | 6,939                | 213.915                      | 700      | 27,125               | 21,305               | 250.663                      |
| 310      | 9,807                | 7,230                | 215.146                      | 710      | 27,622               | 21,719               | 251.368                      |
| 320      | 10,186               | 7,526                | 216.351                      | 720      | 28,121               | 22,134               | 252.065                      |
| 330      | 10,570               | 7,826                | 217.534                      | 730      | 28,622               | 22,522               | 252.755                      |
| 340      | 10,959               | 8,131                | 218.694                      | 740      | 29,124               | 22,972               | 253.439                      |
| 350      | 11,351               | 8,439                | 219.831                      | 750      | 29,629               | 23,393               | 254.117                      |
| 360      | 11,748               | 8,752                | 220.948                      | 760      | 30,135               | 23,817               | 254.787                      |
| 370      | 12,148               | 9,068                | 222.044                      | 770      | 30,644               | 24,242               | 255.452                      |
| 380      | 12,552               | 9,392                | 223.122                      | 780      | 31,154               | 24,669               | 256.110                      |
| 390      | 12,960               | 9,718                | 224.182                      | 790      | 31,665               | 25,097               | 256.762                      |
| 400      | 13,372               | 10,046               | 225.225                      | 800      | 32,179               | 25,527               | 257.408                      |
| 410      | 13,787               | 10,378               | 226.250                      | 810      | 32,694               | 25,959               | 258.048                      |
| 420      | 14,206               | 10,714               | 227.258                      | 820      | 33,212               | 26,394               | 258.682                      |
| 430      | 14,628               | 11,053               | 228.252                      | 830      | 33,730               | 26,829               | 259.311                      |
| 440      | 15,054               | 11,393               | 229.230                      | 840      | 34,251               | 27,267               | 259.934                      |
| 450      | 15,483               | 11,742               | 230.194                      | 850      | 34,773               | 27,706               | 260.551                      |
| 460      | 15,916               | 12,091               | 231.144                      | 860      | 35,296               | 28,125               | 261.164                      |
| 470      | 16,351               | 12,444               | 232.080                      | 870      | 35,821               | 28,588               | 261.770                      |
| 480      | 16,791               | 12,800               | 233.004                      | 880      | 36,347               | 29,031               | 262.371                      |
| 490      | 17,232               | 13,158               | 233.916                      | 890      | 36,876               | 29,476               | 262.968                      |
| 500      | 17,678               | 13,521               | 234.814                      | 900      | 37,405               | 29,922               | 263.559                      |
| 510      | 18,126               | 13,885               | 235.700                      | 910      | 37,935               | 30,369               | 264.146                      |
| 520      | 18,576               | 14,253               | 236.575                      | 920      | 38,467               | 30,818               | 264.728                      |
| 530      | 19,029               | 14,622               | 237.439                      | 930      | 39,000               | 31,268               | 265.304                      |
| 540      | 19,485               | 14,996               | 238.292                      | 940      | 39,535               | 31,719               | 265.877                      |
| 550      | 19,945               | 15,372               | 239.135                      | 950      | 40,070               | 32,171               | 266.444                      |
| 560      | 20,407               | 15,751               | 239.962                      | 960      | 40,607               | 32,625               | 267.007                      |
| 570      | 20,870               | 16,131               | 240.789                      | 970      | 41,145               | 33,081               | 267.566                      |
| 580      | 21,337               | 16,515               | 241.602                      | 980      | 41,685               | 33,537               | 268.119                      |
| 590      | 21,807               | 16,902               | 242.405                      | 990      | 42,226               | 33,995               | 268.670                      |

TABLE A-20

Ideal-gas properties of carbon dioxide, CO<sub>2</sub> (Concluded)

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 1000     | 42,769               | 34,455               | 269.215                      | 1760     | 86,420               | 71,787               | 301.543                      |
| 1020     | 43,859               | 35,378               | 270.293                      | 1780     | 87,612               | 72,812               | 302.217                      |
| 1040     | 44,953               | 36,306               | 271.354                      | 1800     | 88,806               | 73,840               | 302.884                      |
| 1060     | 46,051               | 37,238               | 272.400                      | 1820     | 90,000               | 74,868               | 303.544                      |
| 1080     | 47,153               | 38,174               | 273.430                      | 1840     | 91,196               | 75,897               | 304.198                      |
| 1100     | 48,258               | 39,112               | 274.445                      | 1860     | 92,394               | 76,929               | 304.845                      |
| 1120     | 49,369               | 40,057               | 275.444                      | 1880     | 93,593               | 77,962               | 305.487                      |
| 1140     | 50,484               | 41,006               | 276.430                      | 1900     | 94,793               | 78,996               | 306.122                      |
| 1160     | 51,602               | 41,957               | 277.403                      | 1920     | 95,995               | 80,031               | 306.751                      |
| 1180     | 52,724               | 42,913               | 278.361                      | 1940     | 97,197               | 81,067               | 307.374                      |
| 1200     | 53,848               | 43,871               | 297.307                      | 1960     | 98,401               | 82,105               | 307.992                      |
| 1220     | 54,977               | 44,834               | 280.238                      | 1980     | 99,606               | 83,144               | 308.604                      |
| 1240     | 56,108               | 45,799               | 281.158                      | 2000     | 100,804              | 84,185               | 309.210                      |
| 1260     | 57,244               | 46,768               | 282.066                      | 2050     | 103,835              | 86,791               | 310.701                      |
| 1280     | 58,381               | 47,739               | 282.962                      | 2100     | 106,864              | 89,404               | 312.160                      |
| 1300     | 59,522               | 48,713               | 283.847                      | 2150     | 109,898              | 92,023               | 313.589                      |
| 1320     | 60,666               | 49,691               | 284.722                      | 2200     | 112,939              | 94,648               | 314.988                      |
| 1340     | 61,813               | 50,672               | 285.586                      | 2250     | 115,984              | 97,277               | 316.356                      |
| 1360     | 62,963               | 51,656               | 286.439                      | 2300     | 119,035              | 99,912               | 317.695                      |
| 1380     | 64,116               | 52,643               | 287.283                      | 2350     | 122,091              | 102,552              | 319.011                      |
| 1400     | 65,271               | 53,631               | 288.106                      | 2400     | 125,152              | 105,197              | 320.302                      |
| 1420     | 66,427               | 54,621               | 288.934                      | 2450     | 128,219              | 107,849              | 321.566                      |
| 1440     | 67,586               | 55,614               | 289.743                      | 2500     | 131,290              | 110,504              | 322.808                      |
| 1460     | 68,748               | 56,609               | 290.542                      | 2550     | 134,368              | 113,166              | 324.026                      |
| 1480     | 66,911               | 57,606               | 291.333                      | 2600     | 137,449              | 115,832              | 325.222                      |
| 1500     | 71,078               | 58,606               | 292.114                      | 2650     | 140,533              | 118,500              | 326.396                      |
| 1520     | 72,246               | 59,609               | 292.888                      | 2700     | 143,620              | 121,172              | 327.549                      |
| 1540     | 73,417               | 60,613               | 292.654                      | 2750     | 146,713              | 123,849              | 328.684                      |
| 1560     | 74,590               | 61,620               | 294.411                      | 2800     | 149,808              | 126,528              | 329.800                      |
| 1580     | 76,767               | 62,630               | 295.161                      | 2850     | 152,908              | 129,212              | 330.896                      |
| 1600     | 76,944               | 63,741               | 295.901                      | 2900     | 156,009              | 131,898              | 331.975                      |
| 1620     | 78,123               | 64,653               | 296.632                      | 2950     | 159,117              | 134,589              | 333.037                      |
| 1640     | 79,303               | 65,668               | 297.356                      | 3000     | 162,226              | 137,283              | 334.084                      |
| 1660     | 80,486               | 66,592               | 298.072                      | 3050     | 165,341              | 139,982              | 335.114                      |
| 1680     | 81,670               | 67,702               | 298.781                      | 3100     | 168,456              | 142,681              | 336.126                      |
| 1700     | 82,856               | 68,721               | 299.482                      | 3150     | 171,576              | 145,385              | 337.124                      |
| 1720     | 84,043               | 69,742               | 300.177                      | 3200     | 174,695              | 148,089              | 338.109                      |
| 1740     | 85,231               | 70,764               | 300.863                      | 3250     | 177,822              | 150,801              | 339.069                      |

TABLE A-21

Ideal-gas properties of carbon monoxide, CO

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 600      | 17,611               | 12,622               | 218.204                      |
| 220      | 6,391                | 4,562                | 188.683                      | 610      | 17,915               | 12,843               | 218.708                      |
| 230      | 6,683                | 4,771                | 189.980                      | 620      | 18,221               | 13,066               | 219.205                      |
| 240      | 6,975                | 4,979                | 191.221                      | 630      | 18,527               | 13,289               | 219.695                      |
| 250      | 7,266                | 5,188                | 192.411                      | 640      | 18,833               | 13,512               | 220.179                      |
| 260      | 7,558                | 5,396                | 193.554                      | 650      | 19,141               | 13,736               | 220.656                      |
| 270      | 7,849                | 5,604                | 194.654                      | 660      | 19,449               | 13,962               | 221.127                      |
| 280      | 8,140                | 5,812                | 195.713                      | 670      | 19,758               | 14,187               | 221.592                      |
| 290      | 8,432                | 6,020                | 196.735                      | 680      | 20,068               | 14,414               | 222.052                      |
| 298      | 8,669                | 6,190                | 197.543                      | 690      | 20,378               | 14,641               | 222.505                      |
| 300      | 8,723                | 6,229                | 197.723                      | 700      | 20,690               | 14,870               | 222.953                      |
| 310      | 9,014                | 6,437                | 198.678                      | 710      | 21,002               | 15,099               | 223.396                      |
| 320      | 9,306                | 6,645                | 199.603                      | 720      | 21,315               | 15,328               | 223.833                      |
| 330      | 9,597                | 6,854                | 200.500                      | 730      | 21,628               | 15,558               | 224.265                      |
| 340      | 9,889                | 7,062                | 201.371                      | 740      | 21,943               | 15,789               | 224.692                      |
| 350      | 10,181               | 7,271                | 202.217                      | 750      | 22,258               | 16,022               | 225.115                      |
| 360      | 10,473               | 7,480                | 203.040                      | 760      | 22,573               | 16,255               | 225.533                      |
| 370      | 10,765               | 7,689                | 203.842                      | 770      | 22,890               | 16,488               | 225.947                      |
| 380      | 11,058               | 7,899                | 204.622                      | 780      | 23,208               | 16,723               | 226.357                      |
| 390      | 11,351               | 8,108                | 205.383                      | 790      | 23,526               | 16,957               | 226.762                      |
| 400      | 11,644               | 8,319                | 206.125                      | 800      | 23,844               | 17,193               | 227.162                      |
| 410      | 11,938               | 8,529                | 206.850                      | 810      | 24,164               | 17,429               | 227.559                      |
| 420      | 12,232               | 8,740                | 207.549                      | 820      | 24,483               | 17,665               | 227.952                      |
| 430      | 12,526               | 8,951                | 208.252                      | 830      | 24,803               | 17,902               | 228.339                      |
| 440      | 12,821               | 9,163                | 208.929                      | 840      | 25,124               | 18,140               | 228.724                      |
| 450      | 13,116               | 9,375                | 209.593                      | 850      | 25,446               | 18,379               | 229.106                      |
| 460      | 13,412               | 9,587                | 210.243                      | 860      | 25,768               | 18,617               | 229.482                      |
| 470      | 13,708               | 9,800                | 210.880                      | 870      | 26,091               | 18,858               | 229.856                      |
| 480      | 14,005               | 10,014               | 211.504                      | 880      | 26,415               | 19,099               | 230.227                      |
| 490      | 14,302               | 10,228               | 212.117                      | 890      | 26,740               | 19,341               | 230.593                      |
| 500      | 14,600               | 10,443               | 212.719                      | 900      | 27,066               | 19,583               | 230.957                      |
| 510      | 14,898               | 10,658               | 213.310                      | 910      | 27,392               | 19,826               | 231.317                      |
| 520      | 15,197               | 10,874               | 213.890                      | 920      | 27,719               | 20,070               | 231.674                      |
| 530      | 15,497               | 11,090               | 214.460                      | 930      | 28,046               | 20,314               | 232.028                      |
| 540      | 15,797               | 11,307               | 215.020                      | 940      | 28,375               | 20,559               | 232.379                      |
| 550      | 16,097               | 11,524               | 215.572                      | 950      | 28,703               | 20,805               | 232.727                      |
| 560      | 16,399               | 11,743               | 216.115                      | 960      | 29,033               | 21,051               | 233.072                      |
| 570      | 16,701               | 11,961               | 216.649                      | 970      | 29,362               | 21,298               | 233.413                      |
| 580      | 17,003               | 12,181               | 217.175                      | 980      | 29,693               | 21,545               | 233.752                      |
| 590      | 17,307               | 12,401               | 217.693                      | 990      | 30,024               | 21,793               | 234.088                      |

TABLE A-21

Ideal-gas properties of carbon monoxide, CO (*Concluded*)

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 1000     | 30,355               | 22,041               | 234.421                      | 1760     | 56,756               | 42,123               | 253.991                      |
| 1020     | 31,020               | 22,540               | 235.079                      | 1780     | 57,473               | 42,673               | 254.398                      |
| 1040     | 31,688               | 23,041               | 235.728                      | 1800     | 58,191               | 43,225               | 254.797                      |
| 1060     | 32,357               | 23,544               | 236.364                      | 1820     | 58,910               | 43,778               | 255.194                      |
| 1080     | 33,029               | 24,049               | 236.992                      | 1840     | 59,629               | 44,331               | 255.587                      |
| 1100     | 33,702               | 24,557               | 237.609                      | 1860     | 60,351               | 44,886               | 255.976                      |
| 1120     | 34,377               | 25,065               | 238.217                      | 1880     | 61,072               | 45,441               | 256.361                      |
| 1140     | 35,054               | 25,575               | 238.817                      | 1900     | 61,794               | 45,997               | 256.743                      |
| 1160     | 35,733               | 26,088               | 239.407                      | 1920     | 62,516               | 46,552               | 257.122                      |
| 1180     | 36,406               | 26,602               | 239.989                      | 1940     | 63,238               | 47,108               | 257.497                      |
| 1200     | 37,095               | 27,118               | 240.663                      | 1960     | 63,961               | 47,665               | 257.868                      |
| 1220     | 37,780               | 27,637               | 241.128                      | 1980     | 64,684               | 48,221               | 258.236                      |
| 1240     | 38,466               | 28,426               | 241.686                      | 2000     | 65,408               | 48,780               | 258.600                      |
| 1260     | 39,154               | 28,678               | 242.236                      | 2050     | 67,224               | 50,179               | 259.494                      |
| 1280     | 39,844               | 29,201               | 242.780                      | 2100     | 69,044               | 51,584               | 260.370                      |
| 1300     | 40,534               | 29,725               | 243.316                      | 2150     | 70,864               | 52,988               | 261.226                      |
| 1320     | 41,226               | 30,251               | 243.844                      | 2200     | 72,688               | 54,396               | 262.065                      |
| 1340     | 41,919               | 30,778               | 244.366                      | 2250     | 74,516               | 55,809               | 262.887                      |
| 1360     | 42,613               | 31,306               | 244.880                      | 2300     | 76,345               | 57,222               | 263.692                      |
| 1380     | 43,309               | 31,836               | 245.388                      | 2350     | 78,178               | 58,640               | 264.480                      |
| 1400     | 44,007               | 32,367               | 245.889                      | 2400     | 80,015               | 60,060               | 265.253                      |
| 1420     | 44,707               | 32,900               | 246.385                      | 2450     | 81,852               | 61,482               | 266.012                      |
| 1440     | 45,408               | 33,434               | 246.876                      | 2500     | 83,692               | 62,906               | 266.755                      |
| 1460     | 46,110               | 33,971               | 247.360                      | 2550     | 85,537               | 64,335               | 267.485                      |
| 1480     | 46,813               | 34,508               | 247.839                      | 2600     | 87,383               | 65,766               | 268.202                      |
| 1500     | 47,517               | 35,046               | 248.312                      | 2650     | 89,230               | 67,197               | 268.905                      |
| 1520     | 48,222               | 35,584               | 248.778                      | 2700     | 91,077               | 68,628               | 269.596                      |
| 1540     | 48,928               | 36,124               | 249.240                      | 2750     | 92,930               | 70,066               | 270.285                      |
| 1560     | 49,635               | 36,665               | 249.695                      | 2800     | 94,784               | 71,504               | 270.943                      |
| 1580     | 50,344               | 37,207               | 250.147                      | 2850     | 96,639               | 72,945               | 271.602                      |
| 1600     | 51,053               | 37,750               | 250.592                      | 2900     | 98,495               | 74,383               | 272.249                      |
| 1620     | 51,763               | 38,293               | 251.033                      | 2950     | 100,352              | 75,825               | 272.884                      |
| 1640     | 52,472               | 38,837               | 251.470                      | 3000     | 102,210              | 77,267               | 273.508                      |
| 1660     | 53,184               | 39,382               | 251.901                      | 3050     | 104,073              | 78,715               | 274.123                      |
| 1680     | 53,895               | 39,927               | 252.329                      | 3100     | 105,939              | 80,164               | 274.730                      |
| 1700     | 54,609               | 40,474               | 252.751                      | 3150     | 107,802              | 81,612               | 275.326                      |
| 1720     | 55,323               | 41,023               | 253.169                      | 3200     | 109,667              | 83,061               | 275.914                      |
| 1740     | 56,039               | 41,572               | 253.582                      | 3250     | 111,534              | 84,513               | 276.494                      |

TABLE A-22

Ideal-gas properties of hydrogen,  $H_2$ 

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 1440     | 42,808               | 30,835               | 177.410                      |
| 260      | 7,370                | 5,209                | 126.636                      | 1480     | 44,091               | 31,786               | 178.291                      |
| 270      | 7,657                | 5,412                | 127.719                      | 1520     | 45,384               | 32,746               | 179.153                      |
| 280      | 7,945                | 5,617                | 128.765                      | 1560     | 46,683               | 33,713               | 179.995                      |
| 290      | 8,233                | 5,822                | 129.775                      | 1600     | 47,990               | 34,687               | 180.820                      |
| 298      | 8,468                | 5,989                | 130.574                      | 1640     | 49,303               | 35,668               | 181.632                      |
| 300      | 8,522                | 6,027                | 130.754                      | 1680     | 50,622               | 36,654               | 182.428                      |
| 320      | 9,100                | 6,440                | 132.621                      | 1720     | 51,947               | 37,646               | 183.208                      |
| 340      | 9,680                | 6,853                | 134.378                      | 1760     | 53,279               | 38,645               | 183.973                      |
| 360      | 10,262               | 7,268                | 136.039                      | 1800     | 54,618               | 39,652               | 184.724                      |
| 380      | 10,843               | 7,684                | 137.612                      | 1840     | 55,962               | 40,663               | 185.463                      |
| 400      | 11,426               | 8,100                | 139.106                      | 1880     | 57,311               | 41,680               | 186.190                      |
| 420      | 12,010               | 8,518                | 140.529                      | 1920     | 58,668               | 42,705               | 186.904                      |
| 440      | 12,594               | 8,936                | 141.888                      | 1960     | 60,031               | 43,735               | 187.607                      |
| 460      | 13,179               | 9,355                | 143.187                      | 2000     | 61,400               | 44,771               | 188.297                      |
| 480      | 13,764               | 9,773                | 144.432                      | 2050     | 63,119               | 46,074               | 189.148                      |
| 500      | 14,350               | 10,193               | 145.628                      | 2100     | 64,847               | 47,386               | 189.979                      |
| 520      | 14,935               | 10,611               | 146.775                      | 2150     | 66,584               | 48,708               | 190.796                      |
| 560      | 16,107               | 11,451               | 148.945                      | 2200     | 68,328               | 50,037               | 191.598                      |
| 600      | 17,280               | 12,291               | 150.968                      | 2250     | 70,080               | 51,373               | 192.385                      |
| 640      | 18,453               | 13,133               | 152.863                      | 2300     | 71,839               | 52,716               | 193.159                      |
| 680      | 19,630               | 13,976               | 154.645                      | 2350     | 73,608               | 54,069               | 193.921                      |
| 720      | 20,807               | 14,821               | 156.328                      | 2400     | 75,383               | 55,429               | 194.669                      |
| 760      | 21,988               | 15,669               | 157.923                      | 2450     | 77,168               | 56,798               | 195.403                      |
| 800      | 23,171               | 16,520               | 159.440                      | 2500     | 78,960               | 58,175               | 196.125                      |
| 840      | 24,359               | 17,375               | 160.891                      | 2550     | 80,755               | 59,554               | 196.837                      |
| 880      | 25,551               | 18,235               | 162.277                      | 2600     | 82,558               | 60,941               | 197.539                      |
| 920      | 26,747               | 19,098               | 163.607                      | 2650     | 84,368               | 62,335               | 198.229                      |
| 960      | 27,948               | 19,966               | 164.884                      | 2700     | 86,186               | 63,737               | 198.907                      |
| 1000     | 29,154               | 20,839               | 166.114                      | 2750     | 88,008               | 65,144               | 199.575                      |
| 1040     | 30,364               | 21,717               | 167.300                      | 2800     | 89,838               | 66,558               | 200.234                      |
| 1080     | 31,580               | 22,601               | 168.449                      | 2850     | 91,671               | 67,976               | 200.885                      |
| 1120     | 32,802               | 23,490               | 169.560                      | 2900     | 93,512               | 69,401               | 201.527                      |
| 1160     | 34,028               | 24,384               | 170.636                      | 2950     | 95,358               | 70,831               | 202.157                      |
| 1200     | 35,262               | 25,284               | 171.682                      | 3000     | 97,211               | 72,268               | 202.778                      |
| 1240     | 36,502               | 26,192               | 172.698                      | 3050     | 99,065               | 73,707               | 203.391                      |
| 1280     | 37,749               | 27,106               | 173.687                      | 3100     | 100,926              | 75,152               | 203.995                      |
| 1320     | 39,002               | 28,027               | 174.652                      | 3150     | 102,793              | 76,604               | 204.592                      |
| 1360     | 40,263               | 28,955               | 175.593                      | 3200     | 104,667              | 78,061               | 205.181                      |
| 1400     | 41,530               | 29,889               | 176.510                      | 3250     | 106,545              | 79,523               | 205.765                      |



TABLE A-23

Ideal-gas properties of water vapor,  $H_2O$ 

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 600      | 20,402               | 15,413               | 212.920                      |
| 220      | 7,295                | 5,466                | 178.576                      | 610      | 20,765               | 15,693               | 213.529                      |
| 230      | 7,628                | 5,715                | 180.054                      | 620      | 21,130               | 15,975               | 214.122                      |
| 240      | 7,961                | 5,965                | 181.471                      | 630      | 21,495               | 16,257               | 214.707                      |
| 250      | 8,294                | 6,215                | 182.831                      | 640      | 21,862               | 16,541               | 215.285                      |
| 260      | 8,627                | 6,466                | 184.139                      | 650      | 22,230               | 16,826               | 215.856                      |
| 270      | 8,961                | 6,716                | 185.399                      | 660      | 22,600               | 17,112               | 216.419                      |
| 280      | 9,296                | 6,968                | 186.616                      | 670      | 22,970               | 17,399               | 216.976                      |
| 290      | 9,631                | 7,219                | 187.791                      | 680      | 23,342               | 17,688               | 217.527                      |
| 298      | 9,904                | 7,425                | 188.720                      | 690      | 23,714               | 17,978               | 218.071                      |
| 300      | 9,966                | 7,472                | 188.928                      | 700      | 24,088               | 18,268               | 218.610                      |
| 310      | 10,302               | 7,725                | 190.030                      | 710      | 24,464               | 18,561               | 219.142                      |
| 320      | 10,639               | 7,978                | 191.098                      | 720      | 24,840               | 18,854               | 219.668                      |
| 330      | 10,976               | 8,232                | 192.136                      | 730      | 25,218               | 19,148               | 220.189                      |
| 340      | 11,314               | 8,487                | 193.144                      | 740      | 25,597               | 19,444               | 220.707                      |
| 350      | 11,652               | 8,742                | 194.125                      | 750      | 25,977               | 19,741               | 221.215                      |
| 360      | 11,992               | 8,998                | 195.081                      | 760      | 26,358               | 20,039               | 221.720                      |
| 370      | 12,331               | 9,255                | 196.012                      | 770      | 26,741               | 20,339               | 222.221                      |
| 380      | 12,672               | 9,513                | 196.920                      | 780      | 27,125               | 20,639               | 222.717                      |
| 390      | 13,014               | 9,771                | 197.807                      | 790      | 27,510               | 20,941               | 223.207                      |
| 400      | 13,356               | 10,030               | 198.673                      | 800      | 27,896               | 21,245               | 223.693                      |
| 410      | 13,699               | 10,290               | 199.521                      | 810      | 28,284               | 21,549               | 224.174                      |
| 420      | 14,043               | 10,551               | 200.350                      | 820      | 28,672               | 21,855               | 224.651                      |
| 430      | 14,388               | 10,813               | 201.160                      | 830      | 29,062               | 22,162               | 225.123                      |
| 440      | 14,734               | 11,075               | 201.955                      | 840      | 29,454               | 22,470               | 225.592                      |
| 450      | 15,080               | 11,339               | 202.734                      | 850      | 29,846               | 22,779               | 226.057                      |
| 460      | 15,428               | 11,603               | 203.497                      | 860      | 30,240               | 23,090               | 226.517                      |
| 470      | 15,777               | 11,869               | 204.247                      | 870      | 30,635               | 23,402               | 226.973                      |
| 480      | 16,126               | 12,135               | 204.982                      | 880      | 31,032               | 23,715               | 227.426                      |
| 490      | 16,477               | 12,403               | 205.705                      | 890      | 31,429               | 24,029               | 227.875                      |
| 500      | 16,828               | 12,671               | 206.413                      | 900      | 31,828               | 24,345               | 228.321                      |
| 510      | 17,181               | 12,940               | 207.112                      | 910      | 32,228               | 24,662               | 228.763                      |
| 520      | 17,534               | 13,211               | 207.799                      | 920      | 32,629               | 24,980               | 229.202                      |
| 530      | 17,889               | 13,482               | 208.475                      | 930      | 33,032               | 25,300               | 229.637                      |
| 540      | 18,245               | 13,755               | 209.139                      | 940      | 33,436               | 25,621               | 230.070                      |
| 550      | 18,601               | 14,028               | 209.795                      | 950      | 33,841               | 25,943               | 230.499                      |
| 560      | 18,959               | 14,303               | 210.440                      | 960      | 34,247               | 26,265               | 230.924                      |
| 570      | 19,318               | 14,579               | 211.075                      | 970      | 34,653               | 26,588               | 231.347                      |
| 580      | 19,678               | 14,856               | 211.702                      | 980      | 35,061               | 26,913               | 231.767                      |
| 590      | 20,039               | 15,134               | 212.320                      | 990      | 35,472               | 27,240               | 232.184                      |

TABLE A-23

Ideal-gas properties of water vapor,  $H_2O$  (Continued)

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 1000     | 35,882               | 27,568               | 232.597                      | 1760     | 70,535               | 55,902               | 258.151                      |
| 1020     | 36,709               | 28,228               | 233.415                      | 1780     | 71,523               | 56,723               | 258.708                      |
| 1040     | 37,542               | 28,895               | 234.223                      | 1800     | 72,513               | 57,547               | 259.262                      |
| 1060     | 38,380               | 29,567               | 235.020                      | 1820     | 73,507               | 58,375               | 259.811                      |
| 1080     | 39,223               | 30,243               | 235.806                      | 1840     | 74,506               | 59,207               | 260.357                      |
| 1100     | 40,071               | 30,925               | 236.584                      | 1860     | 75,506               | 60,042               | 260.898                      |
| 1120     | 40,923               | 31,611               | 237.352                      | 1880     | 76,511               | 60,880               | 261.436                      |
| 1140     | 41,780               | 32,301               | 238.110                      | 1900     | 77,517               | 61,720               | 261.969                      |
| 1160     | 42,642               | 32,997               | 238.859                      | 1920     | 78,527               | 62,564               | 262.497                      |
| 1180     | 43,509               | 33,698               | 239.600                      | 1940     | 79,540               | 63,411               | 263.022                      |
| 1200     | 44,380               | 34,403               | 240.333                      | 1960     | 80,555               | 64,259               | 263.542                      |
| 1220     | 45,256               | 35,112               | 241.057                      | 1980     | 81,573               | 65,111               | 264.059                      |
| 1240     | 46,137               | 35,827               | 241.773                      | 2000     | 82,593               | 65,965               | 264.571                      |
| 1260     | 47,022               | 36,546               | 242.482                      | 2050     | 85,156               | 68,111               | 265.838                      |
| 1280     | 47,912               | 37,270               | 243.183                      | 2100     | 87,735               | 70,275               | 267.081                      |
| 1300     | 48,807               | 38,000               | 243.877                      | 2150     | 90,330               | 72,454               | 268.301                      |
| 1320     | 49,707               | 38,732               | 244.564                      | 2200     | 92,940               | 74,649               | 269.500                      |
| 1340     | 50,612               | 39,470               | 245.243                      | 2250     | 95,562               | 76,855               | 270.679                      |
| 1360     | 51,521               | 40,213               | 245.915                      | 2300     | 98,199               | 79,076               | 271.839                      |
| 1380     | 52,434               | 40,960               | 246.582                      | 2350     | 100,846              | 81,308               | 272.978                      |
| 1400     | 53,351               | 41,711               | 247.241                      | 2400     | 103,508              | 83,553               | 274.098                      |
| 1420     | 54,273               | 42,466               | 247.895                      | 2450     | 106,183              | 85,811               | 275.201                      |
| 1440     | 55,198               | 43,226               | 248.543                      | 2500     | 108,868              | 88,082               | 276.286                      |
| 1460     | 56,128               | 43,989               | 249.185                      | 2550     | 111,565              | 90,364               | 277.354                      |
| 1480     | 57,062               | 44,756               | 249.820                      | 2600     | 114,273              | 92,656               | 278.407                      |
| 1500     | 57,999               | 45,528               | 250.450                      | 2650     | 116,991              | 94,958               | 279.441                      |
| 1520     | 58,942               | 46,304               | 251.074                      | 2700     | 119,717              | 97,269               | 280.462                      |
| 1540     | 59,888               | 47,084               | 251.693                      | 2750     | 122,453              | 99,588               | 281.464                      |
| 1560     | 60,838               | 47,868               | 252.305                      | 2800     | 125,198              | 101,917              | 282.453                      |
| 1580     | 61,792               | 48,655               | 252.912                      | 2850     | 127,952              | 104,256              | 283.429                      |
| 1600     | 62,748               | 49,445               | 253.513                      | 2900     | 130,717              | 106,605              | 284.390                      |
| 1620     | 63,709               | 50,240               | 254.111                      | 2950     | 133,486              | 108,959              | 285.338                      |
| 1640     | 64,675               | 51,039               | 254.703                      | 3000     | 136,264              | 111,321              | 286.273                      |
| 1660     | 65,643               | 51,841               | 255.290                      | 3050     | 139,051              | 113,692              | 287.194                      |
| 1680     | 66,614               | 52,646               | 255.873                      | 3100     | 141,846              | 116,072              | 288.102                      |
| 1700     | 67,589               | 53,455               | 256.450                      | 3150     | 144,648              | 118,458              | 288.999                      |
| 1720     | 68,567               | 54,267               | 257.022                      | 3200     | 147,457              | 120,851              | 289.884                      |
| 1740     | 69,550               | 55,083               | 257.589                      | 3250     | 150,272              | 123,250              | 290.756                      |

TABLE A-24

Ideal-gas properties of monatomic oxygen, O

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 2400     | 50,894               | 30,940               | 204.932                      |
| 298      | 6,852                | 4,373                | 160.944                      | 2450     | 51,936               | 31,566               | 205.362                      |
| 300      | 6,892                | 4,398                | 161.079                      | 2500     | 52,979               | 32,193               | 205.783                      |
| 500      | 11,197               | 7,040                | 172.088                      | 2550     | 54,021               | 32,820               | 206.196                      |
| 1000     | 21,713               | 13,398               | 186.678                      | 2600     | 55,064               | 33,447               | 206.601                      |
| 1500     | 32,150               | 19,679               | 195.143                      | 2650     | 56,108               | 34,075               | 206.999                      |
| 1600     | 34,234               | 20,931               | 196.488                      | 2700     | 57,152               | 34,703               | 207.389                      |
| 1700     | 36,317               | 22,183               | 197.751                      | 2750     | 58,196               | 35,332               | 207.772                      |
| 1800     | 38,400               | 23,434               | 198.941                      | 2800     | 59,241               | 35,961               | 208.148                      |
| 1900     | 40,482               | 24,685               | 200.067                      | 2850     | 60,286               | 36,590               | 208.518                      |
| 2000     | 42,564               | 25,935               | 201.135                      | 2900     | 61,332               | 37,220               | 208.882                      |
| 2050     | 43,605               | 26,560               | 201.649                      | 2950     | 62,378               | 37,851               | 209.240                      |
| 2100     | 44,646               | 27,186               | 202.151                      | 3000     | 63,425               | 38,482               | 209.592                      |
| 2150     | 45,687               | 27,811               | 202.641                      | 3100     | 65,520               | 39,746               | 210.279                      |
| 2200     | 46,728               | 28,436               | 203.119                      | 3200     | 67,619               | 41,013               | 210.945                      |
| 2250     | 47,769               | 29,062               | 203.588                      | 3300     | 69,720               | 42,283               | 211.592                      |
| 2300     | 48,811               | 29,688               | 204.045                      | 3400     | 71,824               | 43,556               | 212.220                      |
| 2350     | 49,852               | 30,314               | 204.493                      | 3500     | 73,932               | 44,832               | 212.831                      |

TABLE A-25

Ideal-gas properties of hydroxyl, OH

| $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K | $T$<br>K | $\bar{h}$<br>kJ/kmol | $\bar{u}$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|----------|----------------------|----------------------|------------------------------|----------|----------------------|----------------------|------------------------------|
| 0        | 0                    | 0                    | 0                            | 2400     | 77,015               | 57,061               | 248.628                      |
| 298      | 9,188                | 6,709                | 183.594                      | 2450     | 78,801               | 58,431               | 249.364                      |
| 300      | 9,244                | 6,749                | 183.779                      | 2500     | 80,592               | 59,806               | 250.088                      |
| 500      | 15,181               | 11,024               | 198.955                      | 2550     | 82,388               | 61,186               | 250.799                      |
| 1000     | 30,123               | 21,809               | 219.624                      | 2600     | 84,189               | 62,572               | 251.499                      |
| 1500     | 46,046               | 33,575               | 232.506                      | 2650     | 85,995               | 63,962               | 252.187                      |
| 1600     | 49,358               | 36,055               | 234.642                      | 2700     | 87,806               | 65,358               | 252.864                      |
| 1700     | 52,706               | 38,571               | 236.672                      | 2750     | 89,622               | 66,757               | 253.530                      |
| 1800     | 56,089               | 41,123               | 238.606                      | 2800     | 91,442               | 68,162               | 254.186                      |
| 1900     | 59,505               | 43,708               | 240.453                      | 2850     | 93,266               | 69,570               | 254.832                      |
| 2000     | 62,952               | 46,323               | 242.221                      | 2900     | 95,095               | 70,983               | 255.468                      |
| 2050     | 64,687               | 47,642               | 243.077                      | 2950     | 96,927               | 72,400               | 256.094                      |
| 2100     | 66,428               | 48,968               | 243.917                      | 3000     | 98,763               | 73,820               | 256.712                      |
| 2150     | 68,177               | 50,301               | 244.740                      | 3100     | 102,447              | 76,673               | 257.919                      |
| 2200     | 69,932               | 51,641               | 245.547                      | 3200     | 106,145              | 79,539               | 259.093                      |
| 2250     | 71,694               | 52,987               | 246.338                      | 3300     | 109,855              | 82,418               | 260.235                      |
| 2300     | 73,462               | 54,339               | 247.116                      | 3400     | 113,578              | 85,309               | 261.347                      |
| 2350     | 75,236               | 55,697               | 247.879                      | 3500     | 117,312              | 88,212               | 262.429                      |

TABLE A-26

Enthalpy of formation, Gibbs function of formation, and absolute entropy at 25°C, 1 atm

| Substance          | Formula                             | $\bar{h}_f^\circ$<br>kJ/kmol | $\bar{g}_f^\circ$<br>kJ/kmol | $\bar{s}^\circ$<br>kJ/kmol·K |
|--------------------|-------------------------------------|------------------------------|------------------------------|------------------------------|
| Carbon             | C(s)                                | 0                            | 0                            | 5.74                         |
| Hydrogen           | H <sub>2</sub> (g)                  | 0                            | 0                            | 130.68                       |
| Nitrogen           | N <sub>2</sub> (g)                  | 0                            | 0                            | 191.61                       |
| Oxygen             | O <sub>2</sub> (g)                  | 0                            | 0                            | 205.04                       |
| Carbon monoxide    | CO(g)                               | -110,530                     | -137,150                     | 197.65                       |
| Carbon dioxide     | CO <sub>2</sub> (g)                 | -393,520                     | -394,360                     | 213.80                       |
| Water vapor        | H <sub>2</sub> O(g)                 | -241,820                     | -228,590                     | 188.83                       |
| Water              | H <sub>2</sub> O(l)                 | -285,830                     | -237,180                     | 69.92                        |
| Hydrogen peroxide  | H <sub>2</sub> O <sub>2</sub> (g)   | -136,310                     | -105,600                     | 232.63                       |
| Ammonia            | NH <sub>3</sub> (g)                 | -46,190                      | -16,590                      | 192.33                       |
| Methane            | CH <sub>4</sub> (g)                 | -74,850                      | -50,790                      | 186.16                       |
| Acetylene          | C <sub>2</sub> H <sub>2</sub> (g)   | +226,730                     | +209,170                     | 200.85                       |
| Ethylene           | C <sub>2</sub> H <sub>4</sub> (g)   | +52,280                      | +68,120                      | 219.83                       |
| Ethane             | C <sub>2</sub> H <sub>6</sub> (g)   | -84,680                      | -32,890                      | 229.49                       |
| Propylene          | C <sub>3</sub> H <sub>6</sub> (g)   | +20,410                      | +62,720                      | 266.94                       |
| Propane            | C <sub>3</sub> H <sub>8</sub> (g)   | -103,850                     | -23,490                      | 269.91                       |
| <i>n</i> -Butane   | C <sub>4</sub> H <sub>10</sub> (g)  | -126,150                     | -15,710                      | 310.12                       |
| <i>n</i> -Octane   | C <sub>8</sub> H <sub>18</sub> (g)  | -208,450                     | +16,530                      | 466.73                       |
| <i>n</i> -Octane   | C <sub>8</sub> H <sub>18</sub> (l)  | -249,950                     | +6,610                       | 360.79                       |
| <i>n</i> -Dodecane | C <sub>12</sub> H <sub>26</sub> (g) | -291,010                     | +50,150                      | 622.83                       |
| Benzene            | C <sub>6</sub> H <sub>6</sub> (g)   | +82,930                      | +129,660                     | 269.20                       |
| Methyl alcohol     | CH <sub>3</sub> OH(g)               | -200,670                     | -162,000                     | 239.70                       |
| Methyl alcohol     | CH <sub>3</sub> OH(l)               | -238,660                     | -166,360                     | 126.80                       |
| Ethyl alcohol      | C <sub>2</sub> H <sub>5</sub> OH(g) | -235,310                     | -168,570                     | 282.59                       |
| Ethyl alcohol      | C <sub>2</sub> H <sub>5</sub> OH(l) | -277,690                     | -174,890                     | 160.70                       |
| Oxygen             | O(g)                                | +249,190                     | +231,770                     | 161.06                       |
| Hydrogen           | H(g)                                | +218,000                     | +203,290                     | 114.72                       |
| Nitrogen           | N(g)                                | +472,650                     | +455,510                     | 153.30                       |
| Hydroxyl           | OH(g)                               | +39,460                      | +34,280                      | 183.70                       |

Source of Data: From JANAF, *Thermochemical Tables* (Midland, MI: Dow Chemical Co., 1971); *Selected Values of Chemical Thermodynamic Properties*, NBS Technical Note 270-3, 1968; and *API Research Project 44* (Carnegie Press, 1953).

TABLE A-27

Properties of some common fuels and hydrocarbons

| Fuel (phase)        | Formula                                            | Molar mass, kg/kmol | Density, <sup>1</sup> kg/L | Enthalpy of vaporization, <sup>2</sup> kJ/kg | Specific heat, <sup>1</sup> $c_p$ kJ/kg·K | Higher heating value, <sup>3</sup> kJ/kg | Lower heating value, <sup>3</sup> kJ/kg |
|---------------------|----------------------------------------------------|---------------------|----------------------------|----------------------------------------------|-------------------------------------------|------------------------------------------|-----------------------------------------|
| Carbon (s)          | C                                                  | 12.011              | 2                          | —                                            | 0.708                                     | 32,800                                   | 32,800                                  |
| Hydrogen (g)        | H <sub>2</sub>                                     | 2.016               | —                          | —                                            | 14.4                                      | 141,800                                  | 120,000                                 |
| Carbon monoxide (g) | CO                                                 | 28.013              | —                          | —                                            | 1.05                                      | 10,100                                   | 10,100                                  |
| Methane (g)         | CH <sub>4</sub>                                    | 16.043              | —                          | 509                                          | 2.20                                      | 55,530                                   | 50,050                                  |
| Methanol (ℓ)        | CH <sub>4</sub> O                                  | 32.042              | 0.790                      | 1168                                         | 2.53                                      | 22,660                                   | 19,920                                  |
| Acetylene (g)       | C <sub>2</sub> H <sub>2</sub>                      | 26.038              | —                          | —                                            | 1.69                                      | 49,970                                   | 48,280                                  |
| Ethane (g)          | C <sub>2</sub> H <sub>6</sub>                      | 30.070              | —                          | 172                                          | 1.75                                      | 51,900                                   | 47,520                                  |
| Ethanol (ℓ)         | C <sub>2</sub> H <sub>6</sub> O                    | 46.069              | 0.790                      | 919                                          | 2.44                                      | 29,670                                   | 26,810                                  |
| Propane (ℓ)         | C <sub>3</sub> H <sub>8</sub>                      | 44.097              | 0.500                      | 335                                          | 2.77                                      | 50,330                                   | 46,340                                  |
| Butane (ℓ)          | C <sub>4</sub> H <sub>10</sub>                     | 58.123              | 0.579                      | 362                                          | 2.42                                      | 49,150                                   | 45,370                                  |
| 1-Pentene (ℓ)       | C <sub>5</sub> H <sub>10</sub>                     | 70.134              | 0.641                      | 363                                          | 2.20                                      | 47,760                                   | 44,630                                  |
| Isopentane (ℓ)      | C <sub>5</sub> H <sub>12</sub>                     | 72.150              | 0.626                      | —                                            | 2.32                                      | 48,570                                   | 44,910                                  |
| Benzene (ℓ)         | C <sub>6</sub> H <sub>6</sub>                      | 78.114              | 0.877                      | 433                                          | 1.72                                      | 41,800                                   | 40,100                                  |
| Hexene (ℓ)          | C <sub>6</sub> H <sub>12</sub>                     | 84.161              | 0.673                      | 392                                          | 1.84                                      | 47,500                                   | 44,400                                  |
| Hexane (ℓ)          | C <sub>6</sub> H <sub>14</sub>                     | 86.177              | 0.660                      | 366                                          | 2.27                                      | 48,310                                   | 44,740                                  |
| Toluene (ℓ)         | C <sub>7</sub> H <sub>8</sub>                      | 92.141              | 0.867                      | 412                                          | 1.71                                      | 42,400                                   | 40,500                                  |
| Heptane (ℓ)         | C <sub>7</sub> H <sub>16</sub>                     | 100.204             | 0.684                      | 365                                          | 2.24                                      | 48,100                                   | 44,600                                  |
| Octane (ℓ)          | C <sub>8</sub> H <sub>18</sub>                     | 114.231             | 0.703                      | 363                                          | 2.23                                      | 47,890                                   | 44,430                                  |
| Decane (ℓ)          | C <sub>10</sub> H <sub>22</sub>                    | 142.285             | 0.730                      | 361                                          | 2.21                                      | 47,640                                   | 44,240                                  |
| Gasoline (ℓ)        | C <sub>n</sub> H <sub>1.87n</sub>                  | 100–110             | 0.72–0.78                  | 350                                          | 2.4                                       | 47,300                                   | 44,000                                  |
| Light diesel (ℓ)    | C <sub>n</sub> H <sub>1.8n</sub>                   | 170                 | 0.78–0.84                  | 270                                          | 2.2                                       | 46,100                                   | 43,200                                  |
| Heavy diesel (ℓ)    | C <sub>n</sub> H <sub>1.7n</sub>                   | 200                 | 0.82–0.88                  | 230                                          | 1.9                                       | 45,500                                   | 42,800                                  |
| Natural gas (g)     | C <sub>n</sub> H <sub>3.8n</sub> N <sub>0.1n</sub> | 18                  | —                          | —                                            | 2                                         | 50,000                                   | 45,000                                  |

<sup>1</sup>At 1 atm and 20°C.<sup>2</sup>At 25°C for liquid fuels, and 1 atm and normal boiling temperature for gaseous fuels.<sup>3</sup>At 25°C. Multiply by molar mass to obtain heating values in kJ/kmol.

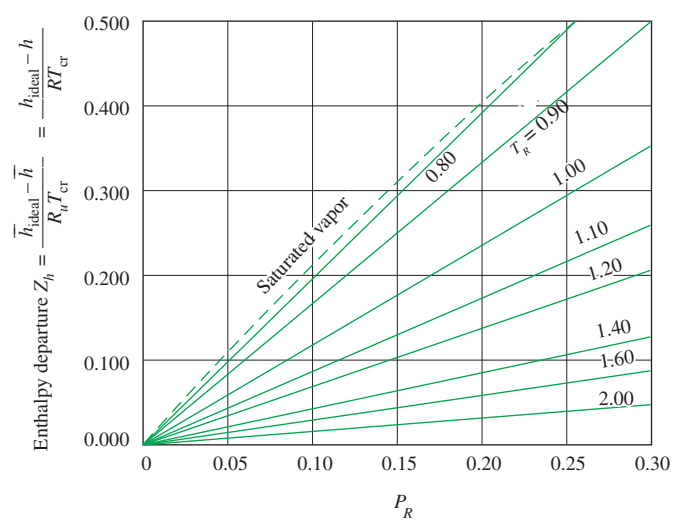
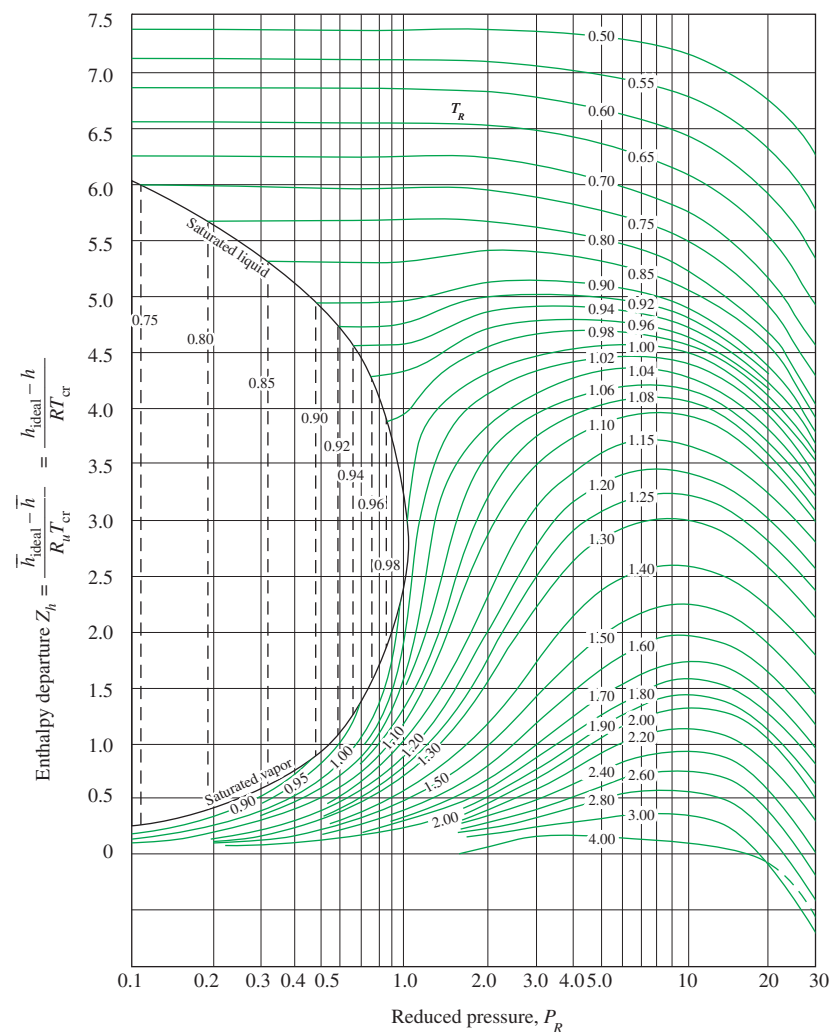
TABLE A-28

Natural logarithms of the equilibrium constant  $K_p$ 

The equilibrium constant  $K_p$  for the reaction  $\nu_A A + \nu_B B \rightleftharpoons \nu_C C + \nu_D D$  is defined as  $K_p \equiv \frac{P_C^{\nu_C} P_D^{\nu_D}}{P_A^{\nu_A} P_B^{\nu_B}}$

| Temp.,<br>K | $\text{H}_2 \rightleftharpoons 2\text{H}$ | $\text{O}_2 \rightleftharpoons 2\text{O}$ | $\text{N}_2 \rightleftharpoons 2\text{N}$ | $\text{H}_2\text{O} \rightleftharpoons \text{H}_2 + \frac{1}{2}\text{O}_2$ | $\text{H}_2\text{O} \rightleftharpoons \frac{1}{2}\text{H}_2 + \text{OH}$ | $\text{CO}_2 \rightleftharpoons \text{CO} + \frac{1}{2}\text{O}_2$ | $\frac{1}{2}\text{N}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{NO}$ |
|-------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| 298         | -164.005                                  | -186.975                                  | -367.480                                  | -92.208                                                                    | -106.208                                                                  | -103.762                                                           | -35.052                                                                      |
| 500         | -92.827                                   | -105.630                                  | -213.372                                  | -52.691                                                                    | -60.281                                                                   | -57.616                                                            | -20.295                                                                      |
| 1000        | -39.803                                   | -45.150                                   | -99.127                                   | -23.163                                                                    | -26.034                                                                   | -23.529                                                            | -9.388                                                                       |
| 1200        | -30.874                                   | -35.005                                   | -80.011                                   | -18.182                                                                    | -20.283                                                                   | -17.871                                                            | -7.569                                                                       |
| 1400        | -24.463                                   | -27.742                                   | -66.329                                   | -14.609                                                                    | -16.099                                                                   | -13.842                                                            | -6.270                                                                       |
| 1600        | -19.637                                   | -22.285                                   | -56.055                                   | -11.921                                                                    | -13.066                                                                   | -10.830                                                            | -5.294                                                                       |
| 1800        | -15.866                                   | -18.030                                   | -48.051                                   | -9.826                                                                     | -10.657                                                                   | -8.497                                                             | -4.536                                                                       |
| 2000        | -12.840                                   | -14.622                                   | -41.645                                   | -8.145                                                                     | -8.728                                                                    | -6.635                                                             | -3.931                                                                       |
| 2200        | -10.353                                   | -11.827                                   | -36.391                                   | -6.768                                                                     | -7.148                                                                    | -5.120                                                             | -3.433                                                                       |
| 2400        | -8.276                                    | -9.497                                    | -32.011                                   | -5.619                                                                     | -5.832                                                                    | -3.860                                                             | -3.019                                                                       |
| 2600        | -6.517                                    | -7.521                                    | -28.304                                   | -4.648                                                                     | -4.719                                                                    | -2.801                                                             | -2.671                                                                       |
| 2800        | -5.002                                    | -5.826                                    | -25.117                                   | -3.812                                                                     | -3.763                                                                    | -1.894                                                             | -2.372                                                                       |
| 3000        | -3.685                                    | -4.357                                    | -22.359                                   | -3.086                                                                     | -2.937                                                                    | -1.111                                                             | -2.114                                                                       |
| 3200        | -2.534                                    | -3.072                                    | -19.937                                   | -2.451                                                                     | -2.212                                                                    | -0.429                                                             | -1.888                                                                       |
| 3400        | -1.516                                    | -1.935                                    | -17.800                                   | -1.891                                                                     | -1.576                                                                    | 0.169                                                              | -1.690                                                                       |
| 3600        | -0.609                                    | -0.926                                    | -15.898                                   | -1.392                                                                     | -1.088                                                                    | 0.701                                                              | -1.513                                                                       |
| 3800        | 0.202                                     | -0.019                                    | -14.199                                   | -0.945                                                                     | -0.501                                                                    | 1.176                                                              | -1.356                                                                       |
| 4000        | 0.934                                     | 0.796                                     | -12.660                                   | -0.542                                                                     | -0.044                                                                    | 1.599                                                              | -1.216                                                                       |
| 4500        | 2.486                                     | 2.513                                     | -9.414                                    | 0.312                                                                      | 0.920                                                                     | 2.490                                                              | -0.921                                                                       |
| 5000        | 3.725                                     | 3.895                                     | -6.807                                    | 0.996                                                                      | 1.689                                                                     | 3.197                                                              | -0.686                                                                       |
| 5500        | 4.743                                     | 5.023                                     | -4.666                                    | 1.560                                                                      | 2.318                                                                     | 3.771                                                              | -0.497                                                                       |
| 6000        | 5.590                                     | 5.963                                     | -2.865                                    | 2.032                                                                      | 2.843                                                                     | 4.245                                                              | -0.341                                                                       |

Source of Data: Gordon J. Van Wylen and Richard E. Sonntag, *Fundamentals of Classical Thermodynamics*, English/SI Version, 3rd ed. (New York: John Wiley & Sons, 1986), p. 723, table A.14. Based on thermodynamic data given in JANAF, *Thermochemical Tables* (Midland, MI: Thermal Research Laboratory, The Dow Chemical Company, 1971).

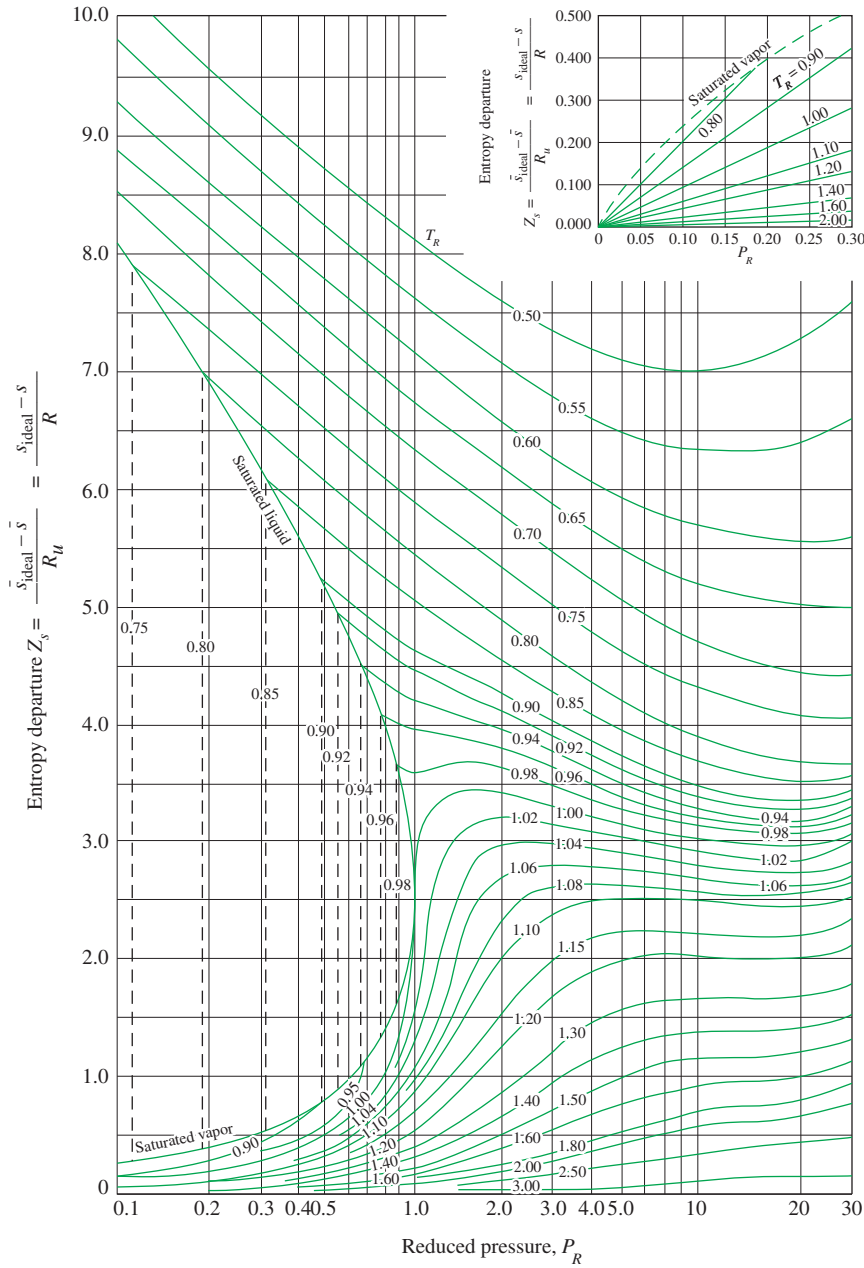


**FIGURE A-29**

Generalized enthalpy departure chart.

Source of Data: Redrawn from Gordon van Wylen and Richard Sontag, *Fundamentals of Classical Thermodynamics, (SI version), 2d ed., Wiley, New York, 1976.*



**FIGURE A-30**

Generalized entropy departure chart.

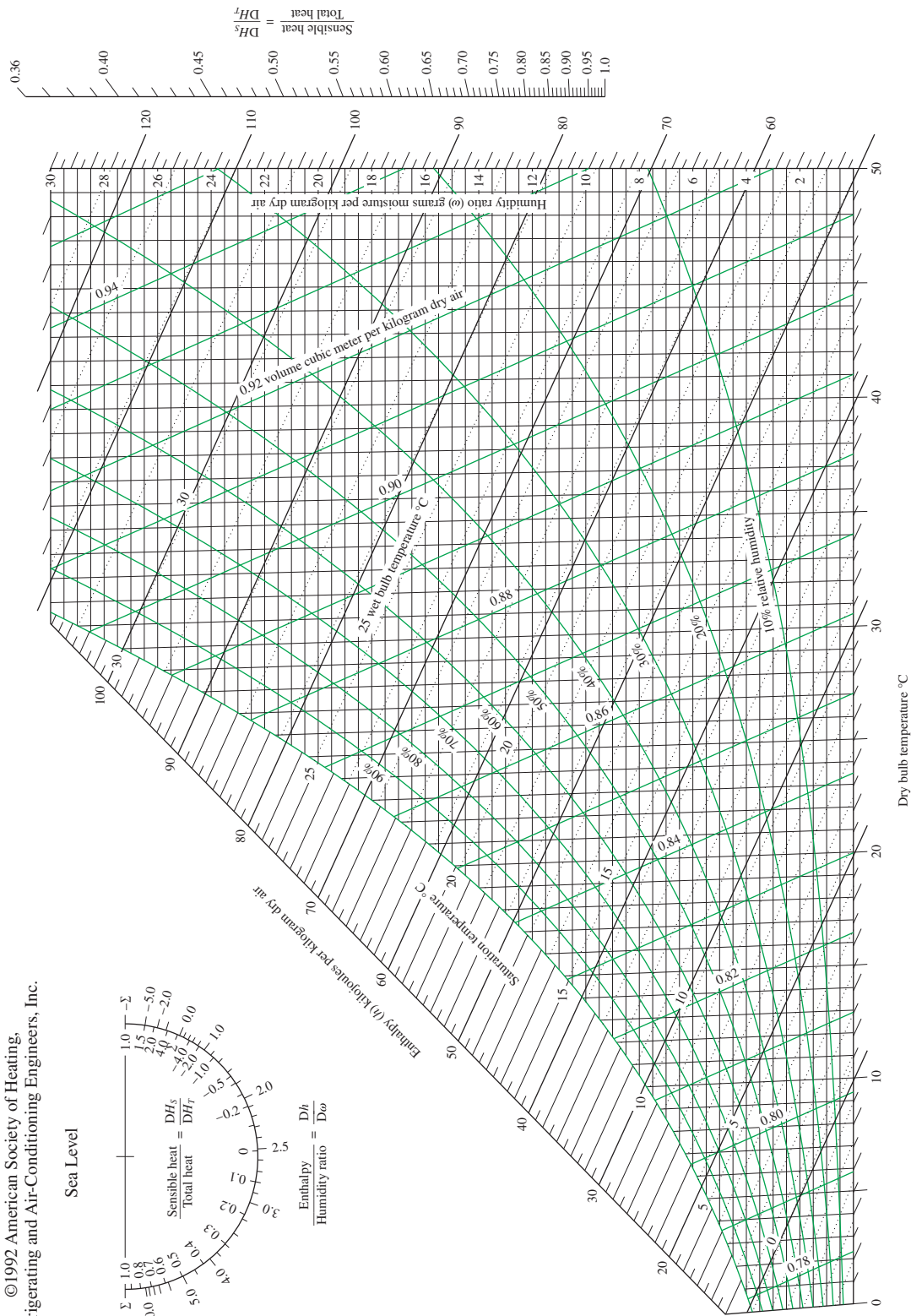
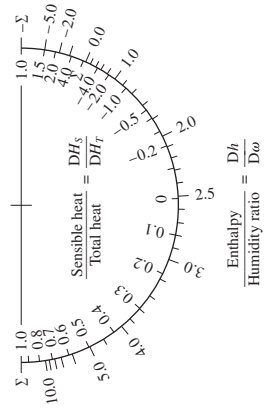
Source of Data: Redrawn from Gordon van Wylen and Richard Sontag, *Fundamentals of Classical Thermodynamics, (SI version), 2d ed.*, Wiley, New York, 1976.



**ASHRAE Psychrometric Chart No. 1**  
Normal Temperature  
Barometric Pressure: 101.325 kPa

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Sea Level



Prepared by Center for Applied Thermodynamic Studies, University of Idaho.

**FIGURE A-31**

Psychrometric chart at 1 atm total pressure.

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TABLE A-32

One-dimensional isentropic compressible-flow functions for an ideal gas with  $k = 1.4$ 

| Ma  | Ma*    | A/A*   | P/P <sub>0</sub> | ρ/ρ <sub>0</sub> | T/T <sub>0</sub> |
|-----|--------|--------|------------------|------------------|------------------|
| 0   | 0      | ∞      | 1.0000           | 1.0000           | 1.0000           |
| 0.1 | 0.1094 | 5.8218 | 0.9930           | 0.9950           | 0.9980           |
| 0.2 | 0.2182 | 2.9635 | 0.9725           | 0.9803           | 0.9921           |
| 0.3 | 0.3257 | 2.0351 | 0.9395           | 0.9564           | 0.9823           |
| 0.4 | 0.4313 | 1.5901 | 0.8956           | 0.9243           | 0.9690           |
| 0.5 | 0.5345 | 1.3398 | 0.8430           | 0.8852           | 0.9524           |
| 0.6 | 0.6348 | 1.1882 | 0.7840           | 0.8405           | 0.9328           |
| 0.7 | 0.7318 | 1.0944 | 0.7209           | 0.7916           | 0.9107           |
| 0.8 | 0.8251 | 1.0382 | 0.6560           | 0.7400           | 0.8865           |
| 0.9 | 0.9146 | 1.0089 | 0.5913           | 0.6870           | 0.8606           |
| 1.0 | 1.0000 | 1.0000 | 0.5283           | 0.6339           | 0.8333           |
| 1.2 | 1.1583 | 1.0304 | 0.4124           | 0.5311           | 0.7764           |
| 1.4 | 1.2999 | 1.1149 | 0.3142           | 0.4374           | 0.7184           |
| 1.6 | 1.4254 | 1.2502 | 0.2353           | 0.3557           | 0.6614           |
| 1.8 | 1.5360 | 1.4390 | 0.1740           | 0.2868           | 0.6068           |
| 2.0 | 1.6330 | 1.6875 | 0.1278           | 0.2300           | 0.5556           |
| 2.2 | 1.7179 | 2.0050 | 0.0935           | 0.1841           | 0.5081           |
| 2.4 | 1.7922 | 2.4031 | 0.0684           | 0.1472           | 0.4647           |
| 2.6 | 1.8571 | 2.8960 | 0.0501           | 0.1179           | 0.4252           |
| 2.8 | 1.9140 | 3.5001 | 0.0368           | 0.0946           | 0.3894           |
| 3.0 | 1.9640 | 4.2346 | 0.0272           | 0.0760           | 0.3571           |
| 5.0 | 2.2361 | 25.000 | 0.0019           | 0.0113           | 0.1667           |
| ∞   | 2.2495 | ∞      | 0                | 0                | 0                |

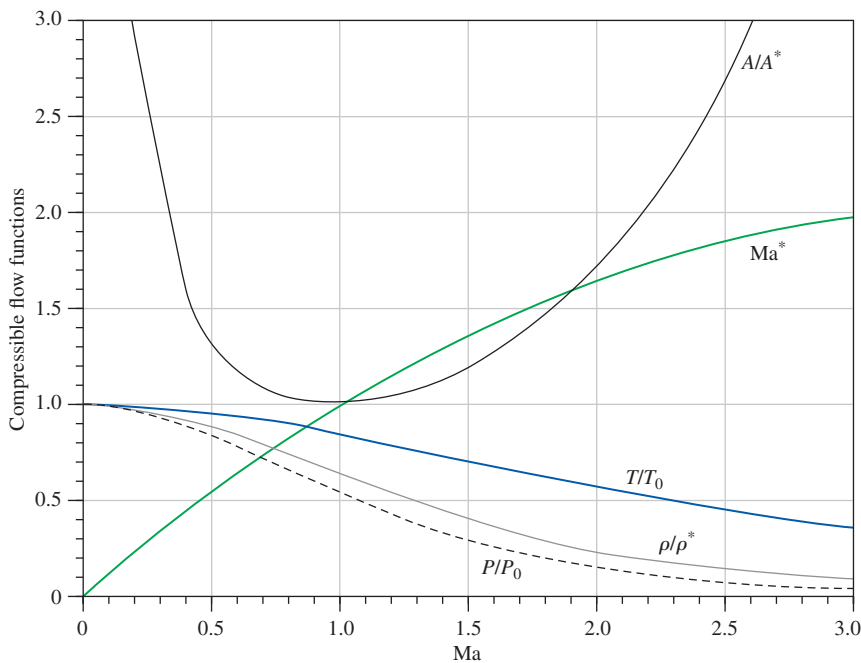
$$\text{Ma}^* = \text{Ma} \sqrt{\frac{k+1}{2+(k-1)\text{Ma}^2}}$$

$$\frac{A}{A^*} = \frac{1}{\text{Ma}} \left( \frac{2}{k+1} \right) \left( 1 + \frac{k-1}{2} \text{Ma}^2 \right)^{\frac{0.5(k+1)}{(k-1)}}$$

$$\frac{P}{P_0} = \left( 1 + \frac{k-1}{2} \text{Ma}^2 \right)^{-k/(k-1)}$$

$$\frac{\rho}{\rho_0} = \left( 1 + \frac{k-1}{2} \text{Ma}^2 \right)^{-1/(k-1)}$$

$$\frac{T}{T_0} = \left( 1 + \frac{k-1}{2} \text{Ma}^2 \right)^{-1}$$



$$T_{01} = T_{02}$$

$$\text{Ma}_2 = \sqrt{\frac{(k-1)\text{Ma}_1^2 + 2}{2k\text{Ma}_1^2 - k + 1}}$$

$$\frac{P_2}{P_1} = \frac{1 + k\text{Ma}_1^2}{1 + k\text{Ma}_2^2} = \frac{2k\text{Ma}_1^2 - k + 1}{k + 1}$$

$$\frac{\rho_2}{\rho_1} = \frac{P_2/P_1}{T_2/T_1} = \frac{(k+1)\text{Ma}_1^2}{2 + (k-1)\text{Ma}_1^2} = \frac{V_1}{V_2}$$

$$\frac{T_2}{T_1} = \frac{2 + \text{Ma}_1^2(k-1)}{2 + \text{Ma}_2^2(k-1)}$$

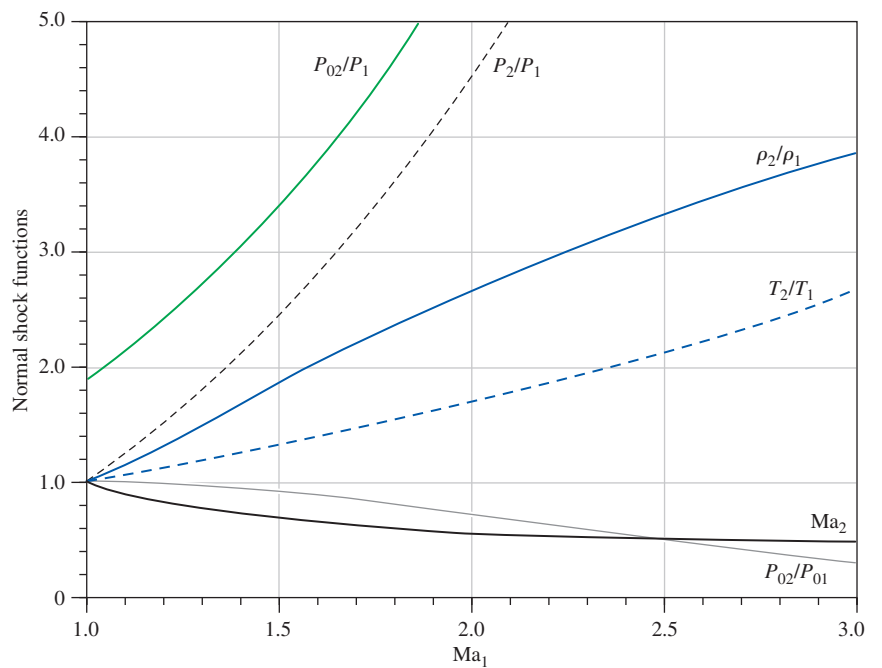
$$\frac{P_{02}}{P_{01}} = \frac{\text{Ma}_1 \left[ 1 + \frac{\text{Ma}_2^2(k-1)}{2} \right]^{(k+1)/[2(k-1)]}}{\text{Ma}_2 \left[ 1 + \frac{\text{Ma}_1^2(k-1)}{2} \right]^{(k+1)/[2(k-1)]}}$$

$$\frac{P_{02}}{P_1} = \frac{(1 + k\text{Ma}_1^2)[1 + \text{Ma}_2^2(k-1)/2]^{k/(k-1)}}{1 + k\text{Ma}_2^2}$$

TABLE A-33

One-dimensional normal-shock functions for an ideal gas with  $k = 1.4$ 

| $\text{Ma}_1$ | $\text{Ma}_2$ | $P_2/P_1$ | $\rho_2/\rho_1$ | $T_2/T_1$ | $P_{02}/P_{01}$ | $P_{02}/P_1$ |
|---------------|---------------|-----------|-----------------|-----------|-----------------|--------------|
| 1.0           | 1.0000        | 1.0000    | 1.0000          | 1.0000    | 1.0000          | 1.8929       |
| 1.1           | 0.9118        | 1.2450    | 1.1691          | 1.0649    | 0.9989          | 2.1328       |
| 1.2           | 0.8422        | 1.5133    | 1.3416          | 1.1280    | 0.9928          | 2.4075       |
| 1.3           | 0.7860        | 1.8050    | 1.5157          | 1.1909    | 0.9794          | 2.7136       |
| 1.4           | 0.7397        | 2.1200    | 1.6897          | 1.2547    | 0.9582          | 3.0492       |
| 1.5           | 0.7011        | 2.4583    | 1.8621          | 1.3202    | 0.9298          | 3.4133       |
| 1.6           | 0.6684        | 2.8200    | 2.0317          | 1.3880    | 0.8952          | 3.8050       |
| 1.7           | 0.6405        | 3.2050    | 2.1977          | 1.4583    | 0.8557          | 4.2238       |
| 1.8           | 0.6165        | 3.6133    | 2.3592          | 1.5316    | 0.8127          | 4.6695       |
| 1.9           | 0.5956        | 4.0450    | 2.5157          | 1.6079    | 0.7674          | 5.1418       |
| 2.0           | 0.5774        | 4.5000    | 2.6667          | 1.6875    | 0.7209          | 5.6404       |
| 2.1           | 0.5613        | 4.9783    | 2.8119          | 1.7705    | 0.6742          | 6.1654       |
| 2.2           | 0.5471        | 5.4800    | 2.9512          | 1.8569    | 0.6281          | 6.7165       |
| 2.3           | 0.5344        | 6.0050    | 3.0845          | 1.9468    | 0.5833          | 7.2937       |
| 2.4           | 0.5231        | 6.5533    | 3.2119          | 2.0403    | 0.5401          | 7.8969       |
| 2.5           | 0.5130        | 7.1250    | 3.3333          | 2.1375    | 0.4990          | 8.5261       |
| 2.6           | 0.5039        | 7.7200    | 3.4490          | 2.2383    | 0.4601          | 9.1813       |
| 2.7           | 0.4956        | 8.3383    | 3.5590          | 2.3429    | 0.4236          | 9.8624       |
| 2.8           | 0.4882        | 8.9800    | 3.6636          | 2.4512    | 0.3895          | 10.5694      |
| 2.9           | 0.4814        | 9.6450    | 3.7629          | 2.5632    | 0.3577          | 11.3022      |
| 3.0           | 0.4752        | 10.3333   | 3.8571          | 2.6790    | 0.3283          | 12.0610      |
| 4.0           | 0.4350        | 18.5000   | 4.5714          | 4.0469    | 0.1388          | 21.0681      |
| 5.0           | 0.4152        | 29.0000   | 5.0000          | 5.8000    | 0.0617          | 32.6335      |
| $\infty$      | 0.3780        | $\infty$  | 6.0000          | $\infty$  | 0               | $\infty$     |



**TABLE A-34**Rayleigh flow functions for an ideal gas with  $k = 1.4$ 

| Ma  | $T_0/T_0^*$ | $P_0/P_0^*$ | $T/T^*$ | $P/P^*$ | $V/V^*$ |
|-----|-------------|-------------|---------|---------|---------|
| 0.0 | 0.0000      | 1.2679      | 0.0000  | 2.4000  | 0.0000  |
| 0.1 | 0.0468      | 1.2591      | 0.0560  | 2.3669  | 0.0237  |
| 0.2 | 0.1736      | 1.2346      | 0.2066  | 2.2727  | 0.0909  |
| 0.3 | 0.3469      | 1.1985      | 0.4089  | 2.1314  | 0.1918  |
| 0.4 | 0.5290      | 1.1566      | 0.6151  | 1.9608  | 0.3137  |
| 0.5 | 0.6914      | 1.1141      | 0.7901  | 1.7778  | 0.4444  |
| 0.6 | 0.8189      | 1.0753      | 0.9167  | 1.5957  | 0.5745  |
| 0.7 | 0.9085      | 1.0431      | 0.9929  | 1.4235  | 0.6975  |
| 0.8 | 0.9639      | 1.0193      | 1.0255  | 1.2658  | 0.8101  |
| 0.9 | 0.9921      | 1.0049      | 1.0245  | 1.1246  | 0.9110  |
| 1.0 | 1.0000      | 1.0000      | 1.0000  | 1.0000  | 1.0000  |
| 1.2 | 0.9787      | 1.0194      | 0.9118  | 0.7958  | 1.1459  |
| 1.4 | 0.9343      | 1.0777      | 0.8054  | 0.6410  | 1.2564  |
| 1.6 | 0.8842      | 1.1756      | 0.7017  | 0.5236  | 1.3403  |
| 1.8 | 0.8363      | 1.3159      | 0.6089  | 0.4335  | 1.4046  |
| 2.0 | 0.7934      | 1.5031      | 0.5289  | 0.3636  | 1.4545  |
| 2.2 | 0.7561      | 1.7434      | 0.4611  | 0.3086  | 1.4938  |
| 2.4 | 0.7242      | 2.0451      | 0.4038  | 0.2648  | 1.5252  |
| 2.6 | 0.6970      | 2.4177      | 0.3556  | 0.2294  | 1.5505  |
| 2.8 | 0.6738      | 2.8731      | 0.3149  | 0.2004  | 1.5711  |
| 3.0 | 0.6540      | 3.4245      | 0.2803  | 0.1765  | 1.5882  |

$$\frac{T_0}{T_0^*} = \frac{(k+1)Ma^2[2 + (k-1)Ma^2]}{(1+kMa^2)^2}$$

$$\frac{P_0}{P_0^*} = \frac{k+1}{1+kMa^2} \left( \frac{2 + (k-1)Ma^2}{k+1} \right)^{k/(k-1)}$$

$$\frac{T}{T^*} = \left( \frac{Ma(1+k)}{1+kMa^2} \right)^2$$

$$\frac{P}{P^*} = \frac{1+k}{1+kMa^2}$$

$$\frac{V}{V^*} = \frac{\rho^*}{\rho} = \frac{(1+k)Ma^2}{1+kMa^2}$$

